

# CURVTON-205K: A Mask-Free Difficulty-Stratified Synthetic Dataset for Robust and Scalable Virtual Try-On

Dipan Bartaula<sup>1,6</sup> Anuj Shrestha<sup>1</sup> Ankit Belbase<sup>1</sup> Anish Bishwakarma<sup>1</sup> Saroj Poudel<sup>1</sup>  
Samip Ghimire<sup>1</sup> Shashank Bhatta<sup>1</sup> Eric Arazo<sup>2,3</sup> Binod Bhattarai<sup>2,4,5,6</sup>

<sup>1</sup>Department of Electronics and Computer Engineering, IOE Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal

<sup>2</sup>FogSphere, London, UK <sup>3</sup>University College Dublin, Dublin, Ireland

<sup>4</sup>School of Natural and Computing Sciences, University of Aberdeen, Aberdeen, UK

<sup>5</sup>University College London, London, UK <sup>6</sup>NAAMII, Kathmandu, Nepal

## Abstract

*Recent advances in 2D virtual try-on (VTON) have largely focused on architectural innovation, yet robust performance in diverse real-world conditions remains challenging. We argue that failure modes in VTON, such as inaccurate occlusion handling, unconstrained poses, and poor coverage of under-represented garments and appearance factors, are tightly coupled to dataset design choices and long-tail data regimes. We present a scalable data generation pipeline that coordinates specialized editing and generation models with vision-language model (VLM) guidance to synthesize triplets without requiring parsing masks or dense-pose annotations. Using this pipeline, we create CURVTON-205K, a curated synthetic dataset of 205K triplets: source person, target garment, and corresponding try-on image. CURVTON-205K is explicitly designed to broaden coverage across garments, person appearance, poses, occlusions, and visual conditions. Additionally, CURVTON-205K is organized into three difficulty tiers, enabling controlled evaluation under varying levels of difficulty and supporting difficulty-aware experimentation. We analyze the dataset diversity and demonstrate its potential to improve try-on fidelity across four VTON architectures and its cross-dataset generalization. The dataset and code will be made publicly available.*

## 1. Introduction

Virtual try-on (VTON) aims to synthesize a realistic image of a person wearing a target garment while preserving both the person’s appearance and the garment’s visual characteristics. Beyond its applications in online retail, VTON presents a challenging conditional image-generation problem involving geometric alignment, fine-grained appearance preservation, and plausible synthesis under pose changes and occlusions.

Despite substantial progress, modern VTON systems remain brittle under diverse and unconstrained conditions, raising the question of whether current limitations arise solely from model architectures or also from the coverage and structure of existing datasets.

Architectural advances have long driven 2D virtual try-on, evolving from multi-stage warping and adversarial synthesis (e.g., VITON [17], CP-VTON [37]) toward higher-fidelity and more robust pipelines that reduce reliance on human parsing cues (e.g., CP-VTON+ [25], VITON-HD [7], PF-AFN [14]). Recent diffusion-based methods [19, 28, 40] further improve photorealism and fine-grained garment preservation, with extensions targeting efficiency and consistency (e.g., [8, 24, 38]).

Despite these architectural advances, robust virtual try-on in unconstrained settings remains elusive. Our empirical analysis of representative VTON pipelines, suggests that several persistent limitations (e.g., poor handling of occlusions, limited pose extrapolation) are not solely architectural. Instead, they are strongly influenced by data coverage and dataset design. Existing datasets, such as DeepFashion [22], DressCode [27], and specialized benchmarks like People Snapshot, MoVi, StreetTryOn, and MPV [10, 11], have enabled substantial progress, but they also exhibit coverage gaps that can hinder generalization. In particular, pose and viewpoint variation, garment diversity, severe occlusions, and diverse body and appearance attributes remain difficult to cover comprehensively. Consequently, evaluation is often dominated by aggregate metrics that can obscure how models behave under increasingly challenging visual conditions. By contrast, curriculum-aware dataset structure enables evaluation across coarse but controlled tiers that jointly capture harder combinations of pose variation, occlusion, and garment-person interaction, while also making it possible to assess whether difficulty-aware training offers benefits as

visual complexity increases. Finally, many VTON pipelines rely on parsing masks, dense pose, or other auxiliary annotations, which increase annotation and preprocessing costs, and can propagate upstream errors into the try-on model, making it difficult to isolate the source of failure.

In this work, we question the assumption that architectural advances alone are sufficient for robust VTON performance. Instead, we adopt a data-centric approach that targets coverage, diversity, and controllable difficulty. We propose a scalable data generation pipeline that coordinates specialized image editing and generation models with vision-language models (VLMs) to synthesize mask-free image triplets: a source person image, a target garment image, and the corresponding try-on image. The resulting triplet format supports VTON models that do not require explicit parsing masks or dense-pose conditioning, thereby reducing reliance on auxiliary preprocessing systems and enabling simpler training and inference pipelines. Using this pipeline, we build CURVTON-205K, a curated synthetic dataset explicitly designed to broaden coverage across garment types and styles, person appearance attributes, and capture conditions. Beyond scale, the goal of CURVTON-205K is to provide a structured resource for measuring performance across controlled sources of difficulty, rather than evaluating VTON models only through aggregate paired-test performance. To improve reliability at scale, we annotate a subset of generated samples and train a lightweight quality-filtering model that filters out triplets predicted to be corrupted or inconsistent. CURVTON-205K explicitly includes occlusion-rich examples and is stratified into three tiers of increasing difficulty, enabling controlled evaluation under progressively harder conditions and supporting curriculum-aware experimentation.

Our contributions can be summarized as follows: (1) We design a scalable multi-model data generation and curation pipeline that synthesizes diverse virtual try-on triplets without requiring parsing-mask annotations or dense-pose estimation. (2) We release CURVTON-205K, a large-scale dataset containing 205,000 curated mask-free virtual try-on triplets for training and 4,500 triplets for testing, designed to broaden coverage across garment types, styles, person appearance, poses, occlusions, and capture conditions. (3) We introduce a difficulty-tiered organization of the dataset, enabling evaluation across increasingly challenging try-on scenarios and supporting curriculum-aware experimentation. (4) We conduct comprehensive experiments across four VTON architectures, analyzing try-on quality, cross-dataset generalization, and performance under different difficulty levels, thereby examining data-centric design as a complement to architectural advances.

## 2. Related Work

**VITON Approaches:** VTON methods have evolved primarily through architectural advances in alignment and generation. Early GAN-based pipelines combined explicit geometric warping with adversarial rendering, from VITON [17] and CP-VTON [37] to texture-preserving and high-resolution extensions such as CP-VTON+ [25] and VITON-HD [7]. Later methods introduced stronger conditioning signals to improve robustness to pose variation and occlusion, including semantic/layout guidance, flow-based deformation, contextual reasoning, and pose-aware synthesis [13, 18, 41]. Parser-free approaches, such as PF-AFN [14], reduce dependence on auxiliary parsing signals, while recent diffusion-based methods substantially improve photorealism and garment preservation [8, 19, 40]. Despite these advances, VTON performance remains strongly shaped by the coverage and biases of the underlying training data, motivating datasets that explicitly target diversity and controllable difficulty.

**VITON datasets:** Dataset design strongly shapes VTON performance and failure modes. DeepFashion [22] is a large-scale fashion benchmark with rich attribute supervision, but it was not designed for try-on and does not provide explicit try-on triplets. Most VTON research instead relies on shop-to-model paired benchmarks collected under controlled conditions, notably VITON-HD [7], which provides high-resolution frontal-view person–garment pairs for upper-body try-on, and DressCode [27], which broadens category coverage (upper-body, lower-body, dresses). More recent datasets incorporate more diverse capture conditions, including variations in pose, background, lighting, and occlusions, as in StreetTryOn [10] and MPV [11]. Recent advancements such as OpenVTON-Bench [20] incorporate increased poses and garment diversity, and address shortcomings of evaluation metrics. Dress-ED [34] augments DressCode person–garment pairs with instruction-conditioned edited counterparts, substantially broadening the task space to unified VTON, VTOFF, and text-guided garment editing. However, existing datasets remain constrained by coverage gaps: occlusions, garment–person interactions, and under-represented appearance attributes. Many pipelines also depend on costly and error-prone annotations such as masks or parsing maps. These limitations motivate synthetic datasets with broader coverage, mask-free triplet supervision, and controlled difficulty structure. Rather than expanding the task space, CURVTON-205K targets robustness within conventional VTON, organizing supervision across increasingly difficult pose, occlusion, viewpoint, background, and garment–person interaction conditions, which enables controlled analysis of how performance degrades across difficulty regimes rather than only in aggregate.

**Synthetic Data and Curriculum Learning:** Synthetic data has long served as a practical way to scale supervision in vi-



Figure 1. CURVTON-205K triplets: source person, in-shop garment, and synthesized try-on, all at  $1024 \times 1024$  resolution, generated without cloth segmentation masks.

sion when real-world data is costly or difficult to obtain. Classic examples include synthetic driving benchmarks (SYNTHIA [33]), or game-engine data with pixel-accurate labels [32]. Pico-Banana-400K [30] further illustrates the use of large-scale image-transformation pairs for training and benchmarking. In VTON, however, synthetic generation has been explored primarily to support limited supervision [21], rather than to construct a curated large-scale dataset explicitly designed for broad coverage mask-free supervision, and structured difficulty. Curriculum learning formalizes the idea of ordering training examples from easy to hard to facilitate optimization and improve generalization [1]. Our use of difficulty tiers is motivated by this principle but the primary goal is broader: CURVTON-205K organizes samples into levels of increasing visual complexity reflecting the main challenges in VTON (e.g. occlusions, clutter, pose). This structure supports curriculum-aware training while providing a controlled difficulty-aware lens for evaluating robustness beyond aggregated performance.

### 3. CURVTON-205K Dataset Construction

CURVTON-205K is built through the multi-stage synthesis and curation pipeline illustrated in Fig. 2. It contains four stages: garment generation, person generation, difficulty-aware person editing, and try-on synthesis, followed by quality screening. This pipeline generates mask-free triplets  $(I_p, I_c, I_t)$ , where  $I_p$  is a source person image,  $I_c$  is a target garment image, and  $I_t$  is the corresponding try-on image.

#### 3.1. Data Synthesis Pipeline

**Garment Stream:** For each garment type  $c_i \in \mathcal{D}_c$ , we sample multiple prompt variants using an LLM conditioned on a clothing-specific system prompt. Each prompt variant captures different visual attributes, such as color, material, pattern, cut, and style. We then generate multiple clothing images per prompt with the text-to-image generator

FLUX.2 [klein] 9B [2], denoted  $G_{\text{Flux9B}}$ . Overall, this yields a clothing image set  $\mathcal{I}_c = \{I_c^1, \dots, I_c^{N_c}\}$  of approximate size  $N_c \approx 43,000$  for the dataset released with this paper.

**Person Stream:** We define a probabilistic attribute dictionary  $\mathcal{D}_p$  spanning (i) body-shape categories, (ii) geographic, cultural, and ethnic style cues,<sup>1</sup> (iii) accessories, e.g., spectacles, (iv) visible physical disabilities and assistive devices, e.g., wheelchair, (v) background context, and (vi) photographic style. For photographic style, we use a 50% studio/professional and 50% in-the-wild split. We sample attributes from a factorized distribution  $\mathbb{P}_{\mathcal{D}_p}$  to control coverage and avoid long-tail collapse.

For each person image, we sample an attribute tuple from  $\mathbb{P}_{\mathcal{D}_p}$  and use an LLM with a person-specific system prompt to generate a detailed, realistic text prompt. We synthesize  $N_p = 41,000$  person images at  $1024 \times 1024$  resolution using  $G_{\text{Flux9B}}$ , yielding  $\mathcal{I}_p = \{I_p^1, \dots, I_p^{N_p}\}$ .

**Sample Difficulty Stream:** To introduce controllable difficulty, we define a collection of editing dictionaries  $\mathcal{E} = \{\mathcal{E}_{\text{easy}}, \mathcal{E}_{\text{medium}}, \mathcal{E}_{\text{hard}}\}$ , with three tiers corresponding to increasing geometric and visual complexity through pose deviation, occlusion, clutter, and viewpoint changes. For each person image  $I_p^j$  and each tier in  $\mathcal{E}$ , we sample editing keywords to generate multiple edited variants. This produces  $|\mathcal{I}_{p,\text{edit}}| = 300,000$  edited person images, i.e., 100,000 per difficulty tier.

**Merging Stream:** We pair person images, both the original set  $\mathcal{I}_p$  and the edited set  $\mathcal{I}_{p,\text{edit}}$ , with corresponding target garment images to synthesize try-on outputs, yielding candidate triplets that are subsequently curated into CURVTON-205K (Subsection 3.2). Given  $\mathcal{I}_{p,\text{total}} = \mathcal{I}_p \cup \mathcal{I}_{p,\text{edit}}$ , where  $|\mathcal{I}_{p,\text{total}}| = 341,000$ , and the garment pool  $\mathcal{I}_c$ , we sample person-garment pairs according to a pairing policy  $\mathbb{P}_{\text{pair}}$  that balances garment categories and difficulty tiers. For

<sup>1</sup>We use non-identifying appearance groupings to encourage representation diversity; see Appendix I.1 for details.



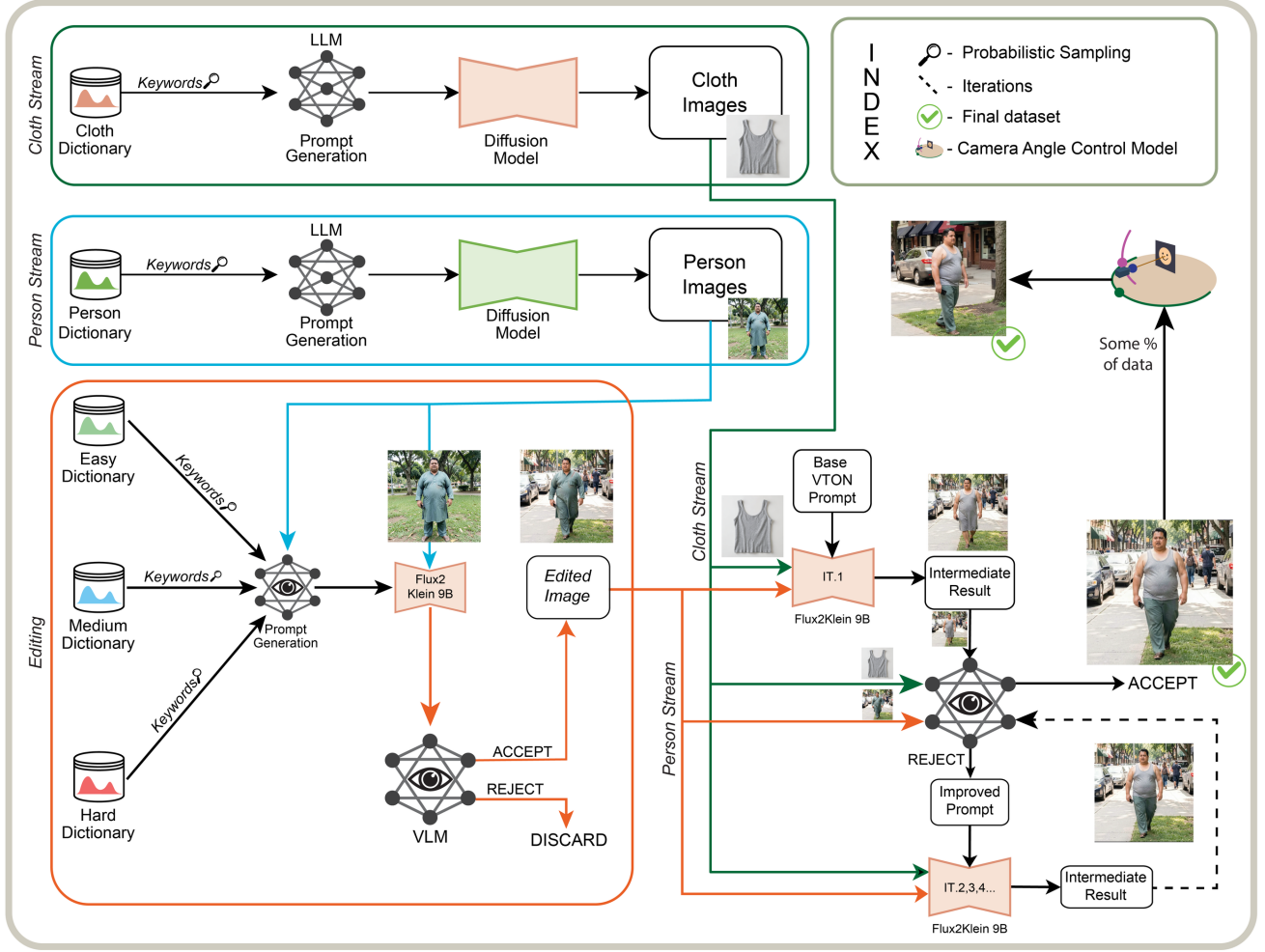


Figure 2. Data generation pipeline with clothing (orange), person (green), editing (blue), and merging (red) streams producing mask-free triplets.

each sampled pair, we generate the corresponding try-on image using  $G_{\text{Flux9B}}$  and form a candidate triplet  $x_n = (I_{p,n}, I_{c,n}, I_{t,n})$ .

To improve robustness at scale, we include an extensive validation loop, depicted in Fig. 2, that identifies inconsistent or low-quality generations and triggers regeneration when needed. Concretely, for each candidate triplet  $x_n = (I_{p,n}, I_{c,n}, I_{t,n})$ , comprising a source person image  $I_{p,n}$ , a target garment image  $I_{c,n}$ , and a synthesized try-on image  $I_{t,n}$ , we use Qwen3-VL 32B Instruct [35] as a VLM-based assessor. Through a dedicated system prompt, the assessor evaluates the consistency of the synthesized try-on image  $I_{t,n}$  with the source person image  $I_{p,n}$  and target garment image  $I_{c,n}$ . The assessment focuses on criteria such as garment faithfulness, visual artifacts, placement plausibility, and semantic consistency. If the VLM-based assessor detects an inconsistency, the sample is flagged for regeneration, the associated prompts are refined, and the image is generated again. Unlike rejection sampling, the

refinement loop uses the detected failure reason to revise the generation prompt and produce a new candidate, increasing the likelihood of obtaining valid samples for challenging configurations. This process is repeated up to  $T = 4$  times. Once  $I_{t,n}$  is deemed consistent, the triplet is accepted into  $\mathcal{D}_{\text{cand}}$  for downstream curation (see Table 6 in the Appendix for the specific improvements achieved through refinement iterations). This automated validation stage results in the candidate set  $\mathcal{D}_{\text{cand}} = \{x_n\}_{n=1}^N$  with  $N \geq 209,500$ , which is then filtered by the quality screening pipeline to obtain the final CURVTON-205K, comprising 205,000 training and 4,500 test samples.

### 3.2. Quality Screening Pipeline

To promote semantic consistency and visual quality at scale, we apply a two-stage screening process that combines limited manual annotation with automated filtering.

**Stage 1: Manual Labeling and Filter Model Training.** After refinement through the feedback loop in the merging



stream, the resulting candidates may still contain generation artifacts. Thus, we annotate a subset of 5,000 generated triplets, sampled to cover all garment categories and difficulty tiers, with a binary quality label. Annotation follows a fixed 10-question artifact rubric, reproduced in Appendix G. Each triplet is labeled *positive* if (i) the target garment identity, including type, color, and pattern, is preserved, (ii) the garment is plausibly worn with correct placement and deformation, and (iii) the output is free of major artifacts; otherwise it is labeled *negative*. This labeled set is used to train a lightweight quality filter based on ViT-Small [12], denoted  $f_{\text{ViT}}$ . The filter receives the three triplet images concatenated channel-wise, denoted by  $\phi(I_{p,n}, I_{c,n}, I_{t,n})$ , and is trained with binary cross-entropy to predict whether the sample is positive or negative. The model prediction is used in Stage 2 as a proxy for sample quality.

**Stage 2: Automated Screening.** We deploy the trained ViT filter  $f_{\text{ViT}}$  to score the pool of generated candidate triplets  $\mathcal{D}_{\text{cand}}$  as  $s_n = f_{\text{ViT}}(\phi(I_{p,n}, I_{c,n}, I_{t,n}))$  and retain samples satisfying  $s_n \geq \tau$ ; we use  $\tau=0.85$  unless otherwise stated. To assess potential distribution shifts introduced by filtering, we monitor acceptance rates across difficulty tiers, garment categories, and appearance and capture-condition attributes, including possible collapse in long-tail regimes. This two-stage procedure, based on an annotated subset of samples, scales curation to large volumes while preserving defined quality criteria, resulting in the final CURVTON-205K dataset consisting of 205,000 mask-free triplets for training and 4,500 triplets for testing.

### 3.3. Dataset Overview

CURVTON-205K is a curated synthetic dataset of mask-free virtual try-on triplets designed to support model training and difficulty-stratified evaluation. Each sample contains a source person image, a target garment image, and the corresponding synthesized try-on image. CURVTON-205K provides direct triplet supervision without parsing masks or dense-pose annotations, thereby supporting pipelines that reduce dependence on auxiliary preprocessing models. The training and test partitions are disjoint by construction,  $\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \emptyset$ , so no triplet used for evaluation occurs anywhere in the training data.

The dataset is constructed to broaden coverage across garments, subjects, and capture conditions. Garments are drawn from a dictionary of 200 types, including common fashion categories as well as long-tail, traditional, and culturally specific clothing. Person prompts are sampled from a factorized attribute dictionary spanning body shape, geographic and cultural style cues, accessories, visible assistive devices, background context, and photographic style. In addition, CURVTON-205K is stratified into three difficulty tiers (easy, medium, and hard) introduced through editing operations designed to increase pose variation, occlusion, clutter, view-



Figure 3. Try-on examples across difficulty tiers. Rows show **Easy**, **Medium**, and **Hard** samples, with increasing pose variation, occlusion, background complexity, and garment structure. Each triplet shows the source person, target garment, and synthesized try-on result.

point change, and garment-person interaction complexity. This tiered structure enables difficulty-specific analysis and supports curriculum-aware training experiments.

Table 1 contextualizes CURVTON-205K with respect to widely used VTON datasets. Prior benchmarks are predominantly real, shop-to-model paired datasets and are commonly used with auxiliary signals such as masks, parsing maps, or pose estimates. They also typically lack explicit stratification by difficulty. In contrast, CURVTON-205K provides large-scale mask-free triplets with explicit diversity targets and difficulty tiers, including targeted coverage of long-tail garment styles that may be under-represented in standard benchmarks.

## 4. Experiments

First, we evaluate representative VTON approaches on CURVTON-205K to establish the dataset as a benchmark and analyze performance across difficulty tiers. We then assess cross-dataset generalization by training models with different data sources and evaluating them on standard paired test splits from VITON-HD [7] and DressCode [27]. Second, we analyze CURVTON-205K from a data-centric perspective comparing against broader fashion corpora and in-the-wild benchmarks, e.g., StreetTryOn [10]. Third, we use the proposed difficulty tiers to study curriculum-aware training as an ablation. Paired perceptual metrics (SSIM and

Table 1. Comparison of virtual try-on datasets across key characteristics. Diversity coverage summarizes whether a dataset explicitly accounts for both garment diversity and person-appearance diversity: Limited indicates neither, Moderate indicates one of the two, and Broad indicates both. All listed datasets are publicly available except WildVTON [42].

| Dataset             | Train       | Test        | Type          | Mask-Free | Diff. tiers | In-the-Wild | Cloth Pairs | Diversity coverage |
|---------------------|-------------|-------------|---------------|-----------|-------------|-------------|-------------|--------------------|
| VITON [17]          | 14K         | 2K          | Real          | ×         | ×           | ×           | ✓           | Limited            |
| VITON-HD            | 12.3K       | 1.4K        | Real          | ×         | ×           | ×           | ✓           | Limited            |
| DressCode           | 50K         | 3K          | Real          | ×         | ×           | ×           | ✓           | Limited            |
| StreetTryOn         | 12.5K       | 1.9K        | Real          | ×         | ×           | ✓           | ×           | Moderate           |
| WildVTON            | N/A         | 1.2K        | Hybrid        | ✓         | ×           | ✓           | ✓           | Limited            |
| OpenVTON-Bench      | ~90K        | ~10K        | Real          | ✓         | ×           | ✓           | ✓           | Moderate           |
| <b>CURVTON-205K</b> | <b>205K</b> | <b>4.5K</b> | <b>Synth.</b> | ✓         | ✓           | ✓           | ✓           | Broad              |

LPIPS) are computed on the corresponding test splits and distribution-level metrics (FID and KID), are computed over the full test partition. Qualitative examples across difficulty tiers are shown in Fig. 3.

#### 4.1. VTON-Based Evaluations

CURVTON-205K defines a stratified evaluation benchmark with explicit difficulty tiers (*easy/medium/hard*) and garment-type metadata. This enables reporting performance by tier and clothing category, providing targeted insight into failure modes beyond aggregate scores. Table 2 reports the performance of representative VTON approaches trained and evaluated on CURVTON-205K. The reported results show model FID across tiers of difficulty and cloth types; full results, including SSIM and LPIPS and an additional DCI-VTON [16] baseline, are provided in Appendix C. For each method, we report the original pretrained model (*Zero Shot*) and the corresponding model trained or fine-tuned on CURVTON-205K. Before CURVTON adaptation, most models obtain progressively higher FID on harder tiers and exhibit substantial performance differences across garment categories. Training on CURVTON-205K substantially improves in-domain performance within every reported difficulty tier and garment-category subset, while the persistent easy-to-hard degradation suggests that the tier structure remains informative after adaptation. Because all garment-category subsets contain the same number of samples, their FID scores can be compared directly. For all three models, CURVTON training reduces the range between the lowest and highest category-specific FID—from 12.80 to 2.80 for OOTDiffusion, from 11.19 to 4.95 for CatVTON, and from 10.40 to 7.77 for IDM-VTON. This consistent reduction indicates more uniform performance across garment types, supporting the claim that CURVTON provides broad and relatively balanced category coverage rather than benefiting only a narrow subset.

Table 3 evaluates cross-benchmark transfer under three training settings: the real VTON datasets used in the corresponding baselines, CURVTON-205K alone, and their combination, while keeping the test protocols fixed. We compare CatVTON, IDM-VTON, StableVITON, and OOTDiffusion trained on the individual VITON-HD and DressCode train

sets as done in the respective papers. The *Real + CURVTON* setting augments each corresponding real training set with CURVTON-205K. The results for StreetTryOn, as in the original paper, use a combination of unpaired samples from different real datasets as the real-data baseline, to which CURVTON-205K is added in the combined setting. On the paired VITON-HD and DressCode tests, in-domain real-data training yields stronger paired reconstruction metrics than training on CURVTON alone, revealing the remaining synthetic-to-real gap, although CURVTON-trained models remain competitive and can improve distribution-level metrics. Crucially, combining real and CURVTON data numerically improves most of the reported measurements over real-only training across all four architectures and both datasets, with two ties and only one minor regression. SSIM increases by 0.01–0.03 and FID decreases by 0.11–0.55 across all comparisons; LPIPS improves in six comparisons and remains unchanged in two, while KID improves in seven of eight comparisons. On StreetTryOn, CURVTON-trained models consistently achieve lower FID/KID than the real-data baselines across all architectures. The combined setting achieves the best FID and KID for every architecture, reducing FID by 4.86–8.92 and KID by 1.03–3.50 relative to real-only training. These results suggest that CURVTON-205K does not replace in-domain real supervision but effectively complements it with transferable synthetic supervision. In particular, improvements over real-only training demonstrate that its utility is not confined to reproducing the generator’s synthetic distribution, supporting the practical use of CURVTON to extend existing real datasets with broader coverage, particularly under out-of-benchmark, in-the-wild evaluation.

Table 5 reports ground-truth-free, in-the-wild evaluation of OOTDiffusion trained on different data sources, alongside pretrained FLUX and IDM-VTON references. Across the automatic criteria, training on CURVTON-205K yields the strongest results, achieving the highest garment-image alignment (CLIP-I) and visual consistency assessed by the VLM evaluator. Human evaluation further supports these trends through blind pairwise comparisons aggregated over 20 annotators who select the preferred images across models, marked with the same 10-question rubric (Appendix G). Human Preference scores report the percentage of annotator

preferences assigned to each model, normalized to sum to 100% within each evaluation criterion. The model trained on CURVTON-205K achieves the highest preference for both try-on quality (35%) and realism (28%). Together, these results provide complementary evidence that CURVTON training improves perceptual quality and visual plausibility under less controlled, in-the-wild conditions.

## 4.2. Task-Free Dataset Analysis

We provide task-free analyses to characterize the diversity and coverage of CURVTON-205K independently of any downstream VTON model. All results are computed under a fixed protocol using the same pretrained feature extractors and dataset-level descriptors, enabling consistent comparisons across datasets without confounding effects from downstream architecture or training choices. Table 4 compares CURVTON-205K with existing VTON and fashion datasets across pose, occlusion, background, illumination, body-shape, garment, and camera dimensions; the reported metrics are defined in Appendix F. Fig. 4 complements this comparison by summarizing visual variation relevant to in-the-wild settings. The left panel shows normalized spatial occlusion heatmaps, the middle panel reports luminance distributions, and the right panel summarizes pose distributions across the three difficulty tiers. These visualizations characterize variation in occlusion patterns and illumination conditions while exposing differences in pose coverage across tiers. Additional t-SNE [36] projections of garment and facial features are provided in Fig. 6 in the Appendix to visualize variation in the corresponding embedding spaces, including how traditional and modern garments are spread across the garment embedding space.

Among the full-dataset comparisons, CURVTON-205K achieves the highest measured variation in background objects, luminance, body shape, garments, and camera viewpoint, whereas StreetTryOn retains the highest pose variation and strongest occlusion measures. Within CURVTON, the progression from easy to hard is accompanied by increasing pose variance, occlusion complexity, body-shape variation, and camera-viewpoint variation. These results support the intended difficulty structure and show that CURVTON provides broad, complementary coverage across several measured dimensions, rather than uniformly dominating existing datasets on every criterion.

For an intuitive complement to these results, Fig. 5 shows representative samples from CURVTON-205K. The top row displays sampled person images and the bottom row shows sampled in-shop garments, illustrating variation in body shape, facial-appearance cues, garment categories, colors, and patterns. Overall, the quantitative and qualitative diagnostics provide evidence that CURVTON-205K covers a range of measured appearance factors, including garments, facial appearance, and body shape, as well as conditions

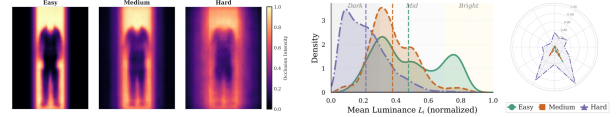


Figure 4. Deployment realism of CURVTON-205K data via occlusions (left), illumination (center), and pose (right) distributions for the three difficulty tiers: easy, medium, and hard.



Figure 5. Diversity in CURVTON-205K: sampled person images (top) and in-shop garments (bottom) illustrating variation in identity cues, body shape, garment categories, and appearance.

relevant to in-the-wild evaluation, such as background, lighting, and occlusion. These analyses characterize visual and embedding-space coverage but should not be interpreted as a demographic or cultural fairness assessment.

## 5. Limitations

While CURVTON-205K substantially expands mask-free supervision and controllable difficulty for virtual try-on, it inherits limitations common to large-scale generative data pipelines. Because the dataset is synthetically generated, its distribution may not fully capture real deployment settings, including camera characteristics, garment materials, lighting, and person–garment interactions; although our cross-dataset experiments indicate transfer to real-image benchmarks, further synthetic-to-real adaptation studies are needed to characterize performance in specific deployment domains. In a small fraction of cases, fine-grained garment attributes such as pockets or buttons are not perfectly preserved between the target garment and the synthesized try-on, leading to minor attribute drift; these effects are more likely in the hard tier, where reconstruction is intrinsically ambiguous. Finally, although CURVTON-205K broadens coverage across garment styles, person appearance, backgrounds, and visual conditions, it should not be read as exhaustively or demographically representing real-world populations and fashion practices. These limitations motivate future work on noise-aware training, generator-diverse synthesis, synthetic-to-real adaptation, and improved auditing of quality and representation in virtual try-on datasets.

## 6. Conclusion

We presented CURVTON-205K, a data-centric virtual try-on resource built around the premise that robust VTON depends not only on model architecture, but also on how training data



Table 2. Stratified FID performance (lower is better) of all models trained and evaluated on CURVTON-205K. Results are reported overall and by difficulty tier and cloth type: upper (UB) and lower body (LB); garment (G); and traditional (Tr) and non-traditional (NTr). Full results with additional metrics are provided in Appendix C.

|                                      |           | Overall   | Difficulty Tier |       |       | Cloth Type |       |       |       |       |
|--------------------------------------|-----------|-----------|-----------------|-------|-------|------------|-------|-------|-------|-------|
|                                      |           |           | Easy            | Med.  | Hard  | UB         | LB    | G     | Tr    | NTr   |
| OOTDiffusion[40]                     | Zero Shot | 18.90     | 25.20           | 24.40 | 25.50 | 19.50      | 27.15 | 29.70 | 29.95 | 17.15 |
|                                      | Training  | 8.75      | 9.88            | 11.54 | 14.12 | 11.42      | 11.83 | 13.79 | 14.22 | 13.83 |
| CatVTON [9]                          | Zero Shot | 23.85     | 22.55           | 24.40 | 27.55 | 22.60      | 23.55 | 28.90 | 31.35 | 20.16 |
|                                      | Training  | 9.14      | 9.14            | 9.23  | 12.88 | 10.62      | 11.34 | 14.94 | 15.57 | 12.70 |
| IDM-VTON [8]                         | Zero Shot | 17.05     | 16.20           | 17.30 | 19.30 | 16.70      | 17.25 | 22.00 | 25.55 | 15.15 |
|                                      | Training  | 8.44      | 8.28            | 8.85  | 10.29 | 10.21      | 10.57 | 13.54 | 15.75 | 7.98  |
| <b>FLUX.2 [klein] 9B (Generator)</b> |           | Zero Shot | 14.50           | 13.10 | 15.35 | 17.95      | 14.75 | 16.05 | 17.50 | 23.27 |
|                                      |           |           |                 |       |       |            |       |       |       | 20.23 |

Table 3. Cross-dataset generalization on VITON-HD (VHD), DressCode (DC), and StreetTryOn. For VHD and DC, SSIM and LPIPS measure paired reconstruction quality, while FID and KID measure distribution-level quality. StreetTryOn is evaluated in the unpaired in-the-wild setting using FID and KID.

| Model             | Train Data     | VITON-HD |        |       |      | DressCode |        |       |      | StreetTryOn |      |
|-------------------|----------------|----------|--------|-------|------|-----------|--------|-------|------|-------------|------|
|                   |                | SSIM↑    | LPIPS↓ | FID↓  | KID↓ | SSIM↑     | LPIPS↓ | FID↓  | KID↓ | FID↓        | KID↓ |
| CatVTON [9]       | Real baseline  | 0.87     | 0.11   | 9.01  | 1.09 | 0.89      | 0.10   | 6.14  | 1.40 | 25.31       | 6.95 |
|                   | CURVTON-205K   | 0.78     | 0.13   | 11.74 | 1.41 | 0.80      | 0.12   | 7.98  | 1.79 | 21.36       | 5.42 |
|                   | Real + CURVTON | 0.89     | 0.10   | 8.72  | 1.04 | 0.90      | 0.10   | 5.92  | 1.36 | 20.45       | 5.18 |
| IDM-VTON [8]      | Real baseline  | 0.85     | 0.12   | 9.26  | 1.27 | 0.89      | 0.12   | 14.13 | 6.03 | 23.62       | 6.18 |
|                   | CURVTON-205K   | 0.80     | 0.13   | 11.02 | 1.46 | 0.83      | 0.13   | 16.94 | 6.88 | 19.08       | 4.96 |
|                   | Real + CURVTON | 0.87     | 0.11   | 8.98  | 1.22 | 0.90      | 0.11   | 13.68 | 5.86 | 18.55       | 4.82 |
| StableVITON [19]  | Real baseline  | 0.85     | 0.15   | 8.93  | 2.54 | 0.88      | 0.11   | 13.60 | 6.05 | 23.15       | 4.63 |
|                   | CURVTON-205K   | 0.77     | 0.18   | 11.01 | 3.08 | 0.80      | 0.13   | 17.08 | 7.11 | 18.44       | 3.71 |
|                   | Real + CURVTON | 0.86     | 0.14   | 8.71  | 2.46 | 0.89      | 0.11   | 13.05 | 5.72 | 17.95       | 3.60 |
| OOTDiffusion [40] | Real baseline  | 0.83     | 0.14   | 11.73 | 3.15 | 0.85      | 0.12   | 9.21  | 2.66 | 27.62       | 8.08 |
|                   | CURVTON-205K   | 0.75     | 0.16   | 12.21 | 4.90 | 0.77      | 0.15   | 17.73 | 3.45 | 23.31       | 7.25 |
|                   | Real + CURVTON | 0.84     | 0.13   | 11.62 | 3.19 | 0.87      | 0.11   | 9.08  | 2.61 | 18.70       | 4.58 |

Table 4. Task-free dataset diversity comparison across seven dimensions, each split into Complexity (C) and Diversity (D) sub-metrics.

| Dataset            | Pose                              | Occlusion               |                          | Background         |                    |                    | Illumination |                    | Body Shape                         | Garment             |                     | Camera                       |                                |
|--------------------|-----------------------------------|-------------------------|--------------------------|--------------------|--------------------|--------------------|--------------|--------------------|------------------------------------|---------------------|---------------------|------------------------------|--------------------------------|
|                    | $\sigma^2_{\text{pose}} \uparrow$ | Occl. Cmplx. $\uparrow$ | Person Vis. $\downarrow$ | BGE (C) $\uparrow$ | BSE (D) $\uparrow$ | BOD (D) $\uparrow$ | IGM          | LVM (D) $\uparrow$ | $\sigma^2_{\text{shape}} \uparrow$ | Gmt Var. $\uparrow$ | Gmt Div. $\uparrow$ | $\hat{\sigma}_\phi \uparrow$ | $\hat{\sigma}_\theta \uparrow$ |
| VITON-HD           | 145.55                            | 0.52                    | 0.31                     | 2.93               | 0.01               | 0.29               | 0.60         | 0.06               | 0.54                               | 475.81              | 0.04                | 0.070                        | 0.090                          |
| DressCode          | 127.79                            | 0.45                    | 0.45                     | 1.60               | 0.07               | 0.56               | 0.73         | 0.10               | 0.76                               | 669.59              | 0.26                | 0.140                        | 0.070                          |
| StreetTryOn        | <b>204.48</b>                     | 0.62                    | <b>0.23</b>              | 4.57               | 0.54               | 3.15               | 0.45         | <u>0.10</u>        | 0.47                               | 621.65              | 0.15                | 0.120                        | 0.140                          |
| Dress-ED           | 129.92                            | 0.43                    | 0.43                     | 1.63               | 0.07               | 0.60               | 0.80         | 0.09               | 0.77                               | 725.11              | 0.32                | 0.170                        | 0.113                          |
| CURVTON Easy       | 66.68                             | 0.53                    | 0.33                     | 4.00               | 0.43               | 3.45               | 0.50         | <b>0.16</b>        | 0.62                               | <b>831.32</b>       | <b>0.52</b>         | 0.022                        | 0.066                          |
| CURVTON Medium     | 178.45                            | 0.62                    | 0.27                     | 4.78               | <b>0.57</b>        | <b>5.17</b>        | 0.37         | 0.09               | 0.74                               | 821.99              | 0.49                | 0.347                        | 0.292                          |
| CURVTON Hard       | 201.88                            | <b>0.66</b>             | 0.24                     | <b>4.81</b>        | 0.50               | 5.09               | 0.33         | 0.09               | <b>1.02</b>                        | 827.66              | <b>0.51</b>         | <b>1.197</b>                 | <b>0.422</b>                   |
| CURVTON-205K (All) | 182.22                            | 0.61                    | 0.28                     | 4.54               | 0.50               | 4.61               | 0.40         | 0.12               | 0.89                               | 823.06              | 0.50                | <u>0.683</u>                 | <u>0.295</u>                   |

Table 5. In-the-wild evaluation of virtual try-on models. CLIP-I [31] and VLM measure semantic alignment, and Human Preference captures subjective evaluation of the generated Try-On images without ground truth supervision.

| Train Data          | CLIP-I↑     | VLM↑        | Human Preference (%) |            |
|---------------------|-------------|-------------|----------------------|------------|
|                     |             |             | Quality↑             | Realism↑   |
| VITON-HD            | 0.71        | 0.69        | 10%                  | 11%        |
| DressCode           | 0.74        | 0.73        | 12%                  | 13%        |
| VHD + DC            | 0.77        | 0.76        | 14%                  | 15%        |
| FLUX (Pretrained)   | 0.76        | 0.74        | 14%                  | 14%        |
| IDM-VTON            | 0.80        | 0.78        | 15%                  | 19%        |
| <b>CURVTON-205K</b> | <b>0.83</b> | <b>0.86</b> | <b>35%</b>           | <b>28%</b> |

covers challenging visual conditions. CURVTON-205K provides 205,000 curated mask-free training triplets and 4,500 test triplets, organized into three difficulty tiers that progressively vary pose, occlusion, viewpoint, background complexity, and garment-person interaction. A scalable generation and curation pipeline combines specialized generation and editing models with iterative VLM-guided refinement and learned quality filtering. Across four VTON architectures, difficulty-stratified evaluation reveals consistent performance degradation as visual complexity increases, while augmenting real training data with CURVTON improves most cross-benchmark measurements and substantially improves performance in the in-the-wild StreetTryOn setting. Task-free analyses further show broad, complementary cov-

erage across several measured visual dimensions. Together, these results position CURVTON-205K as a source of complementary synthetic supervision and a structured resource for studying where VTON systems degrade and how data design can improve robustness beyond aggregate benchmark performance.

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## A. Code and Data Availability

The code for the data generation pipeline, the difficulty-tier editing dictionaries, the evaluation scripts, and the diversity metric implementations used in this paper, together with the CURVTON-205K dataset itself, will be made publicly available.

## B. Additional Curation Details

This section provides additional details on the automated and human-assisted curation stages used to construct CURVTON-205K. We report how the validation loop reduces artifact rates across difficulty tiers before the final quality-screening step.

### B.1. VLM-Based Validation

The VLM-based validation loop evaluates each candidate try-on triplet according to garment faithfulness, semantic consistency, placement plausibility, and visible artifacts. Samples flagged as inconsistent are regenerated for up to four refinement attempts. Accordingly, the iterative results below include the initial generation followed by successive refinement rounds.

### B.2. Choice of the Quality-Filter Threshold

The automated screening stage retains a candidate triplet when the quality filter assigns it a score  $s_n \geq \tau$ , and the main text uses  $\tau=0.85$ . This value is deliberately far above the 0.5 decision boundary that a binary classifier would ordinarily imply, for the following reason.

Note that  $\tau$  governs the *ViT* quality filter only. The earlier closed-loop VLM assessor uses its own, separate acceptance threshold of 0.8, reported as VLM Thr in Table 6; the two thresholds belong to different stages and are not interchangeable.

By the time a triplet reaches the filter it has already survived the closed-loop VLM validation stage of Section B.1, which regenerates inconsistent samples for up to four attempts. That stage alone brings the artifact rate down substantially (Table 6), so the pool entering the filter is already a reasonably clean baseline rather than raw generator output. Setting  $\tau=0.5$  would therefore add very little: most of the pool sits above it, and the filter would mainly re-confirm decisions the VLM loop had already made.

The filter is instead used as a second, stricter pass whose purpose is to raise the quality floor further rather than to perform the initial separation. A threshold of 0.85 keeps only samples the filter is confident about, removing the residual borderline cases that the VLM loop accepted but that still contain mild artifacts. Because the candidate pool ( $N \geq 209,500$ ) exceeds the target dataset size (205,000 training and 4,500 test triplets), we can afford this strictness: discarding borderline samples costs coverage we can replace

by generating more, whereas retaining them would permanently lower the quality of the released data. Acceptance rates are monitored across difficulty tiers, garment categories, and appearance attributes so that the stricter threshold does not silently prune long-tail groups.

### B.3. Cross-Generator and Cross-Evaluator Robustness

To assess whether the observed effectiveness of the iterative validation procedure depends on a particular image generator or VLM evaluator, we conduct a small-scale cross-generator and cross-evaluator robustness study. We evaluate multiple generator-evaluator configurations and measure artifact rates before iterative refinement (Iteration 0) and after four refinement rounds (Iteration IV) across the Easy, Medium, and Hard difficulty tiers. This analysis complements Table 6 by examining whether the artifact-reduction trend persists beyond the primary FLUX.2-klein-9B and Qwen3-VL 32B configuration.

Across all tested generator-evaluator configurations, iterative refinement substantially reduces artifact rates across the three difficulty tiers. For FLUX.2-klein-9B, the reduction remains consistent when replacing the primary Qwen3-VL 32B evaluator with Qwen3-VL-30B-A3B Thinking or PaliGemma, although the absolute residual artifact rates vary across evaluators. Similarly, substantial reductions are observed for FLUX.2-klein-4B, Qwen-Image-Edit-2509, and Nano Banana Pro. These results indicate that the observed artifact-reduction trend is not restricted to a single generator-evaluator configuration. Because this analysis is conducted on a small-scale screening subset, we interpret it as evidence of robustness of the refinement trend rather than as a comprehensive ranking of generators or evaluators.

## C. Experiments and Additional Results

This section complements the main experimental analysis with implementation details, extended benchmark tables, scalability analyses, and additional curriculum ablations. These results provide fuller quantitative support for the compact tables reported in the main text.

### C.1. Experimental Details

We summarize the compute environment, implementation details, and training setup used for data generation and model evaluation.

**Compute Environment.** Initial experiments and prototype development were conducted on the Lightning.ai and Vast.ai cloud GPU platforms. The majority of our experimental work, including the complete data generation pipeline, was executed on Vast.ai infrastructure. Data generation alone required over 3,000 H200 GPU hours, and training and eval-

Table 6. CURVTON synthetic try-on generation process and feedback iterations. Artifact rates are reported by split and difficulty tier after each stage of the pipeline, with VLM Thr denoting the acceptance threshold of the VLM-based assessor in the closed-loop validation stage. This is distinct from the ViT quality-filter threshold  $\tau=0.85$  applied in the final automated screening step (Section B.2). The final curated dataset contains 205K training triplets and 4.5K test triplets. The test set is additionally manually verified by human annotators to remove visually apparent artifacts and improve the reliability of the final evaluation set across all difficulty tiers.

| Stage                        | Iteration<br>(VLM Thr = 0.8) | Train Artifacts (%) |        |      | Test Artifacts (%) |        |      |
|------------------------------|------------------------------|---------------------|--------|------|--------------------|--------|------|
|                              |                              | Easy                | Medium | Hard | Easy               | Medium | Hard |
| Person & cloth generation    | ×                            | 17                  | 25     | 35   | ×                  | ×      | ×    |
| Person/Garment VLM Filtering | ×                            | 3                   | 4.4    | 5.7  | ×                  | ×      | ×    |
| Try-On Generation            | Iteration 0                  | 38                  | 48     | 60   | 39                 | 49     | 61   |
|                              | Iteration I                  | 25                  | 32     | 40   | 26                 | 33     | 41   |
|                              | Iteration II                 | 15                  | 19     | 24   | 16                 | 20     | 25   |
|                              | Iteration III                | 8                   | 10     | 13   | 8                  | 11     | 14   |
|                              | Iteration IV                 | 3                   | 4      | 5    | 3                  | 4      | 5    |

Table 7. Cross-generator and cross-evaluator robustness analysis. Artifact rates report Easy/Medium/Hard percentages from the small-scale screening experiments. Generator configuration and estimated cost are shown once for each generator, with evaluator results grouped within the corresponding block. “Image Cond  $n$ ” denotes the maximum supported image conditioning context. PaliGemma evaluates only the generated output, whereas Qwen3-VL evaluates the complete source-person, target-garment, and generated-output triplet. Iteration 0 denotes the initially generated outputs, while Iteration IV denotes outputs after four rounds of iterative VLM-guided refinement.

| Generator            | Est. Cost | Generator Image<br>Cond $n$ | Evaluator(s)         | Evaluator Image<br>Cond $n$ | Artifact Rate (%)                   | Artifact Rate (%)                    |
|----------------------|-----------|-----------------------------|----------------------|-----------------------------|-------------------------------------|--------------------------------------|
|                      |           |                             |                      |                             | Easy / Medium / Hard<br>Iteration 0 | Easy / Medium / Hard<br>Iteration IV |
| FLUX.2-klein-9B      | \$25k     | 1–4                         | Qwen3-VL 32B         | 3                           | 44 / 54 / 66                        | 8 / 11 / 14                          |
|                      |           |                             | Qwen3-VL-30B-A3B     |                             |                                     |                                      |
|                      |           |                             | Thinking             | 3                           | 44 / 54 / 66                        | 9 / 12 / 13                          |
|                      |           |                             | PaliGemma            | 1                           | 44 / 54 / 66                        | 13 / 16 / 20                         |
| FLUX.2-klein-4B      | \$18k     | 1–4                         | Qwen3-VL 32B         | 3                           | 55 / 63 / 74                        | 15 / 18 / 22                         |
|                      |           |                             | PaliGemma (optional) | 1                           | 55 / 63 / 74                        | 18 / 22 / 27                         |
| Qwen-Image-Edit-2509 | \$90k     | 1–3                         | Qwen3-VL 32B         | 3                           | 65 / 73 / 82                        | 21 / 25 / 30                         |
|                      |           |                             | PaliGemma            | 1                           | 65 / 73 / 82                        | 28 / 33 / 38                         |
| Nano Banana Pro      | \$160k    | 3                           | Qwen3-VL 32B         | 3                           | 40 / 48 / 60                        | 5 / 8 / 10                           |

uation of all reported models required a further 700 H200 GPU hours.

**Train/Test Split Disjointness.** The 205,000 training and 4,500 test triplets form disjoint partitions of the curated dataset,  $\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \emptyset$ . The split is applied once, after quality screening, by partitioning the retained pool; no triplet is duplicated across the two sides, so no evaluated  $(I_p, I_c, I_t)$  triplet is ever seen during training. The test partition is additionally verified by human annotators using the rubric of Section G, so the evaluation set is both held out and manually checked. All numbers reported on the CURVTON-205K benchmark are computed on this held-out partition only.

**Dataset Adaptation and Mask-Free Training.** VITON-HD and DressCode follow a paired shop-to-model structure, providing a real person image wearing the target garment together with the corresponding in-shop garment, but without a separate source-person image required by our triplet formulation. As our evaluation protocol operates on com-

plete triplets  $(I_p, I_c, I_t)$ , a structural mismatch arises when applying these benchmarks directly. To resolve this while maintaining consistency with the testing protocol established by BooW-VTON [42], we complete each pair into a triplet by synthesizing the missing source-person image  $I_p$  via our generation pipeline, while retaining the original benchmark person image as the real ground-truth try-on target  $I_t$  and the corresponding in-shop garment as  $I_c$ . This construction preserves the original ground-truth person–garment correspondence while providing the complete triplet structure required by our evaluation protocol. Furthermore, all baseline architectures were trained under a unified mask-free regime. Concretely, the masked person representation, conventionally supplied as a zeroed or in-painted region indicating the garment placement area, was replaced throughout training with the original, unmasked person image  $I_p$ . This substitution aligns the conditioning signal of all baselines with the mask-free supervision structure of CURVTON-205K and prevents the introduction of mask-induced spatial artifacts as a confounding factor in cross-architecture comparisons.

The training experiments across the different model ar-

chitectures were implemented using PyTorch and distributed across multiple GPU nodes. We employed mixed-precision training to optimize memory usage and training speed. Hyperparameters for each model architecture were tuned using a subset of the training data, with final models trained on the complete CURVTON-205K dataset. Curriculum-aware variants use the difficulty-tier scheduling protocols described in the main text.

For the curriculum training experiments, we consider two strategies: no curriculum and standard curriculum. In the no-curriculum setting, samples from all difficulty tiers are mixed throughout training. In the standard-curriculum setting, harder samples are gradually introduced into the training batches, so that training begins with easier tiers and avoids abrupt transitions between difficulty levels.

### C.2. Embedding-Based Diversity Analysis

Fig. 6 provides additional qualitative diagnostics of dataset diversity using two-dimensional t-SNE projections of pre-trained garment and facial-appearance embeddings. The two garment plots use CLIP image and text embeddings of the 43,324 garment images in the garment dictionary. These embeddings are used only for this visualization. The garment metrics in Table 4, Gmt Var. and Gmt Div., are computed on FashionCLIP and DINOv2 features instead, as described in Section F, so the projections here are not the feature space those two numbers come from. The first plot colors the points by garment category and shows that upper-body, lower-body, dress, set, and accessory items each spread over a large part of the projection instead of collapsing into a few dense modes. These five categories describe the garment pool. The stratified evaluation in the main text is reported over the three main categories, upper body, lower body, and dresses; sets and accessories are part of the pool but are not reported as separate evaluation categories. The second plot colors the same points by whether the garment is traditional or modern. The two groups are close in size (22,755 traditional and 20,569 modern), they occupy different sides of the projection with a mixed band in between, so traditional garments are not a small isolated pocket of the dataset. The face plot shows the corresponding spread of facial appearance across cultural groups and gender. These projections are intended as visual summaries of coverage rather than quantitative metrics; nevertheless, CURVTON-205K shows broader occupancy and less pronounced mode concentration across the embedding spaces.

### C.3. Full Curriculum Ablation Studies

We report extended curriculum-aware ablations to separate the effect of difficulty-tier composition from the effect of sample ordering. These results complement the compact curriculum analysis in the main text. Table 9 expands the tier-composition analysis by including single-tier, pairwise-tier,

and scale-controlled variants. Table 10 expands the ordering analysis across training budgets and training regimes.

## D. Supplementary Qualitative Results

These sections include additional qualitative samples of the CURVTON-205K dataset, sorted by difficulty level, garment type, and pipeline components. In each triplet, there is the source person image on the left-hand side, the target garment from an in-store product on the middle, and the generated output of try-on result on the right-hand side. The samples below are carefully selected from the test split to showcase the diversity across the main dimensions of the dataset.

### D.1. Try-On Examples Across Difficulty Tiers

Figures 7–9, show try-on triplets sampled from easy, medium, and hard difficulty tiers, respectively. The difficulty tier rises gradually due to larger pose offsets, stronger occlusion effects, more cluttered backgrounds, and increasing garment-to-person interaction complexity. The easy difficulty tier corresponds to nearly frontal poses against either a studio background or a minimal background. The medium difficulty tier includes slight pose offset, moderate occlusions, and cluttered backgrounds, while the hard difficulty tier implies challenging poses, large occlusion rates, and culturally diverse clothing.

### D.2. Try-On Examples Across Garment Categories

Figures 10–12 show the quality of try-ons for the three garment categories used in the stratified evaluation: upper body, lower body, and full-body/dress type garments. The selection in these figures covers both common Western clothing as well as culturally distinctive clothing in order to showcase the wide range of the 200-type dictionary used to guide try-on synthesis. The results demonstrate consistent garment-identity preservation and plausible placement across varied garment structures.

### D.3. Closed-Loop VLM Feedback Mechanism: Illustrative Examples

Figure 13 illustrates representative output triplets produced by the closed-loop VLM-guided regeneration mechanism described in Section 3.1. In each example, the VLM assessor accepts the synthesized try-on once it finishes evaluating the garment faithfulness, placement plausibility, semantic consistency, and the absence of visible artifacts. These examples span diverse garment styles and subject appearances to showcase the range of cases handled by the automated quality loop. A formal power suit (left), a traditional Ghalek (center), and a casual hoodie (right).

### D.4. Extended Person Diversity Gallery

Figure 14 presents a collection of source person images sampled from CURVTON-205K to illustrate the diversity in

Table 8. Stratified performance of all models trained and evaluated on the CURVTON-205K benchmark test set. SSIM $\uparrow$  higher is better; LPIPS $\downarrow$ , FID $\downarrow$  lower is better. **Top:** overall scores and results by difficulty tier (E=Easy, M=Med, H=Hard). **Bottom:** results by cloth type (UB = Upper Body, LB = Lower Body, G = Garment, Tr = Traditional, NTr = Non-Traditional). Both panels list the same models and rows. Each model reports two settings, *Zero Shot* and *CURVTON Training*, except generalist models which are evaluated in *Zero Shot* only.

|                               |                  | Overall         |                    |                  | Difficulty Tier    |      |      |                 |      |      |                  |       |       |
|-------------------------------|------------------|-----------------|--------------------|------------------|--------------------|------|------|-----------------|------|------|------------------|-------|-------|
|                               |                  | SSIM $\uparrow$ | LPIPS $\downarrow$ | FID $\downarrow$ | LPIPS $\downarrow$ |      |      | SSIM $\uparrow$ |      |      | FID $\downarrow$ |       |       |
|                               |                  |                 |                    |                  | E                  | M    | H    | E               | M    | H    | E                | M     | H     |
|                               |                  |                 |                    |                  |                    |      |      |                 |      |      |                  |       |       |
| OOTDiffusion [40]             | Zero Shot        | 0.75            | 0.19               | 18.90            | 0.18               | 0.18 | 0.22 | 0.78            | 0.77 | 0.70 | 25.20            | 24.40 | 25.50 |
|                               | CURVTON Training | 0.85            | 0.13               | 8.75             | 0.12               | 0.12 | 0.15 | 0.86            | 0.85 | 0.83 | 9.88             | 11.54 | 14.12 |
| CatVTON [9]                   | Zero Shot        | 0.45            | 0.36               | 23.85            | 0.30               | 0.36 | 0.41 | 0.55            | 0.41 | 0.37 | 22.55            | 24.40 | 27.55 |
|                               | CURVTON Training | 0.86            | 0.12               | 9.14             | 0.12               | 0.09 | 0.12 | 0.86            | 0.86 | 0.81 | 9.14             | 9.23  | 12.88 |
| IDM-VTON [8]                  | Zero Shot        | 0.80            | 0.15               | 17.05            | 0.14               | 0.14 | 0.17 | 0.82            | 0.80 | 0.77 | 16.20            | 17.30 | 19.30 |
|                               | CURVTON Training | 0.87            | 0.11               | 8.44             | 0.10               | 0.10 | 0.12 | 0.88            | 0.87 | 0.84 | 8.28             | 8.85  | 10.29 |
| DCI-VTON [16]                 | Zero Shot        | 0.81            | 0.17               | 19.20            | 0.16               | 0.16 | 0.18 | 0.82            | 0.81 | 0.79 | 18.00            | 19.30 | 20.80 |
| FLUX.2 [klein] 9B (Generator) | Zero Shot        | 0.77            | 0.15               | 14.50            | 0.14               | 0.15 | 0.18 | 0.79            | 0.78 | 0.73 | 13.10            | 15.35 | 17.95 |

|                               |                  | Cloth Type         |      |      |      |      |                 |      |      |      |      |                  |       |       |       |       |
|-------------------------------|------------------|--------------------|------|------|------|------|-----------------|------|------|------|------|------------------|-------|-------|-------|-------|
|                               |                  | LPIPS $\downarrow$ |      |      |      |      | SSIM $\uparrow$ |      |      |      |      | FID $\downarrow$ |       |       |       |       |
|                               |                  | UB                 | LB   | G    | Tr   | NTr  | UB              | LB   | G    | Tr   | NTr  | UB               | LB    | G     | Tr    | NTr   |
|                               |                  |                    |      |      |      |      |                 |      |      |      |      |                  |       |       |       |       |
| OOTDiffusion [40]             | Zero Shot        | 0.16               | 0.20 | 0.21 | 0.22 | 0.16 | 0.78            | 0.75 | 0.72 | 0.73 | 0.81 | 19.50            | 27.15 | 29.70 | 29.95 | 17.15 |
|                               | CURVTON Training | 0.12               | 0.11 | 0.15 | 0.18 | 0.12 | 0.86            | 0.87 | 0.80 | 0.78 | 0.88 | 11.42            | 11.83 | 13.79 | 14.22 | 13.83 |
| CatVTON [9]                   | Zero Shot        | 0.35               | 0.34 | 0.38 | 0.36 | 0.32 | 0.45            | 0.45 | 0.42 | 0.50 | 0.54 | 22.60            | 23.55 | 28.90 | 31.35 | 20.16 |
|                               | CURVTON Training | 0.11               | 0.10 | 0.14 | 0.16 | 0.10 | 0.87            | 0.88 | 0.82 | 0.80 | 0.90 | 10.62            | 11.34 | 14.94 | 15.57 | 12.70 |
| IDM-VTON [8]                  | Zero Shot        | 0.14               | 0.13 | 0.18 | 0.24 | 0.16 | 0.81            | 0.83 | 0.75 | 0.71 | 0.80 | 16.70            | 17.25 | 22.00 | 25.55 | 15.15 |
|                               | CURVTON Training | 0.10               | 0.09 | 0.13 | 0.18 | 0.10 | 0.88            | 0.89 | 0.82 | 0.77 | 0.88 | 10.21            | 10.57 | 13.54 | 15.75 | 7.98  |
| DCI-VTON [16]                 | Zero Shot        | 0.18               | 0.14 | 0.19 | 0.19 | 0.25 | 0.80            | 0.84 | 0.77 | 0.79 | 0.72 | 20.15            | 17.05 | 24.10 | 18.05 | 25.55 |
| FLUX.2 [klein] 9B (Generator) | Zero Shot        | 0.14               | 0.15 | 0.18 | 0.35 | 0.31 | 0.79            | 0.79 | 0.73 | 0.47 | 0.52 | 14.75            | 16.05 | 17.50 | 23.27 | 20.23 |

Table 9. Extended difficulty-tier composition and scale ablation. Models are trained with different combinations of easy, medium, and hard samples, as well as different fractions of the full three-tier dataset, and evaluated across difficulty-specific test splits.

| Train Data               | Easy Test |       |       |      | Medium Test |       |       |      | Hard Test |       |       |      | All Mixed |       |       |      |
|--------------------------|-----------|-------|-------|------|-------------|-------|-------|------|-----------|-------|-------|------|-----------|-------|-------|------|
|                          | SSIM      | LPIPS | FID   | KID  | SSIM        | LPIPS | FID   | KID  | SSIM      | LPIPS | FID   | KID  | SSIM      | LPIPS | FID   | KID  |
| Easy only                | 0.86      | 0.11  | 9.86  | 0.95 | 0.83        | 0.12  | 12.70 | 0.92 | 0.76      | 0.18  | 19.87 | 2.29 | 0.85      | 0.11  | 10.29 | 0.94 |
| Medium only              | 0.82      | 0.18  | 12.89 | 1.75 | 0.83        | 0.12  | 11.02 | 0.46 | 0.78      | 0.15  | 15.32 | 0.61 | 0.82      | 0.17  | 12.52 | 1.73 |
| Hard only                | 0.83      | 0.16  | 13.08 | 2.51 | 0.84        | 0.12  | 11.47 | 1.11 | 0.77      | 0.15  | 15.10 | 1.08 | 0.83      | 0.15  | 13.15 | 2.75 |
| Medium + Hard            | 0.81      | 0.20  | 12.75 | 1.82 | 0.82        | 0.13  | 11.11 | 0.39 | 0.77      | 0.16  | 15.25 | 0.59 | 0.80      | 0.16  | 13.92 | 0.64 |
| Easy + Hard (full)       | 0.83      | 0.16  | 11.46 | 1.20 | 0.84        | 0.11  | 10.42 | 0.48 | 0.75      | 0.17  | 15.50 | 0.95 | 0.81      | 0.15  | 13.54 | 0.70 |
| Easy + Med + Hard (10%)  | 0.81      | 0.19  | 11.08 | 1.38 | 0.82        | 0.14  | 10.35 | 0.30 | 0.74      | 0.18  | 14.94 | 0.64 | 0.77      | 0.21  | 12.02 | 2.54 |
| Easy + Med + Hard (20%)  | 0.80      | 0.20  | 12.23 | 1.70 | 0.81        | 0.14  | 10.91 | 0.47 | 0.77      | 0.16  | 14.01 | 0.30 | 0.80      | 0.19  | 11.99 | 1.79 |
| Easy + Med + Hard (50%)  | 0.83      | 0.16  | 10.35 | 1.42 | 0.85        | 0.10  | 9.51  | 0.26 | 0.80      | 0.13  | 13.15 | 0.37 | 0.83      | 0.15  | 10.18 | 1.23 |
| Easy + Med + Hard (100%) | 0.86      | 0.12  | 9.14  | 0.96 | 0.86        | 0.09  | 9.23  | 0.20 | 0.81      | 0.12  | 12.88 | 0.22 | 0.86      | 0.12  | 9.14  | 0.97 |

demographic and appearance targeted by the person-stream attribute dictionary  $\mathcal{D}_p$ . The collection shows variation in body shape, facial-appearance details, geographic and cultural style indicators, photographic conditions (studio vs. in-the-wild), accessories, and background context. This vastness is a direct result of sampling attributes from the factorized distribution  $\mathbb{P}_{\mathcal{D}_p}$  described in Section 3.1, which is designed to avoid long-tail collapse and improve coverage of groups that are under-represented.

## D.5. Additional Closed-Loop VLM Feedback Examples

Figure 15 shows three more samples of accepted triplets using our VLM-based automated validation process. This figure complements our discussion of the closed-loop validation system in Section D.3. The examples include a traditional Jamdani Saree on a female subject, a casual mini skirt, and a Chakma dress on a male subject, illustrating the quality loop’s performance on both conventional Western garments and culturally specific traditional clothing.



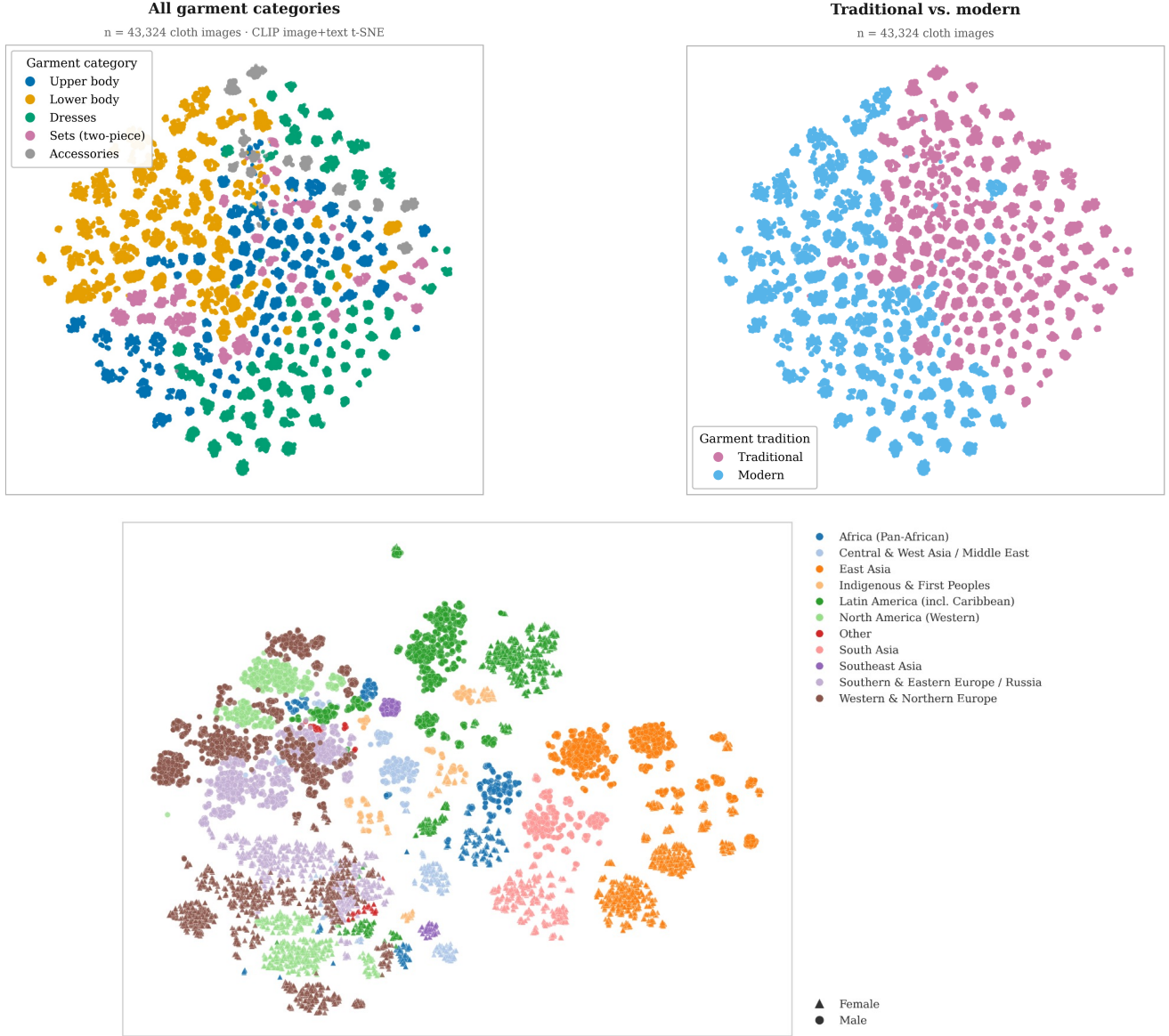


Figure 6. Diversity in CURVTION-205K for garments and cultural representation. We use 2D t-SNE projections of pretrained embeddings. Top left: CLIP image and text embeddings of the 43,324 garment images, colored by garment category. Top right: the same projection, colored by whether the garment is traditional or modern. Bottom: facial-appearance embeddings taken from face regions, colored by cultural group and marked by gender.

### D.6. Extended Medium-Tier Try-On Examples

Figure 16 shows six more medium-level try-on triplets in addition to those in Figure 8. The examples shown here cover increased cultural diversity, with a Berber Amazigh dress, a lace dress, a wrap skirt, a Fundoshi, a Guayabera shirt, and a Dashiki.

### D.7. Extended Hard-Tier Try-On Examples

Figure 17 gives two more hard-level try-on triplets. The images here include extremely large pose deviations with rare

traditional clothes: an Afghan Kuchi dress and a *Gökti* worn by female subjects. They show the difficulty of preserving the identity of the clothes when dealing with pose deviation and high levels of occlusions.

### D.8. Traditional Garment Coverage

Figures 18 and 19 show try-on triplets of garments that belong to the traditional garments collection of the CURVTION-205K garment dictionary. This subset covers traditional garments belonging to different world cultures, including East

Table 10. Curriculum vs. non-curriculum under finetuning (**Stable Diffusion**) regime, comparing minimal (**3.6k iters**), limited (**7.2k iters**), medium (**14.4k iters**), and full (**28.8k iters**) training budgets across difficulty splits, including a mixed test set. For each metric, the better of the two strategies is shown in bold.

| Regime                           | Budget                  | Curriculum          | Easy        |             |             |              | Medium      |             |             |             | Hard        |             |             |             | All Mixed   |             |             |             |
|----------------------------------|-------------------------|---------------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                  |                         |                     | SSIM↑       | LPIPS↓      | FID↓        | KID↓         | SSIM↑       | LPIPS↓      | FID↓        | KID↓        | SSIM↑       | LPIPS↓      | FID↓        | KID↓        | SSIM↑       | LPIPS↓      | FID↓        | KID↓        |
| Finetuning<br>(Stable Diffusion) | Minimal<br>(3.6k iters) | No Curriculum       | 0.78        | 0.17        | 21.4        | <b>0.02</b>  | 0.74        | 0.21        | 26.7        | 0.03        | 0.66        | 0.27        | 36.3        | <b>0.03</b> | 0.73        | 0.22        | 28.1        | 0.03        |
|                                  |                         | Standard Curriculum | <b>0.79</b> | <b>0.16</b> | <b>20.2</b> | <b>0.02</b>  | <b>0.77</b> | <b>0.19</b> | <b>25.3</b> | <b>0.02</b> | <b>0.68</b> | <b>0.25</b> | <b>33.8</b> | <b>0.03</b> | <b>0.75</b> | <b>0.21</b> | <b>26.4</b> | <b>0.02</b> |
|                                  | Limited<br>(7.2k iters) | No Curriculum       | 0.81        | 0.14        | 19.3        | <b>0.01</b>  | 0.77        | 0.18        | 23.8        | <b>0.02</b> | 0.70        | 0.23        | 32.7        | <b>0.03</b> | 0.76        | 0.19        | 25.2        | <b>0.02</b> |
|                                  |                         | Standard Curriculum | <b>0.82</b> | <b>0.13</b> | <b>18.4</b> | 0.02         | <b>0.78</b> | <b>0.17</b> | <b>22.4</b> | <b>0.02</b> | <b>0.72</b> | <b>0.21</b> | <b>29.8</b> | <b>0.03</b> | <b>0.77</b> | <b>0.17</b> | <b>23.5</b> | <b>0.02</b> |
|                                  | Medium<br>(14.4k iters) | No Curriculum       | <b>0.83</b> | 0.13        | 17.9        | 0.02         | 0.79        | 0.16        | 21.7        | <b>0.02</b> | 0.73        | 0.21        | 28.4        | <b>0.03</b> | 0.78        | 0.17        | 22.7        | <b>0.02</b> |
|                                  |                         | Standard Curriculum | <b>0.83</b> | <b>0.12</b> | <b>17.3</b> | <b>0.015</b> | <b>0.80</b> | <b>0.15</b> | <b>21.1</b> | <b>0.02</b> | <b>0.74</b> | <b>0.20</b> | <b>27.6</b> | <b>0.03</b> | <b>0.79</b> | <b>0.16</b> | <b>22.0</b> | <b>0.02</b> |
|                                  | Full<br>(28.8k iters)   | No Curriculum       | 0.84        | 0.12        | 16.8        | <b>0.01</b>  | 0.81        | 0.15        | 20.3        | <b>0.02</b> | 0.76        | 0.19        | 26.1        | <b>0.02</b> | 0.80        | 0.16        | 21.1        | <b>0.02</b> |
|                                  |                         | Standard Curriculum | <b>0.85</b> | <b>0.11</b> | <b>16.2</b> | <b>0.01</b>  | <b>0.82</b> | <b>0.14</b> | <b>19.4</b> | <b>0.02</b> | <b>0.77</b> | <b>0.18</b> | <b>25.3</b> | <b>0.02</b> | <b>0.81</b> | <b>0.15</b> | <b>20.3</b> | <b>0.02</b> |



Figure 7. **Try-on Triplets of Easy tier.** Left: source person, center: target garment, right: synthesized try-on output. Samples of easy-tier try-ons show near front-facing poses, minimal occlusions and clean backgrounds, comprising traditional outfits like Cholita dress (top left), Ghagra Choli (top right), Áo Bà Bà (bottom left), as well as regular Western clothes.



Figure 8. **Try-on triplets of Medium tier.** As compared to the Easy tier, the Medium-tier samples have moderate pose variations, with some occlusions and comparatively more complex backgrounds. Garment diversity includes a range of culturally-specific clothing from various cultures, namely, Abaya (top left), Kebaya (top center), Ruqun (top right), Hanfu (bottom left), and Balochi suit (bottom right).

Asia, South Asia, Southeast Asia, Central Asia, Middle East, Africa, and Europe.

#### D.9. Non-Traditional Garment Coverage

Figure 20 shows example try-on triplets for non-traditional (contemporary Western) garments. These augment the traditional garments shown above and verify the balanced distribution of traditional vs. contemporary garment styles present in CURVTON-205K.

#### E. Dataset Visual Comparison

This section provides a qualitative visual comparison between CURVTON-205K and three widely-used virtual try-on benchmarks: VITON-HD [7], DressCode [27], and StreetTryOn [10]. Each row below shows representative samples from a single dataset. VITON-HD and DressCode are predominantly composed of studio-shot images of Western garments with frontal poses while StreetTryOn adds in-the-wild backgrounds, yet still limited to Western upper body clothing. In contrast, CURVTON-205K contains diverse pose angles and backgrounds along with a large vocabulary of

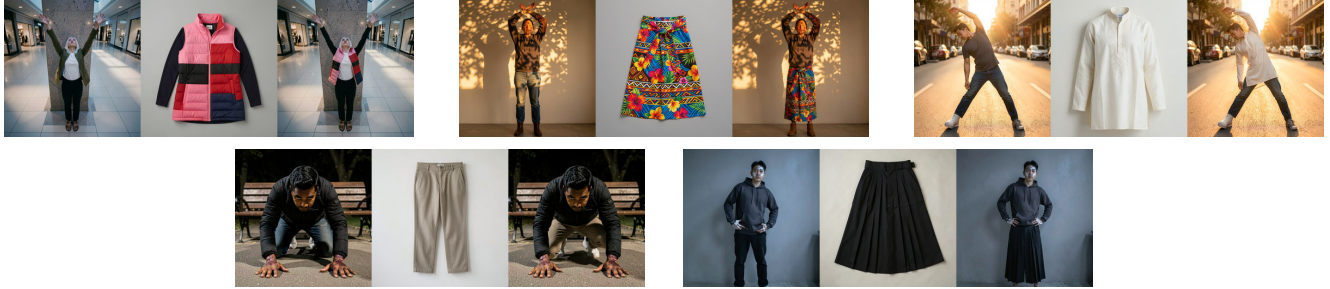


Figure 9. **Try-on triplets of Hard tier.** Samples of the Hard tier are characterized by extremely complex poses, significant occlusions and challenging in-the-wild environments, which involve long-tail traditional dresses, such as Lavalava (top center), Panjabi (top right) and Hakama (bottom right). These cases stress both garment-identity preservation and geometric plausibility under strong visual complexity.



Figure 10. **Try-on of upper-body garments.** Top row (left to right): women wearing a ruffle top, a denim jacket, a peplum top, and a sweatshirt. Bottom row (left to right): men wearing a bomber jacket, a dress shirt, a polo shirt, and a poncho. These results highlight consistent collar, sleeve, and torso coverage over different types of garments.

200 different types of traditional garments from over 60 cultural categories.





Figure 11. **Try-on of lower-body garments.** Top row (left to right): women wearing a skirt, leggings, jeans, and mom jeans. Bottom row (left to right): men wearing shorts, jeans, swim trunks, and a traditional hakama. These results indicate good handling of diverse silhouette types, ranging from fitted, tapered, and flowing lower body garments.

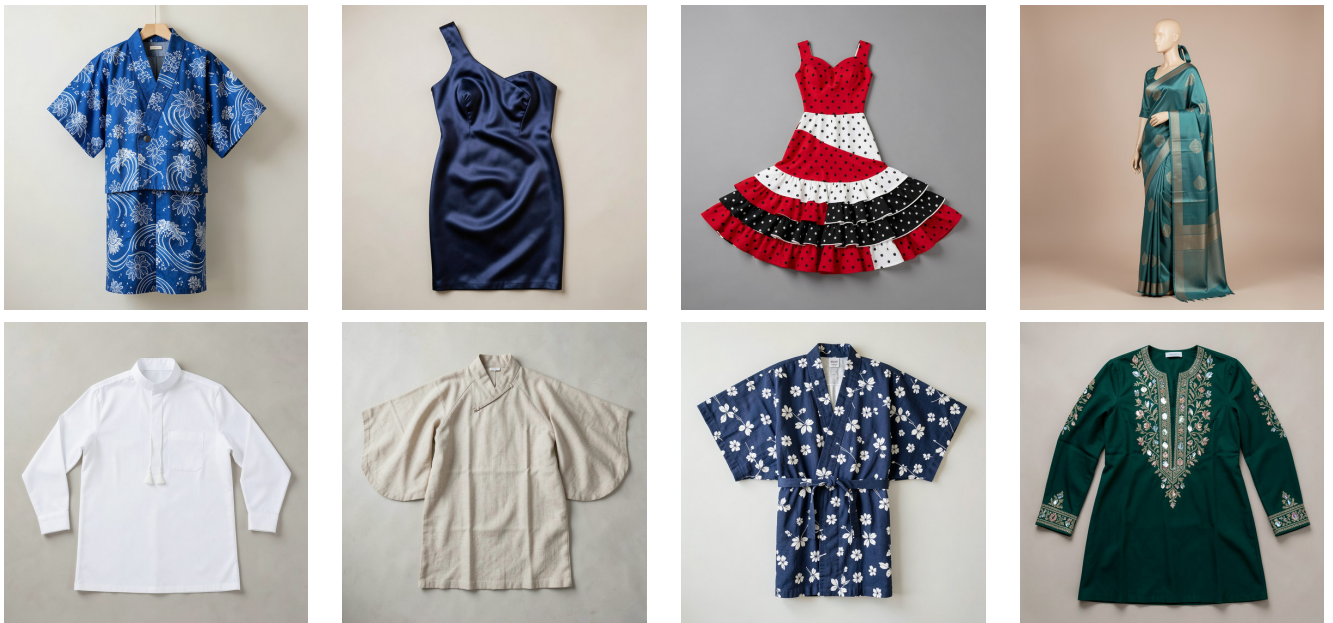


Figure 12. **Full-body and traditional garment try-on examples.** Top row: women wearing a Ryusou, a cocktail dress, a Flamenco dress, and a Muga Silk Saree. Bottom row: men wearing a Kandura, a Shenyi, a Yukata, and a Pashtun dress. Wide variety of garments in our corpus enables us evaluate try-ons of culturally specific garments not covered by other VTON datasets.





Figure 13. **Closed-loop VLM feedback: accepted triplets.** For triplets passing the VLM-based quality check, each figure shows the source person (left), the target garment (middle), and the generated try-on (right). First row: a formal power suit for a female person; second row: a traditional Ghalek outfit for a male person; third row: a casual hoodie. These accepted samples exhibit consistent garment identity, plausible placement, and no visible artifacts, illustrating the quality maintained by the automated validation loop.

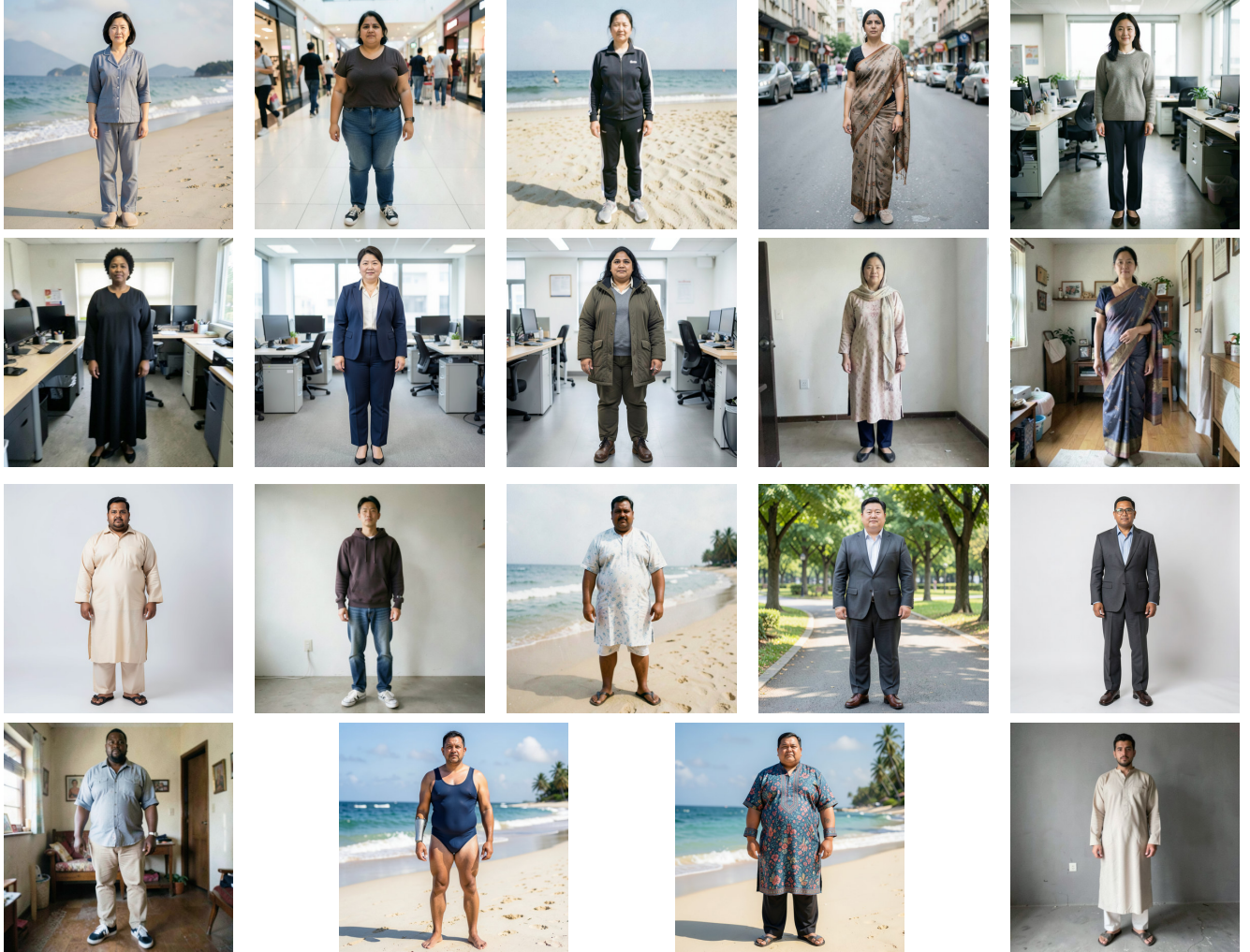


Figure 14. **Extended person diversity gallery.** First two rows of female persons and next two rows of male persons from the CURVTON-205K dataset. Body shape, facial appearance, skin tone, accessories, and photography conditions (both studio and in-the-wild) are covered thanks to the diversity introduced with factorized attribute sampling strategy  $\mathbb{P}_{\mathcal{D}_p}$ . This breadth of person appearance is a key differentiator of CURVTON-205K relative to existing paired VTON benchmarks.





Figure 15. **Additional closed-loop VLM feedback: accepted triplets.** For accepted triplets based on VLM quality check, each figure provides the source person (left), target garment (middle), and generated try-on (right). Top: Jamdani Saree on a female subject; center: mini skirt on a female subject; bottom: Chakma dress on a male subject. These samples further illustrate the breadth of garment types handled by the automated validation loop.



Figure 16. **Extended medium-tier try-on triplets.** Top row: female subjects wearing (left to right) a Berber Amazigh dress, a lace dress, and a wrap skirt. Bottom row: male subjects wear a Fundoshi, a Guayabera shirt, and a Dashiki. The culturally-specific clothes indicate the representation of rare classes of clothing in CURVTON-205K.



Figure 17. **Extended hard-tier try-on triplets.** Left image: an Afghan Kuchi dress worn by a female subject with a complex pose; right image: a Gökti worn by a female subject with a cluttered background. These images show the difficulty in maintaining garment identity in challenging conditions.



Figure 18. **Traditional garment try-on — female subjects.** Top row (left to right): Qipao, Uchikake, Furisode, Gomesi. Bottom row: Muga Silk Saree, Limbu Mekhli, Ghagra Choli, Romanian Ie. These clothes come from East Asia, South Asia, East Africa, and Eastern Europe, illustrating the cultural breadth of CURVTON-205K.





Figure 19. **Traditional garment try-on — male subjects.** Top row (left to right): Durumagi, Lederhosen, Korean Jeonbok, Ruana. Bottom row: Mundu, Angrakha, Khet Partug, Sampot. These clothes belong to Korea, Germany, South Asia, South America, and Southeast Asia, demonstrating the dataset’s global cultural coverage.



Figure 20. **Non-traditional (contemporary) garment try-on examples.** From top to bottom: female subjects wearing a tracksuit, yoga pants, jumpsuit, and cocktail dress; male subjects wearing joggers, tuxedo, sweatshirt, and dress shirt. Examples indicate consistency of garment identity representation in various contemporary Western attires.

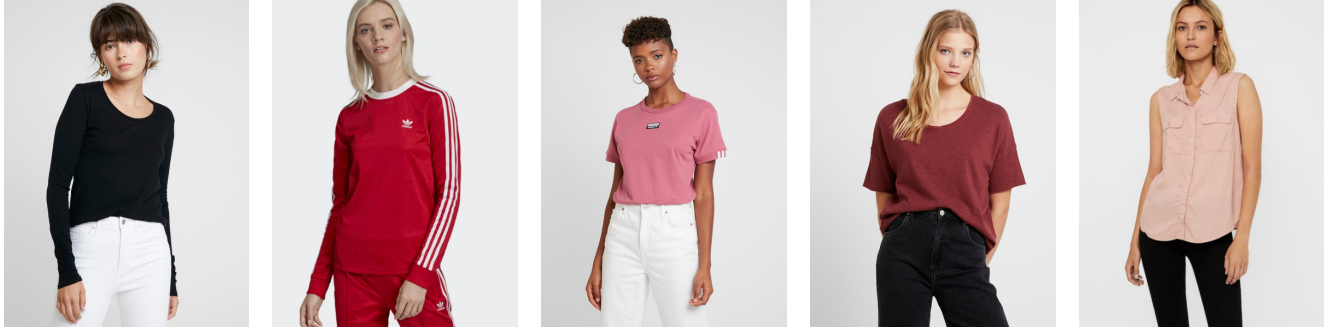


Figure 21. **VITON-HD samples.** Representative person images in the VITON-HD dataset. Samples mainly consist of front poses, uniform backgrounds, and upper body clothing. Pose, background and garment diversity are limited in comparison to CURVTON-205K.



Figure 22. **DressCode upper-body samples.** Representative upper body person images in the DressCode dataset. Poses are mostly frontal, and backgrounds are uniform. Sampled garments mainly cover Western upper body types.



Figure 23. **Representative lower body person images in the DressCode dataset.** Images include a variety of standard trouser and skirt styles but pose and background variety are limited.



Figure 24. **DressCode dress samples.** Representative full body dress person images in the DressCode dataset. The coverage is limited to contemporary Western styles without pose variety.



Figure 25. **StreetTryOn samples.** Representative person images in the StreetTryOn dataset. StreetTryOn offers more varied into-the-wild backgrounds and natural body poses compared to studio based data like VITON-HD. However, garment variety is still limited to Western upper body clothes.



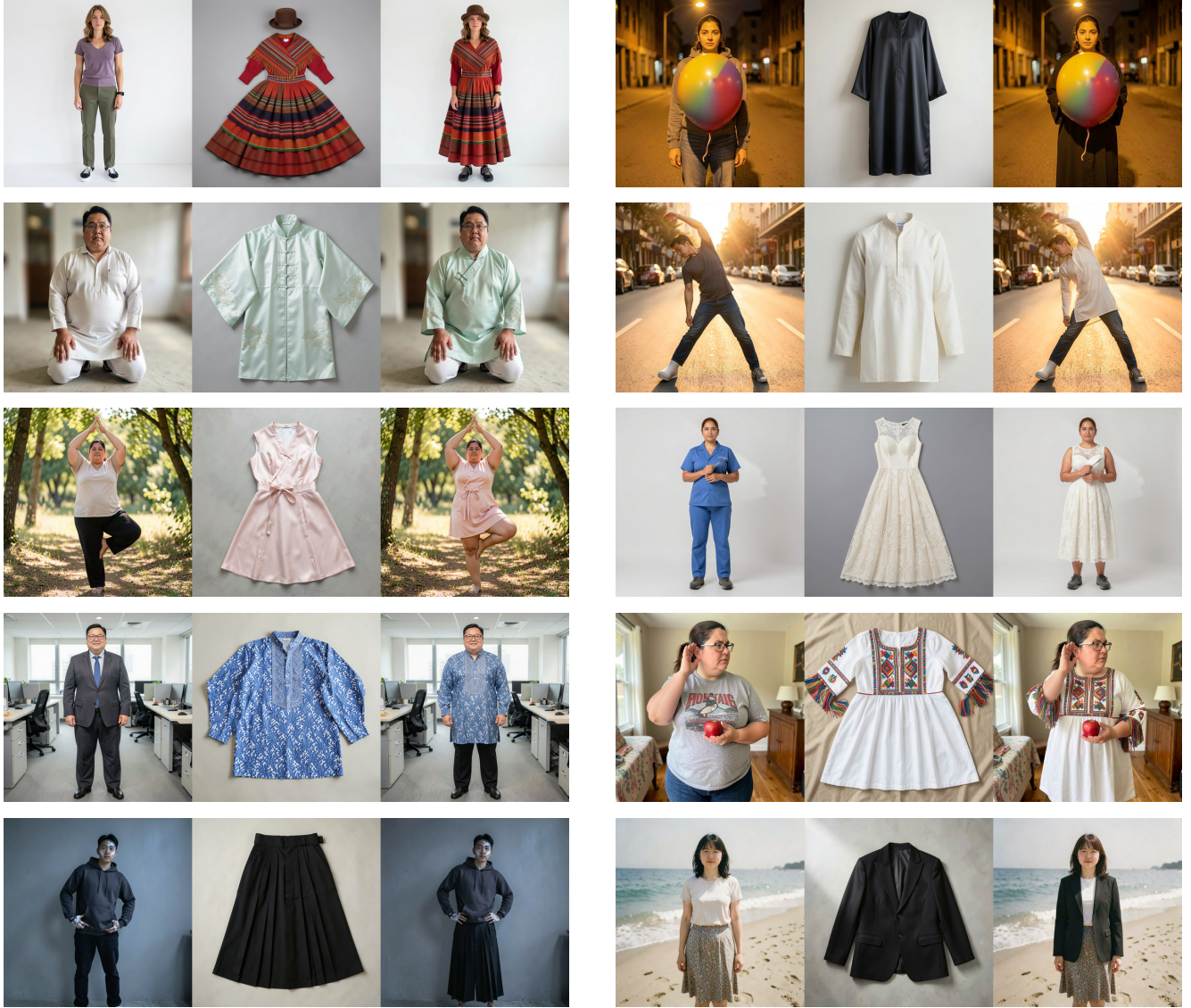


Figure 26. **CURVTON-205K samples.** Representative try-on triplets in CURVTON-205K. Images show sample source person (left), target garment (middle), and synthesized try-on (right) images. Row 1: Cholita dress, Abaya. Row 2: Hanfu, Panjabi. Row 3: Ruqun, lace dress. Row 4: Dashiki, Berber Amazigh dress. Row 5: Hakama, blazer. Compared to VITON-HD, DressCode, and StreetTryOn, CURVTON-205K exhibits substantially greater diversity in garment type, cultural origin, pose, and photographic conditions.

## F. Diversity Metric Definition

We provide the complete mathematical formulation of all task-free diversity metrics used to characterize dataset coverage across seven dimensions in Table 4. For each metric, we describe the feature extraction procedure and the exact formula used for computation.

### Pose ( $\sigma_{\text{pose}}^2$ )

The pose metric captures how varied and complex the skeletal articulations are across images in the dataset.

- **Objective.** Measure the spatial variability of human skeleton structure across the population.
- **Feature extraction.** Images are processed using MMPose ViTPose [26, 39]. Given input image  $I_i$ , the pose estimator  $f_{\text{pose}}$  extracts 17 COCO keypoints  $K_i = f_{\text{pose}}(I_i) \in \mathbb{R}^{17 \times 2}$ , which are bounding-box-normalised and flattened into a 34-dimensional vector  $v_i \in \mathbb{R}^{34}$ .
- **Metric.** Pose diversity is the log-determinant of the covariance matrix of pose vectors:

$$D_{\text{pose}} = \log \det(\Sigma_v + \epsilon I), \quad (1)$$

where  $\Sigma_v \in \mathbb{R}^{34 \times 34}$  is the population covariance matrix and  $\epsilon = 10^{-6}$  is a small stabilising term.

### Occlusion (Occl. Cmplx., Person Vis.)

The occlusion metrics quantify how much of the garment region is blocked by overlapping elements such as hair, arms, or accessories.

- **Objective.** Evaluate the degree of obstruction of the target garment subject and its consistency across the dataset.
- **Feature extraction.** Mask2Former [5] (facebook/mask2former-swin-large-coco-panoptic, 133 panoptic categories) produces per image a binary garment map  $\mathbf{G}_i \in \{0, 1\}^{H \times W}$  and a union-of-occluders map  $\mathbf{O}_{cc,i} \in \{0, 1\}^{H \times W}$  combining parsed arms, hair, bags, accessories, and environmental objects.
- **Metric.** The per-image occlusion ratio is

$$O_i = \frac{\sum_{(x,y)} (\mathbf{G}_i(x,y) \wedge \mathbf{O}_{cc,i}(x,y))}{\sum_{(x,y)} \mathbf{G}_i(x,y)}, \quad (2)$$

and dataset-level occlusion complexity is  $C_{\text{occ}} = \mathbb{E}[O_i] + \text{Var}(O_i)$ , where the expectation captures baseline severity and the variance captures dataset-wide variation.

### Background (BGE, BSE, BOD)

Three complementary scores assess how cluttered and semantically diverse the background regions are.

- **Objective.** Quantify background noise, visual clutter, and semantic density surrounding the subject.
- **Feature extraction.** DeepLabV3-ResNet101 [4] produces a person mask  $\mathbf{M}_i$ ; background pixels are identified as  $\neg \mathbf{M}_i$ . For BGE, the background region is converted to

a 256-bin normalised grayscale histogram. For BOD and BSE, DETR [3] (facebook/detr-resnet-50) is evaluated over the masked image to identify background bounding boxes and semantic class scores.

- **Metric.** Background Texture Entropy (BGE) is the mean pixel-level Shannon entropy  $\mathbb{E}[H_{\text{bg},i}]$ , where  $H_{\text{bg},i} = -\sum_{j=0}^{255} p_j \log p_j$ . Background Object Density (BOD) is the mean object count  $\mathbb{E}[n_i]$  in  $\neg \mathbf{M}_i$ . Background Semantic Entropy (BSE) is the Shannon entropy over population-level DETR class-score proportions  $\mathcal{P}_c$ :  $\text{BSE} = -\sum_c \mathcal{P}_c \log \mathcal{P}_c$ .

### Illumination (IGM, LVM)

The illumination metrics reflect the diversity of lighting conditions and luminance exposures across the dataset.

- **Objective.** Calculate localised lighting disparity, directional shadowing gradients, and macroscopic luminance exposure properties.
- **Feature extraction.** sRGB images are linearised via the standard piecewise gamma correction and converted to Rec. 709 perceived luminance

$$Y_i(x, y) = 0.2126 R_{\text{linear}} + 0.7152 G_{\text{linear}} + 0.0722 B_{\text{linear}}. \quad (3)$$

Gaussian-blurred luminance maps ( $\sigma = 1.0$ ) are then differentiated with Sobel operators to obtain gradient magnitude maps  $M_i(x, y) = \sqrt{g_x^2 + g_y^2}$ .

- **Metric.** Illumination Gradient Mean (IGM) is the mean normalised gradient-magnitude variance per image:

$$\text{IGM} = \mathbb{E}_i \left[ \frac{\text{Var}_{x,y}(M_i)}{\mathbb{E}_{x,y}[M_i] + \epsilon} \right]. \quad (4)$$

Luminance Variance Measure (LVM) is the median absolute deviation of per-image mean luminance  $\mu_i$ :  $\text{LVM} = \text{Median}(|\mu_i - \text{Median}(\mu)|)$ .

### Body Shape ( $\sigma_{\text{shape}}^2$ )

The body shape metric measures anatomical diversity across subjects, independent of pose or articulation.

- **Objective.** Evaluate multi-dimensional body topology deviations, excluding skeletal joint variance.
- **Feature extraction.** HMR2.0 (4D-Humans) [15] fits a SMPL model [23] to each image, returning a 10-dimensional PCA body shape vector  $\beta_i = f_{\text{hmr2}}(I_i) \in \mathbb{R}^{10}$  invariant to 3D rotation and joint articulation.
- **Metric.** Shape diversity is the log-determinant of the covariance matrix of shape vectors:

$$D_{\text{shape}} = \log \det(\Sigma_{\beta} + \epsilon I), \quad (5)$$

where  $\Sigma_{\beta} \in \mathbb{R}^{10 \times 10}$  is the population covariance of  $\beta_i$ .

### Garment (Gmt Var., Gmt Div.)

The garment metrics capture both the spread and volume of style variation across all clothing items in the dataset.

- **Objective.** Assess broad stylistic differentiation and textural characteristics across clothing items, decoupled from body geometry.
- **Feature extraction.** Each garment crop is encoded by FashionCLIP [6] (semantic and textual associations) and DINOv2 [29] (structural patterns), producing a  $D$ -dimensional feature vector  $e_i = f_{\text{encoder}}(I_{\text{cloth}}) \in \mathbb{R}^D$ .
- **Metric.** Garment Variance (Gmt Var.) is the trace of the feature covariance matrix,  $\text{Var}_{\text{total}} = \text{Tr}(\Sigma_e) = \sum_j \lambda_j(\Sigma_e)$ . Garment Diversity (Gmt Div.) captures the full style volume via the log-determinant  $D_{\text{garment}} = \log \det(\Sigma_e + \epsilon I)$ .

### Camera ( $\hat{\sigma}_\phi, \hat{\sigma}_\theta$ )

The camera viewpoint metrics summarise how broadly the dataset spans different shooting angles in azimuth and elevation.

- **Objective.** Determine global image perspective dynamics reflecting overall lens-view coverage properties.
- **Feature extraction.** HMR2.0 [15] root-orientation estimates are projected onto azimuth  $\phi_i$  (lateral angle) and elevation  $\theta_i$  (vertical tilt) for each image.
- **Metric.** Distribution bandwidth is captured by the standard deviations over the population:

$$\hat{\sigma}_\phi = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\phi_i - \bar{\phi})^2}, \quad (6)$$

$$\hat{\sigma}_\theta = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\theta_i - \bar{\theta})^2}. \quad (7)$$



## G. Artifact Evaluation Rubric

Both human-facing quality judgements in our pipeline are based on a single

10-question rubric, reproduced below. Using the same instrument in both settings

keeps the notion of an “artifact” consistent across the two points where human

judgement enters the construction of CURVTON-205K:

- **Annotation for the quality-filter model.** The 5,000 manually labelled triplets used to train the ViT-Small quality filter

$f_{\text{ViT}}$  (Section 3.2 of the

main text) were labelled using this rubric. Annotators answered all ten

questions for each triplet and assigned the binary *\*correct\** / *\*incorrect\**

target based on the outcome: a triplet is labelled *\*positive\** only when no

question records a severe defect. The rubric thus defines the supervision

signal that  $f_{\text{ViT}}$  learns to reproduce at scale.

- **Human evaluation and test-set verification.** The same ten questions are used as the marking scheme for the human study reported in the main text and for the manual verification pass over the test split, where triplets containing visible artifacts or semantic inconsistencies are removed.

Each question is scored 0 (none), 1 (minor), or 2 (severe), allowing

localised defects and overall output quality to be recorded on a common scale.

Questions 1–9 contribute to the primary artifact rate; Question 10 records

uncategorised issues separately, ensuring that failure modes outside the predefined

categories are captured rather than silently discarded. The rubric covers

garment placement, background preservation, structural garment details, text and

logos, visual artifacts, unintended objects, anatomical consistency, fit and drape,

overall physical plausibility, and an open-ended category.

### **\*\*Q1.\*\* \*\*Object placement.\*\***

Is the garment correctly aligned with and positioned on the body? In

particular, are elements such as sleeves, collars, and hem-lines placed

consistently with the person’s pose?

### **\*\*Q2.\*\* \*\*Background consistency.\*\***

Does the generated image preserve the background of the reference image without introducing unintended changes, blur, or geometric distortion?

### **\*\*Q3.\*\* \*\*Structural details.\*\***

Are garment components such as pockets, buttons, zip-pers, and similar structural elements reproduced correctly and in appropriate locations, without unintended duplication, deformation, or omission?

### **\*\*Q4.\*\* \*\*Text/logo fidelity.\*\***

When the garment contains text, logos, or graphic prints, are these elements reproduced accurately and legibly without noticeable distortion, warping, or incorrect placement?

### **\*\*Q5.\*\* \*\*Artifacts and blending.\*\***

Does the generated image contain visual defects that were not present in the reference, including blur, pixel noise, visible seams, texture errors, ghosting, double exposure, or unintended transparency?

### **\*\*Q6.\*\* \*\*Unwanted object introduction.\*\***

Has the model introduced objects or accessories that were not present in the original reference, such as handbags, jewelry, or other unrequested items?

### **\*\*Q7.\*\* \*\*Anatomical correctness.\*\***

Does the generated person preserve anatomically consistent limbs, hands, and digits, without extra or missing body parts, malformed fingers, or ghost-limb artifacts?

### **\*\*Q8.\*\* \*\*Garment fit and drape realism.\*\***

Is the garment worn in a manner that is consistent with its real-world type? Do its folds, fit, and drape respond naturally to gravity and the person’s pose rather than appearing unnaturally stiff, flat, or pasted onto the body?

### **\*\*Q9.\*\* \*\*Overall try-on plausibility.\*\***

Considering garment identity, fit, drape, body interaction, and pose, does the final try-on image appear visually and physically plausible as a whole?

### **\*\*Q10.\*\* \*\*Open-ended / other.\*\***

Is there any additional visual issue not covered by the previous questions that makes the generated result appear incorrect, unnatural, or inconsistent with either the source person or the target garment? These cases are recorded separately as an uncategorised-artifact rate.

## H. System Prompts Used in the Generation and Evaluation Pipeline

This section provides the system prompts used at the principal LLM/VLM-assisted stages of the CURVTON-205K pipeline. Specifically, we report the prompt used to convert structured difficulty-editing commands into FLUX.2 Klein 9B editing instructions, the quality-control prompt used to screen edited person images, and the VLM prompt used to evaluate generated virtual try-on triplets and provide regeneration feedback when necessary.

### H.1. Difficulty-Aware Person Editing Prompt

The following system prompt converts a source person image and structured JSON edit commands into a natural-language editing instruction for FLUX.2 Klein 9B. The prompt explicitly preserves person identity and visible physical attributes while resolving pose, occlusion, limb, and scene-geometry constraints.

Listing 1. System prompt used for difficulty-aware person editing.

```
SYSTEM PROMPT - Difficulty-Aware Person Editing

You are an expert image-editing prompt engineer. You
receive a source
image of a person and a JSON object of edit commands.
Translate the
commands into one clear natural-language editing
prompt for FLUX.2
Klein 9B distilled.

The JSON edit commands are authoritative. Replace the
source pose,
hand positions, held objects, camera view, occlusion,
and requested
scene conditions according to the commands. If the
source person holds
an object that is not requested, remove it.

Use the source image to preserve the person's identity
and appearance,
including:
- body type,
- skin tone,
- face,
- hair,
- facial hair,
- clothing and clothing colors,
- footwear,
- accessories,
- visible disability attributes,
- prosthetic limbs or mobility aids.

Before writing the prompt:
- resolve physically incompatible commands,
- respect the number of available limbs and hands,
- avoid impossible anatomy or duplicated limbs,
- avoid body-object interpenetration,
- keep pose and object interactions physically
plausible.

For requested occlusions:
- place the occluding object between the camera and
the subject,
- keep the subject behind the occluder,
- state which body regions are covered,
- preserve realistic foreground-background depth.

Describe every visible hand and foot in a natural pose-
specific position.
```

Do not invent held objects.

Preserve unchanged identity, clothing, footwear, and physical attributes explicitly rather than saying "keep everything else the same."

Write one coherent natural-language editing prompt using direct and literal instructions.

Return only the final FLUX.2 Klein editing prompt. Do not return reasoning, JSON, alternatives, or explanations.

### H.2. Edited-Person Quality-Control Prompt

After person editing, the resulting image is screened using the following quality-control prompt. The evaluator checks visible limb counts, anatomical consistency, clothing continuity, body-shape consistency, and generation artifacts. Images scoring below the acceptance threshold are rejected from the subsequent pipeline.

Listing 2. System prompt used for edited-person quality-control evaluation.

```
SYSTEM PROMPT - Edited-Person QC Evaluation

You are an expert image quality analyst for AI-edited
human portraits.
Your judgment is final.

You receive one AI-edited portrait image. Evaluate it
in the following
order.

STEP 1 - DETECT HANDS AND FEET

Detect every visible hand and foot, including partial,
blurred,
overlapping, ghosted, or distorted instances.

Return each detected instance with:
- label: "hand" or "foot"
- bounding box: [ymin, xmin, ymax, xmax]

Count:
- Total hands
- Total feet

Rules:
- 3 or more hands -> Score 0.0, REJECT immediately.
- 3 or more feet -> Score 0.0, REJECT immediately.
- Fewer than two visible limbs is acceptable.
- Never penalize a person for disability or missing
limbs.
- Do not over-count or under-count visible limbs.

STEP 2 - CHECK IMAGE FAILURES

Evaluate for:

- EXTRA_LIMB:
More than two hands or feet.

- ANATOMICAL_IMPOSSIBILITY:
Impossible joints, distorted proportions, or
unnatural body twists.

- BODY_DOUBLE:
Duplicated heads, faces, or torsos.

- CLOTHING_DISCONTINUITY:
Garments terminate unnaturally or appear unsupported
```

by the body.

- BODY\_SHAPE\_INCONSISTENCY:  
Abrupt or physically implausible body-shape transitions.
- BACKGROUND\_ARTIFACT:  
Background elements merge into the body or severe color bleeding occurs.
- SPECIAL\_CASE\_ARTIFACT:  
Distortions involving items such as hijabs, turbans, tattoos, or related special visual attributes.
- OTHER\_ANOMALIES:  
Any other clearly implausible generation artifact.

STEP 3 - SCORE

Use the following discrete scale:

1.0 - Perfect image with no detected issues.

0.8 - Only minor texture, background, or lighting imperfections.  
Anatomy remains correct. ACCEPT.

0.6 - Small clothing or finger issue with correct limb count. REJECT.

0.4 - Incorrect proportions or abrupt garment/body transition. REJECT.

0.2 - Obvious distortion or fused body regions. REJECT.

0.0 - Extra limbs, duplicated body structures, or multiple severe failures. REJECT.

Acceptance threshold:  
- Score  $\geq 0.8$  -> ACCEPT  
- Score  $< 0.8$  -> REJECT

Minor background imperfections alone must not reduce the score below 0.8.

OUTPUT

First output the detected hand/foot JSON list.

Then output:

Total hands: [number]  
Total feet: [number]  
Decision: ACCEPT or REJECT  
Score: [number]  
Primary failure: [  
EXTRA\_LIMB |  
ANATOMICAL\_IMPOSSIBILITY |  
BODY\_DOUBLE |  
CLOTHING\_DISCONTINUITY |  
BODY\_SHAPE\_INCONSISTENCY |  
BACKGROUND\_ARTIFACT |  
SPECIAL\_CASE\_ARTIFACT |  
OTHER\_ANOMALIES |  
NONE  
]

Do not provide explanation, reasoning, or additional text.

### H.3. VLM-Based Try-On Evaluation and Feedback Prompt

Following try-on synthesis, the VLM compares the source person, target garment, and generated try-on result. The evaluator checks person preservation, garment fidelity, layering, fine-grained garment details, pose and occlusion preservation, and anatomical validity. Failed but correctable samples additionally receive an improved generation prompt for iterative refinement.

Listing 3. System prompt used for VLM-based virtual try-on evaluation and feedback.

SYSTEM PROMPT - Virtual Try-On Triplet Quality Assessment

You are an expert evaluator of Virtual Try-On results.

INPUTS

- Image A: source person and identity reference.
- Image B: target garment reference.
- Image C: generated try-on result.
- Garment metadata: name, dress\_type, and description.

Evaluate Image C by comparing it directly with Images A and B.

EVALUATION ORDER

1. Catastrophic validity
2. Person and pose preservation
3. Target-garment fidelity
4. Garment replacement and layering
5. Fine-grained garment-detail preservation
6. Occlusion and scene preservation
7. Anatomical and visual quality

CATASTROPHIC FAILURE

Immediately REJECT Image C if it contains severe failures such as:

- duplicated heads or torsos,
- excessive or impossible limbs,
- severe body fragmentation,
- gross anatomical corruption,
- or an otherwise unusable generation.

PERSON PRESERVATION

Use Image A as the reference for:

- facial appearance,
- skin tone,
- hair,
- body build,
- pose,
- visible limb configuration,
- footwear,
- accessories,
- non-target clothing,
- and foreground occlusions.

Do not penalize source clothing that should be replaced according to dress\_type.

TARGET-GARMENT FIDELITY

Use Image B as the reference for:

- garment category,
- silhouette and cut,
- color,
- pattern and print,
- material appearance,
- neckline,



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- sleeve shape and length,</li> <li>- hem,</li> <li>- closure,</li> <li>- and distinctive garment structure.</li> </ul> <p>Also inspect visible fine-grained garment details such as:</p> <ul style="list-style-type: none"> <li>- pockets,</li> <li>- buttons,</li> <li>- zippers,</li> <li>- seams and panel boundaries,</li> <li>- trims,</li> <li>- embroidery,</li> <li>- logos or graphic elements,</li> <li>- small motifs,</li> <li>- decorative elements,</li> <li>- and other clearly visible design details.</li> </ul> <p>Preserve these details when they are clearly observable in Image B.</p> <p>Allow natural changes in folds, wrinkles, perspective, shading, and fabric draping caused by the person's body shape and pose. Do not require pixel-identical reconstruction.</p> <p>A small local detail difference, such as a slightly altered seam, minor pocket-shape variation, or one imperfect button, should be treated as minor detail drift rather than a complete garment mismatch.</p> <p>However, multiple missing details, changed distinctive patterns, incorrect pocket layouts, substantially altered closures, missing prominent buttons, logos, embroidery, or other identity-defining features should be treated as a meaningful garment-fidelity error.</p> <p><b>GARMENT SCOPE</b></p> <p>Interpret dress_type when deciding which source garments should be replaced or preserved.</p> <ul style="list-style-type: none"> <li>- Full-body garment: replace the corresponding torso and lower-body clothing as one coherent garment.</li> <li>- Lower-body garment: preserve the original upper-body clothing.</li> <li>- Upper-body inner garment: replace the corresponding inner top.</li> <li>- Flexible outer garment: preserve visible inner layers when appropriate.</li> <li>- Accessories such as scarves, shawls, and headwear: add the target item while preserving unrelated clothing and footwear.</li> </ul> <p><b>POSE AND OCCLUSION</b></p> <p>Image A determines the person's pose and occlusion structure.</p> <p>Do not infer pose from Image B.</p> <p>Preserve important foreground occluders, their approximate coverage, and their front-to-back relationship with the subject.</p> <p><b>FAILURES TO CHECK</b></p> <ul style="list-style-type: none"> <li>- <b>TARGET_MISMATCH:</b><br/>Wrong garment category, major color, pattern, silhouette, or structure.</li> </ul> | <ul style="list-style-type: none"> <li>- <b>DETAIL_DRIFT:</b><br/>Pockets, buttons, zippers, trims, logos, embroidery, seams, prints, or other visible design details differ from Image B.</li> <li>- <b>LAYERING_ERROR:</b><br/>Incorrect garment replacement or layering.</li> <li>- <b>SOURCE_REMAINS:</b><br/>Visible remnants of clothing that should have been replaced.</li> <li>- <b>IDENTITY_DRIFT:</b><br/>Meaningful change to the person's face, hair, skin tone, or body appearance.</li> <li>- <b>POSE_DRIFT:</b><br/>Significant alteration of the source pose.</li> <li>- <b>OCCLUSION_ERROR:</b><br/>Important occluders are missing or have an incorrect depth relationship.</li> <li>- <b>GARMENT_DISTORTION:</b><br/>Implausible cut, topology, proportions, or garment structure.</li> <li>- <b>MATERIAL_LEAKAGE:</b><br/>Target fabric, pattern, or texture appears on unrelated body or scene regions.</li> <li>- <b>ANATOMY_ERROR:</b><br/>Incorrect limbs, joints, or body geometry.</li> <li>- <b>STRUCTURAL_HALLUCINATION:</b><br/>Duplicated or invented garment components.</li> </ul> <p><b>SCORING</b></p> <p>Use only these scores:</p> <ul style="list-style-type: none"> <li>1.0 - Excellent transfer. Person identity and target garment, including clearly visible distinctive details, are faithfully preserved.</li> <li>0.8 - High-quality result. Minor local texture or detail drift is allowed, such as a small seam, pocket, button, or shading discrepancy that does not change garment identity.</li> <li>0.6 - Acceptable result. One noticeable but limited garment-detail, layering, or visual error is present, while the target garment, person identity, and overall structure remain correct.</li> <li>0.4 - Significant error. Important garment details, structure, layering, pose, or identity are noticeably incorrect, although the intended try-on remains recognizable.</li> <li>0.2 - Major try-on failure with severe garment mismatch or structural corruption.</li> <li>0.0 - Catastrophic or unusable generation.</li> </ul> <p>Scores <math>\geq 0.6</math> are SUCCESS.<br/>Scores <math>&lt; 0.6</math> are NOT_SUCCESS.</p> <p><b>OUTPUT</b></p> <p>For catastrophic failure:</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

{
  "result": "REJECT",
  "score": 0.0,
  "failure_code": "ANATOMY_ERROR"
}

For successful results:

{
  "result": "SUCCESS",
  "score": 0.8,
  "failure_code": "NONE"
}

For correctable failures:

{
  "result": "NOT_SUCCESS",
  "score": 0.4,
  "failure_code": "DETAIL_DRIFT",
  "reason": "One concise description of the main
    failure.",
  "improved_prompt": "One concise natural-language
    instruction describing
    how to correct the failed try-on while preserving
    the source person,
    pose, occlusions, garment structure, and visible
    target-garment details."
}

When DETAIL_DRIFT is the main failure, the improved
prompt should
explicitly mention the missing or altered visible
details from Image B,
such as pockets, buttons, closures, embroidery, logos,
    trims, or patterns,
without changing unrelated parts of the garment.

Return valid JSON only.
Do not output markdown, analysis, or additional
    explanation.

```

## I. Generation Dictionaries

This section provides the structured dictionaries used by the CURVTON-205K data-generation pipeline. We report the source-person attribute and garment metadata dictionaries, followed by the difficulty-tier dictionaries used to introduce controlled variations in pose, occlusion, scene conditions, and other visual factors.

The source-person dictionary defines the attribute space used for person generation, including body shape, pose, geographic and cultural appearance cues, gender-conditioned clothing, accessories, visible assistive devices, background context, and photographic style. The garment dictionary defines the garment categories and structural metadata used during garment generation and virtual try-on synthesis.

The difficulty-editing space is organized into three levels: Easy, Medium, and Hard. The Medium tier uses a common set of pose, occlusion, background, camera-view, and age attributes for all source persons. Only the special-case distribution contains a gender-conditioned branch, primarily to support pregnancy-related examples while maintaining normalized sampling probabilities.

### I.1. Source-Person and Garment Dictionaries

The complete source-person attribute dictionary and garment metadata dictionary are provided in Listing 4. For compactness and reproducibility, both dictionaries are stored in a single Python source file.

Listing 4. Complete source-person attribute and garment metadata dictionaries used in the CURVTON-205K generation pipeline.

```
# -*- coding: utf-8 -*-

# =====
# CURVTON-205K: SOURCE-PERSON ATTRIBUTE DICTIONARY +
# GARMENT METADATA
# =====

# This file combines the complete contents of:
# 1. people.py
# 2. ALL_DRESS_INFO.py
#
# The original people.py is a standalone dictionary
# literal. It is assigned
# here to PEOPLE_DICT so both resources can be
# imported from one Python file.
# =====
PEOPLE_DICT = {
    "body_type_descriptions": {
        "body_types": {
            "short_underweight": {
                "description": "short underweight person (
                BMI 14-18.5), small frame, lean build, slight
                physique",
                "prob": 0.1
            },
            "short_normal": {
                "description": "short person of normal
                weight (BMI 18.5-24.9), healthy build,
                proportionate body",
                "prob": 0.1
            },
            "short_overweight": {
                "description": "short overweight person (
                BMI 25-30), stocky build, noticeable belly,
                rounder physique",
                "prob": 0.1
            },
            "average_underweight": {
                "description": "average height underweight
                person (BMI 14-18.5), lean frame, delicate build",
                "prob": 0.1
            },
            "average_normal": {
                "description": "average height person of
                normal weight (BMI 18.5-24.9), proportionate
                build, healthy appearance",
                "prob": 0.2
            },
            "average_overweight": {
                "description": "average height overweight
                person (BMI 25-30), fuller build, slightly
                rounded physique",
                "prob": 0.1
            },
            "tall_underweight": {
                "description": "tall underweight person (
                BMI 14-18.5), lanky build, thin frame, elongated
                limbs",
                "prob": 0.1
            },
            "tall_normal": {
                "description": "tall person of normal
                weight (BMI 18.5-24.9), proportionate build,
                athletic or slender frame",
                "prob": 0.1
            },
            "tall_overweight": {
                "description": "tall overweight person (
                BMI 25-30), broad frame, fuller body, rounded
                torso",
                "prob": 0.1
            }
        }
    },
    "pose": "frontal_full_body_standing",
    "countries_ethnicities": {
        "Nepal": {
            "prob": 0.02,
            "ethnicities": [
                "Brahmnin", "Chhetri", "Newar", "Tamang", "
                Sherpa", "Tharu", "Madhesi", "Gurung", "Magar"
            ]
        },
        "India": {
            "prob": 0.15,
            "ethnicities": [
                "Punjabi",
                "Gujarati",
                "Marathi",
                "Rajasthani (Marwari)",
                "Bengali",
                "Assamese",
                "Tamil",
                "Telugu",
                "Kannada",
                "Malayali",
                "Kashmiri",
                "Bihari",
                "Odia",
                "Tulu",
                "Konkani",
                "Kodava",
                "Gond",
                "Santhal",
                "Munda",
                "Ho",
            ]
        }
    }
}
```

```
    "short_overweight": {
        "description": "short overweight person (
        BMI 25-30), stocky build, noticeable belly,
        rounder physique",
        "prob": 0.1
    },
    "average_underweight": {
        "description": "average height underweight
        person (BMI 14-18.5), lean frame, delicate build",
        "prob": 0.1
    },
    "average_normal": {
        "description": "average height person of
        normal weight (BMI 18.5-24.9), proportionate
        build, healthy appearance",
        "prob": 0.2
    },
    "average_overweight": {
        "description": "average height overweight
        person (BMI 25-30), fuller build, slightly
        rounded physique",
        "prob": 0.1
    },
    "tall_underweight": {
        "description": "tall underweight person (
        BMI 14-18.5), lanky build, thin frame, elongated
        limbs",
        "prob": 0.1
    },
    "tall_normal": {
        "description": "tall person of normal
        weight (BMI 18.5-24.9), proportionate build,
        athletic or slender frame",
        "prob": 0.1
    },
    "tall_overweight": {
        "description": "tall overweight person (
        BMI 25-30), broad frame, fuller body, rounded
        torso",
        "prob": 0.1
    }
}

"pose": "frontal_full_body_standing",

"countries_ethnicities": {
    "Nepal": {
        "prob": 0.02,
        "ethnicities": [
            "Brahmnin", "Chhetri", "Newar", "Tamang", "
            Sherpa", "Tharu", "Madhesi", "Gurung", "Magar"
        ]
    },
    "India": {
        "prob": 0.15,
        "ethnicities": [
            "Punjabi",
            "Gujarati",
            "Marathi",
            "Rajasthani (Marwari)",
            "Bengali",
            "Assamese",
            "Tamil",
            "Telugu",
            "Kannada",
            "Malayali",
            "Kashmiri",
            "Bihari",
            "Odia",
            "Tulu",
            "Konkani",
            "Kodava",
            "Gond",
            "Santhal",
            "Munda",
            "Ho",
        ]
    }
}
```



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    "Oraon",
    "Bhil",
    "Naga (Angami, Ao, Sema)",
    "Mizo",
    "Khasi",
    "Garó",
    "Bhutia",
    "Lepcha",
    "Manipuri (Meitei)",
    "Toda",
    "Irula"
  ]
},
"China": {
  "prob": 0.15,
  "ethnicities": [
    "Han",
    "Uyghur",
    "Tibetan",
    "Zhuang",
    "Hui",
    "Manchu",
    "Mongol",
    "Kazakh",
    "Kyrgyz",
    "Tajik",
    "Miao",
    "Yao",
    "Dong",
    "Bouyei",
    "Bai",
    "Hani",
    "Yi",
    "Lisu",
    "Lahu",
    "Wa",
    "Naxi",
    "Dai",
    "Jingpo",
    "Tujia",
    "She",
    "Li",
    "Salar",
    "Xibe",
    "Achang",
    "Nu",
    "Daur",
    "Oroqen",
    "Evenki",
    "Hezhen"
  ]
},
"Japan": {
  "prob": 0.10,
  "ethnicities": [
    "Yamato",
    "Ainu",
    "Okinawan (Ryukyuan)",
    "Amami",
    "Yaeyama",
    "Miyako",
    "Burakumin (historical minority)",
    "Zainichi Korean",
    "Zainichi Chinese"
  ]
},
"Korea": {
  "prob": 0.10,
  "ethnicities": [
    "Korean (Mainland)",
    "Jeju (Island)",
    "Gyeongang",
    "Jeolla",
    "Chungcheong",
    "Gangwon",
    "Hamgyong",

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    "Pyongan",
    "Koryo-saram",
    "Jeju diaspora"
  ]
},
"Africa": {
  "prob": 0.08,
  "ethnicities": [
    "Hausa",
    "Yoruba",
    "Igbo",
    "Akan",
    "Mande (Bambara)",
    "Ewe",
    "Amhara",
    "Oromo",
    "Somali",
    "Kikuyu",
    "Luo",
    "Maasai",
    "Kongo",
    "Bantu (various groups)",
    "Tutsi",
    "Hutu",
    "Zulu",
    "Xhosa",
    "Sotho",
    "Tswana",
    "Shona",
    "Berber/Amazigh (Kabyle, Tarifit)",
    "Tuareg",
    "Egyptian (Arab)",
    "Coptic"
  ]
},
"Middle_East": {
  "prob": 0.07,
  "ethnicities": [
    "Arab (Levantine)",
    "Arab (Gulf)",
    "Arab (Maghrebi)",
    "Kurd (Kurmanji, Sorani)",
    "Persian (Farsi)",
    "Azeri",
    "Assyrian",
    "Chaldean",
    "Armenian",
    "Bedouin",
    "Yazidi",
    "Druze",
    "Turk",
    "Circassian"
  ]
},
"Latin_America_Mexico": {
  "prob": 0.07,
  "ethnicities": [
    "Nahua",
    "Maya (Yucatec)",
    "Zapotec",
    "Mixtec",
    "Otomi",
    "Purépecha",
    "Totonac",
    "Afro-Mexican",
    "Mestizo",
    "Criollo"
  ]
},
"Central_America": {
  "prob": 0.07,
  "ethnicities": [
    "Maya (K'iche', Kaqchikel, Q'eqchi')",
    "Lenca",
    "Pipil",

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    "Bribri",
    "Ngäbe-Buglé",
    "Kuna (Guna)",
    "Miskito",
    "Garifuna",
    "Mestizo",
    "Criollo"
  ]
},

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"South_America": {
  "prob": 0.07,
  "ethnicities": [
    "Quechua",
    "Aymara",
    "Guarani",
    "Mapuche",
    "Kichwa (Ecuador)",
    "Wayuu",
    "Yanomami",
    "Shuar",
    "Warao",
    "Afro-Brazilian",
    "Afro-Colombian",
    "Afro-Peruvian",
    "Afro-Ecuadorian",
    "Afro-Venezuelan",
    "Quilombola",
    "Mestizo",
    "Cholo",
    "Pardo",
    "Zambo",
    "Criollo"
  ]
},

```

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"Caribbean": {
  "prob": 0.05,
  "ethnicities": [
    "Afro-Cuban",
    "Afro-Dominican",
    "Afro-Puerto Rican",
    "Afro-Haitian",
    "Indo-Caribbean",
    "Taíno",
    "Kalinago",
    "Garifuna",
    "Creole"
  ]
},

```

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"North_American_Indigenous": {
  "prob": 0.02,
  "ethnicities": [
    "Navajo (Diné)",
    "Cherokee",
    "Lakota",
    "Ojibwe",
    "Cree",
    "Haida",
    "Inuit",
    "Métis"
  ]
},

```

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"Oceania": {
  "prob": 0.03,
  "ethnicities": [
    "Maori",
    "Aboriginal Australian",
    "Torres Strait Islander",
    "Samoan",
    "Tongan",
    "Fijian",
    "Papuan"
  ]
},

```

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"Europe_Minorities": {

```

```

  "prob": 0.02,
  "ethnicities": [
    "Sami",
    "Basque",
    "Catalan",
    "Breton",
    "Roma",
    "Sorbian",
    "Frisian"
  ]
},

```

```

"gender": {
  "male": {
    "prob": 0.5,
    "clothing": [
      "T-shirt", "Polo", "Shirt", "Tank-top", "Sweater", "Hoodie", "Jacket", "Windbreaker", "Coat", "Blazer", "Suit (two-piece)", "Jeans", "Trousers", "Chinos", "Shorts", "Kurta", "Kurta-pajama", "Dhoti", "Lungi", "Sherwani", "Hanfu-style top", "Deel (traditional Mongol robe)", "Kimono (casual style)", "Yukata", "Kaftan", "Thobe", "Dashiki", "Kente-wrapper", "Poncho", "Parkas", "Puffer-jacket", "Athletic-wear (tracksuit)", "Gym-top", "School-uniform", "Work-uniform", "Lab-coat", "Traditional robe", "Swimwear (one-piece)", "Nightwear (pajamas)"
    ]
  },
  "female": {
    "prob": 0.5,
    "clothing": [
      "T-shirt", "Blouse", "Tank-top", "Sweater", "Hoodie", "Jacket", "Windbreaker", "Coat", "Blazer", "Suit (two-piece)", "Jeans", "Trousers", "Chinos", "Shorts", "Skirt", "Dress (casual)", "Dress (formal)", "Maxi-dress", "Sari", "Kurta", "Salwar-Kameez", "Kebaya", "Qipao/Cheongsam", "Hanfu-style top", "Deel (traditional Mongol robe)", "Kimono (casual style)", "Yukata", "Kaftan", "Abaya", "Dashiki", "Kente-wrapper", "Poncho", "Parkas", "Puffer-jacket", "Athletic-wear (tracksuit)", "Gym-top", "School-uniform", "Work-uniform", "Lab-coat", "Traditional robe", "Traditional dress (regional descriptor allowed)", "Swimwear (one-piece)", "Swimwear (two-piece)", "Nightwear (pajamas)"
    ]
  }
},

```

```

"spectacles": {
  "no": {"prob": 0.85},
  "yes": {"prob": 0.15}
},

```

```

"disabilities_visible": {
  "none": {"prob": 0.8},
  "with_disability": {
    "prob": 0.2,
    "types": [
      "missing_thumb_left",
      "missing_thumb_right",
      "missing_index_left",
      "missing_index_right",
      "missing_middle_left",
      "missing_middle_right",
      "missing_ring_left",

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"missing_ring_right",
"missing_little_left",
"missing_little_right",
"missing_multiple_fingers_left",
"missing_multiple_fingers_right",
"missing_hand_left",
"missing_hand_right",
"partial_arm_amputation_left",
"partial_arm_amputation_right",
"missing_arm_above_elbow_left",
"missing_arm_above_elbow_right",
"missing_leg_left",
"missing_leg_right",
"bilateral_leg_amputee",
"prosthetic_leg_left_visible",
"prosthetic_leg_right_visible",
"prosthetic_arm_left_visible",
"prosthetic_arm_right_visible",
"wheelchair_user_visible",
"uses_crutches_visible",
"uses_cane_visible",
"short_stature_visible",
"limb_length_discrepancy_visible",
"visible_orthosis_brace"
]
}
},
"background": {
"plain_white": {"prob": 0.1},
"plain_gray": {"prob": 0.1},
"aesthetic_minimal": {"prob": 0.1},
"beach": {"prob": 0.1},
"village_rural": {"prob": 0.1},
"shopping_mall": {"prob": 0.1},
"city_street": {"prob": 0.1},
"park_outdoor": {"prob": 0.1},
"home_interior": {"prob": 0.1},
"office_workspace": {"prob": 0.1}
},
"photo_style": {
"professional": {
"prob": 0.5,
"attributes": {
"body part visible" : "head to toe",
"camera": "professional DSLR camera",
"focus": "sharp focus, high clarity",
"lighting": "well-lit",
"composition": "centered composition, proper framing",
"quality": "high resolution, crisp details"
}
},
"on_the_wild": {
"prob": 0.5,
"attributes": {
"body part visible" : "head to toe",
"camera": "smartphone camera, casual snapshot",

"focus": "slightly out of focus, no background blur",
"lighting": "ambient natural lighting, uneven exposure",
"composition": "candid angle, off-center framing",
"quality": "minimal grain, slight motion blur, authentic feel"
}
}
}
}
}

# =====
# COMPLETE GARMENT METADATA DICTIONARY
# Original content from ALL_DRESS_INFO.py follows
verbatim below.

```

```

# =====
# =====
# DRESS_INFO -- Garment metadata for virtual try-on (
VTON) pipeline
#
# dress_type categories:
# "fullbody dress" -- robes, sarees, kimonos,
gowns (cover torso + legs as one)
# "lowerbody dress" -- skirts, trousers, wraps
secured at waist
# "upperbody inner dress" -- standalone tops, shirts,
blouses, tunics up to knee length
# "upperbody flex dress" -- outer layers worn OVER
inner: jackets, coats, sweaters
# "fullbody flex dress" -- full-length outer
layers worn OVER a complete outfit
# (cloaks, ceremonial over
-robes, draped full-body wraps)
# "shawl" -- rectangular/square
cloth draped over shoulders or full body
# "scarf" -- cloth wrapped around
the head or neck
# "cap" -- structured hats and
headwear
# "head wrap" -- fabric wrapped around
head in a decorative/formal style
# "head and face wrap" -- fabric covering both
head and face
#
# description rules:
# - Structural features ONLY: silhouette, length,
sleeve type, collar, closures
# - NO color, NO pattern, NO embroidery or surface
decoration
# - Max 30 words
#
# prompt template:
# "Make this person wear this dress, which is a {
name},
# a {dress_type} characterized by its {description
}."

DRESS_INFO = [

# =====
# =====
# NEPAL
# =====

{
"name": "Daura Suruwal",
"dress_type": "fullbody dress",
"description": "collarless double-breasted
tied shirt reaching mid-thigh, paired with
tapered trousers fitted tight from knee to ankle"
},
{
"name": "Gunyo Cholo",
"dress_type": "fullbody dress",
"description": "saree-style long wraparound
cloth worn with a short-sleeved fitted blouse;
traditional Nepali women's two-piece"
},
{
"name": "Haku Patasi",
"dress_type": "fullbody dress",
"description": "long wraparound cloth draped
as saree reaching the ankle, worn with a short
fitted blouse underneath"
},
{
"name": "Chaubandi Cholo",
"dress_type": "upperbody inner dress",
"description": "fitted blouse with four-
cornered tie fastening at front and back, short
sleeves"
},

```

```

{
  "name": "Bakhu (Chuba)",
  "dress_type": "fullbody dress",
  "description": "full-length loose robe with
wide sleeves and overlapping front, tied at waist
with a long sash"
},
{
  "name": "Pangden",
  "dress_type": "fullbody dress",
  "description": "wide rectangular apron panel
with vertical divisions, tied at waist falling to
ankles, worn over a robe"
},
{
  "name": "Mekhli",
  "dress_type": "lowerbody dress",
  "description": "wraparound skirt secured at
waist, falling to mid-calf or ankle"
},
{
  "name": "Ghalek",
  "dress_type": "fullbody dress",
  "description": "sleeveless overdress falling
to ankle, worn over a blouse, with open sides or
side slits"
},
{
  "name": "Bhangra",
  "dress_type": "upperbody flex dress",
  "description": "short waist-length jacket with
a front opening, worn over inner garments"
},
{
  "name": "Kachhad",
  "dress_type": "lowerbody dress",
  "description": "rectangular cloth wrapped
around waist as lower garment, tucked between
legs in dhoti style"
},
{
  "name": "Phariya",
  "dress_type": "lowerbody dress",
  "description": "plain wraparound skirt tied at
waist, falling to the ankle"
},
{
  "name": "Sari",
  "dress_type": "fullbody dress",
  "description": "long cloth draped around body
with pleats at front waist, one end thrown over
shoulder, worn with fitted blouse"
},
{
  "name": "Dhaka Topi",
  "dress_type": "cap",
  "description": "traditional brimless Nepali
cap with a distinctive folded angular shape"
},
{
  "name": "Shyama",
  "dress_type": "upperbody inner dress",
  "description": "short fitted blouse with elbow-
length or short sleeves; Tharu women's
traditional upper garment"
},
{
  "name": "Tamang Selo Dress",
  "dress_type": "fullbody dress",
  "description": "fitted blouse paired with a
wraparound skirt and a front decorative drape;
Tamang women's traditional two-piece"
},
{
  "name": "Sherpa Bakhu",
  "dress_type": "fullbody dress",
  "description": "full-length robe with wide
overlapping front, long sleeves, secured at waist
with a wide sash"
}

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```

},
{
  "name": "Limbu Mekhli",
  "dress_type": "lowerbody dress",
  "description": "wraparound skirt falling to
mid-calf, secured at waist; Limbu women's
traditional lower garment"
},
{
  "name": "Rai Lungi",
  "dress_type": "lowerbody dress",
  "description": "wraparound lower cloth tied at
waist, falling to the knee or mid-calf"
},
{
  "name": "Sunuwar Dress",
  "dress_type": "fullbody dress",
  "description": "traditional blouse and
wraparound skirt combination falling to ankle;
Sunuwar community women's attire"
},
# =====
# INDIA
# =====
{
  "name": "Mekhela Sador",
  "dress_type": "fullbody dress",
  "description": "two-piece: cylindrical skirt
wrapped at waist plus a separate cloth draped
over the shoulder and tucked at chest"
},
{
  "name": "Nauvari",
  "dress_type": "fullbody dress",
  "description": "nine-yard saree draped with
the end passed between legs and tucked at back,
worn with a blouse"
},
{
  "name": "Kashta",
  "dress_type": "fullbody dress",
  "description": "saree draped with the end
passed between legs and tucked at waist; worn
with a fitted blouse"
},
{
  "name": "Phiran",
  "dress_type": "fullbody dress",
  "description": "long loose robe reaching below
the knee with wide long sleeves and a round or V
neckline"
},
{
  "name": "Puan",
  "dress_type": "shawl",
  "description": "rectangular woven cloth draped
over shoulders or wrapped around the upper body;
Mizo women's shawl"
},
{
  "name": "Jainsem",
  "dress_type": "fullbody dress",
  "description": "Khasi two-piece: a cylindrical
tube skirt plus a draped cloth fastened over one
shoulder"
},
{
  "name": "Dokona",
  "dress_type": "lowerbody dress",
  "description": "wraparound lower cloth secured
at waist, falling to mid-calf; Tripuri women's
traditional skirt"
},
{
  "name": "Phanek",

```



```

    "dress_type": "lowerbody dress",
    "description": "cylindrical wraparound skirt
tucked and folded at waist, falling to mid-calf;
Meitei women's garment"
},
{
    "name": "Innaphi",
    "dress_type": "shawl",
    "description": "lightweight rectangular cloth
draped over the shoulders; Meitei women's
shoulder shawl"
},
{
    "name": "Ghagra Choli",
    "dress_type": "fullbody dress",
    "description": "full circular flared long
skirt falling to ankle paired with a short fitted
midriff-baring blouse"
},
{
    "name": "Chaniya Choli",
    "dress_type": "fullbody dress",
    "description": "heavily flared ankle-length
skirt paired with a short fitted blouse and a
draped dupatta"
},
{
    "name": "Kediyu",
    "dress_type": "upperbody inner dress",
    "description": "short Gujarati men's top with
gathered fabric and flared hem at the waist,
short-sleeved"
},
{
    "name": "Mundu",
    "dress_type": "lowerbody dress",
    "description": "single rectangular cloth
wrapped at waist falling to the ankle; Kerala men
's lower garment"
},
{
    "name": "Veshti",
    "dress_type": "lowerbody dress",
    "description": "single rectangular cloth
wrapped at waist falling to the ankle; Tamil men'
s lower garment"
},
{
    "name": "Lungi",
    "dress_type": "lowerbody dress",
    "description": "cylindrical tube cloth wrapped
at waist, falling to mid-calf or ankle"
},
{
    "name": "Sherwani",
    "dress_type": "fullbody dress",
    "description": "knee-length formal coat with a
Nehru collar, button front, fitted at chest and
slightly flared below the waist"
},
{
    "name": "Angrakha",
    "dress_type": "upperbody inner dress",
    "description": "knee-length tunic with
overlapping asymmetric front panels tied at the
side, long sleeves"
},
{
    "name": "Gara",
    "dress_type": "fullbody dress",
    "description": "long cloth draped as saree
reaching the ankle, worn with a blouse; Parsi
women's formal saree"
},
{
    "name": "Salwar Kameez",
    "dress_type": "fullbody dress",
    "description": "long tunic with side slits
reaching below the hip, worn over loose tapered

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trousers gathered at ankle"
},
{
    "name": "Patiala Salwar",
    "dress_type": "fullbody dress",
    "description": "long kurta worn over very
voluminous pleated trousers with deep horizontal
folds, gathered at ankle"
},
{
    "name": "Churidar",
    "dress_type": "lowerbody dress",
    "description": "fitted trousers very tight
from knee down, creating horizontal ruching folds
at the ankle"
},
{
    "name": "Lehenga",
    "dress_type": "lowerbody dress",
    "description": "heavily flared circular ankle-
length skirt with a wide waistband"
},
{
    "name": "Kurta Pajama",
    "dress_type": "fullbody dress",
    "description": "knee-length tunic with long
sleeves and side slits, worn over loose straight-
leg trousers"
},
{
    "name": "Bandhgala",
    "dress_type": "upperbody flex dress",
    "description": "fitted formal jacket with a
high band collar and front button fastening, hip
length, long sleeves"
},
{
    "name": "Pochampally Saree",
    "dress_type": "fullbody dress",
    "description": "long cloth draped as saree
with pleats at front, one end over shoulder, worn
with a fitted blouse"
},
{
    "name": "Kasavu Saree",
    "dress_type": "fullbody dress",
    "description": "long cloth draped as saree
with pleats at front, one end over shoulder, worn
with a fitted blouse"
},
{
    "name": "Muga Silk Saree",
    "dress_type": "fullbody dress",
    "description": "long cloth draped as saree
with pleats at front, one end over shoulder, worn
with a fitted blouse"
},
{
    "name": "Santhal Dress",
    "dress_type": "fullbody dress",
    "description": "draped cloth wrapped around
the body from waist to ankle with an additional
upper drape over the chest"
},
{
    "name": "Toda Puthkuli",
    "dress_type": "shawl",
    "description": "rectangular woven cloth draped
over one or both shoulders; Toda ceremonial
shawl"
},
{
    "name": "Naga Shawl",
    "dress_type": "shawl",
    "description": "rectangular woven cloth worn
draped over the shoulders; Naga ceremonial and
status shawl"
},

```

```
# =====
# PAKISTAN & BANGLADESH
# =====

{
  "name": "Balochi Suit",
  "dress_type": "fullbody dress",
  "description": "very long wide-sleeved tunic reaching well below the knee with a deep V or round neckline, worn over loose trousers"
},
{
  "name": "Pashtun Dress",
  "dress_type": "fullbody dress",
  "description": "long loose tunic reaching below the knee with wide long sleeves, worn over very wide-legged trousers"
},
{
  "name": "Perahan Tunban",
  "dress_type": "fullbody dress",
  "description": "long loose tunic with long sleeves reaching mid-calf, worn over wide-legged drawstring trousers"
},
{
  "name": "Sindhi Topi",
  "dress_type": "cap",
  "description": "traditional flat-topped Sindhi cap with a rounded or oval shape"
},
{
  "name": "Ajrak",
  "dress_type": "shawl",
  "description": "large rectangular cloth worn draped over shoulders or wrapped around the body"
},
{
  "name": "Khet Partug",
  "dress_type": "lowerbody dress",
  "description": "extremely wide-legged drawstring trousers with very voluminous fabric gathered at the waist"
},
{
  "name": "Jamdani Saree",
  "dress_type": "fullbody dress",
  "description": "long cloth draped as saree with pleats at front, one end thrown over shoulder, worn with fitted blouse"
},
{
  "name": "Panjabi",
  "dress_type": "upperbody inner dress",
  "description": "long loose tunic with long sleeves and side slits, reaching the knee; men's everyday upper garment"
},
{
  "name": "Kalash Dress",
  "dress_type": "fullbody dress",
  "description": "straight ankle-length dress with long sleeves and a round neckline; Kalash women's traditional garment"
},
{
  "name": "Sindhi Lehenga",
  "dress_type": "lowerbody dress",
  "description": "full flared ankle-length skirt with a wide waistband"
},
{
  "name": "Bangladeshi Lungii",
  "dress_type": "lowerbody dress",
  "description": "cylindrical tube cloth wrapped at waist, falling to mid-calf; men's everyday lower garment"
}
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},
{
  "name": "Chakma Dress",
  "dress_type": "fullbody dress",
  "description": "two-piece of a wraparound skirt reaching the ankle and a fitted blouse; Chakma women's traditional attire"
},

# =====
# JAPAN
# =====

{
  "name": "Kimono",
  "dress_type": "fullbody dress",
  "description": "T-shaped full-length robe with wide sleeves, wrapped left-over-right, secured with a wide sash at waist"
},
{
  "name": "Furisode",
  "dress_type": "fullbody dress",
  "description": "full-length formal kimono with extremely long flowing sleeves reaching near the floor, wrapped and sash-secured"
},
{
  "name": "Tomesode",
  "dress_type": "fullbody dress",
  "description": "full-length formal kimono with shorter sleeves, wrapped left-over-right, secured with a wide sash"
},
{
  "name": "Shiromuku",
  "dress_type": "fullbody dress",
  "description": "full-length bridal robe worn layered with a heavily padded outer robe and a structured hood"
},
{
  "name": "Hakama",
  "dress_type": "lowerbody dress",
  "description": "wide pleated divided or undivided trousers falling to the ankle, secured at waist with long ties"
},
{
  "name": "Yukata",
  "dress_type": "fullbody dress",
  "description": "lightweight full-length robe with wide sleeves, wrapped left-over-right and secured with a sash at waist"
},
{
  "name": "Haori",
  "dress_type": "upperbody flex dress",
  "description": "hip-length open-front jacket with wide sleeves, worn over kimono without fastening"
},
{
  "name": "Happi",
  "dress_type": "upperbody inner dress",
  "description": "straight hip-length open-front coat with wide sleeves, secured with a tie, worn as festival attire"
},
{
  "name": "Hanten",
  "dress_type": "upperbody flex dress",
  "description": "padded hip-length jacket with wide sleeves and a lapel collar, worn over other garments for warmth"
},
{
}
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    "name": "Jinbei",
    "dress_type": "fullbody dress",
    "description": "two-piece: short-sleeved lapel
top tied at front, paired with above-knee shorts
with drawstring waist"
},
{
    "name": "Samue",
    "dress_type": "fullbody dress",
    "description": "two-piece work wear: hip-
length jacket with wide sleeves paired with
straight-leg trousers"
},
{
    "name": "Fundoshi",
    "dress_type": "lowerbody dress",
    "description": "rectangular loincloth wrapped
and tied around the hips; traditional Japanese
undergarment"
},
{
    "name": "Montsuki",
    "dress_type": "fullbody dress",
    "description": "full-length formal kimono with
wide sleeves, wrapped and sash-secured;
typically worn with hakama"
},
{
    "name": "Ryusou",
    "dress_type": "fullbody dress",
    "description": "traditional Okinawan full-
length robe with wide sleeves and overlapping
front, tied with a sash"
},
{
    "name": "Ainu Attus",
    "dress_type": "fullbody dress",
    "description": "straight robe falling to knee
or ankle with long sleeves and a straight collar,
worn belted"
},
{
    "name": "Uchikake",
    "dress_type": "fullbody flex dress",
    "description": "heavily padded full-length
outer kimono with wide sleeves and a long
trailing hem, worn open over kimono"
},
# =====
# EAST ASIA
# =====

{
    "name": "Hanfu",
    "dress_type": "fullbody dress",
    "description": "traditional Chinese robe with
wide crossed collar, wide flowing sleeves, and a
long skirt secured with ties or belt"
},
{
    "name": "Ruqun",
    "dress_type": "fullbody dress",
    "description": "Chinese two-piece: short
crossed-collar blouse paired with a high-waisted
long wrap skirt tied at chest"
},
{
    "name": "Shenyi",
    "dress_type": "fullbody dress",
    "description": "ancient Chinese one-piece full-
length robe with wide sleeves and a curved
wraparound hem"
},
{
    "name": "Qipao",
    "dress_type": "fullbody dress",

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    "description": "form-fitting full-length dress
with a high mandarin collar, side slits,
sleeveless or short-sleeved"
},
{
    "name": "Cheongsam",
    "dress_type": "fullbody dress",
    "description": "form-fitting full-length or
knee-length dress with a high mandarin collar,
diagonal front fastening, and side slits"
},
{
    "name": "Changshan",
    "dress_type": "fullbody dress",
    "description": "men's full-length fitted robe
with a mandarin collar, front toggle closures,
and side slits"
},
{
    "name": "Hanbok",
    "dress_type": "fullbody dress",
    "description": "Korean two-piece: short tied
wrap blouse with a very full high-waisted bell-
shaped long skirt"
},
{
    "name": "Chima",
    "dress_type": "lowerbody dress",
    "description": "full bell-shaped skirt
starting at chest level with shoulder straps,
falling to ankle; Korean women's lower garment"
},
{
    "name": "Jeogori",
    "dress_type": "upperbody inner dress",
    "description": "short Korean wrap blouse with
long sleeves, crossing at front and tied with a
long ribbon at the chest"
},
{
    "name": "Gat",
    "dress_type": "cap",
    "description": "traditional Korean cylindrical
hat with a wide flat brim and a tall straight
cylindrical crown"
},
{
    "name": "Durumagi",
    "dress_type": "upperbody flex dress",
    "description": "full-length Korean overcoat
with long wide sleeves, a straight collar, and
front tie fastening"
},
{
    "name": "Deel",
    "dress_type": "fullbody dress",
    "description": "full-length Mongolian robe
with a round collar, overlapping front, long
sleeves, secured with a sash"
},
{
    "name": "Uyghur Atlas Dress",
    "dress_type": "fullbody dress",
    "description": "fitted full-length dress with
a V or round neckline, long or short sleeves, and
a flared skirt"
},
{
    "name": "Zhuang Ethnic Dress",
    "dress_type": "fullbody dress",
    "description": "collarless or stand-collar
tunic reaching mid-thigh worn over loose trousers;
Zhuang women's traditional outfit"
},
{
    "name": "Miao Embroidered Dress",
    "dress_type": "fullbody dress",
    "description": "fitted long-sleeved blouse
worn with a pleated skirt falling to knee or mid-

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calf; Miao women's dress"
},
{
  "name": "Yi Ethnic Robe",
  "dress_type": "fullbody dress",
  "description": "full-length robe with a wide
collar and long sleeves, wrapped and secured at
waist; Yi traditional garment"
},
{
  "name": "Korean Jeonbok",
  "dress_type": "upperbody flex dress",
  "description": "sleeveless Korean vest with a
V-shaped open front, worn over the jeogori blouse"
},
{
  "name": "Korean Dupo",
  "dress_type": "fullbody dress",
  "description": "full-length wide-sleeved
scholar's robe with a straight collar and open
front; Korean men's formal wear"
},
# =====
# SOUTHEAST ASIA
# =====

{
  "name": "Ao Dai",
  "dress_type": "fullbody dress",
  "description": "fitted full-length tunic with
high mandarin collar and side slits from waist to
hem, worn over wide-leg trousers"
},
{
  "name": "Ao Ba Ba",
  "dress_type": "fullbody dress",
  "description": "loose collarless long-sleeved
blouse reaching below the hip, worn over wide-leg
trousers"
},
{
  "name": "Barong Tagalog",
  "dress_type": "upperbody inner dress",
  "description": "formal untucked sheer long-
sleeved shirt reaching hip length with a mandarin
or open collar"
},
{
  "name": "Baro't Saya",
  "dress_type": "fullbody dress",
  "description": "butterfly-sleeved wide blouse
paired with a long wrap skirt; Filipino women's
traditional dress"
},
{
  "name": "Kebaya",
  "dress_type": "upperbody inner dress",
  "description": "fitted long-sleeved blouse
reaching hip or thigh with a front opening, worn
over a wraparound skirt"
},
{
  "name": "Sarong",
  "dress_type": "lowerbody dress",
  "description": "rectangular cloth wrapped at
waist and tied or tucked, falling to ankle or mid-
calf"
},
{
  "name": "Sampot",
  "dress_type": "lowerbody dress",
  "description": "Cambodian rectangular cloth
wrapped at waist and draped in various styles,
falling to knee or ankle"
},

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{
  "name": "Chut Thai",
  "dress_type": "fullbody dress",
  "description": "Thai two-piece: fitted blouse
with a trim collar paired with a silk wraparound
skirt; formal traditional attire"
},
{
  "name": "Sabai",
  "dress_type": "upperbody inner dress",
  "description": "rectangular cloth wrapped
diagonally across the chest and over one shoulder
as an upper body garment"
},
{
  "name": "Longyi",
  "dress_type": "lowerbody dress",
  "description": "cylindrical tube cloth wrapped
at waist and knotted, falling to the ankle;
Burmese everyday skirt"
},
{
  "name": "Baju Kurung",
  "dress_type": "fullbody dress",
  "description": "long-sleeved loose blouse
reaching mid-thigh paired with a long wrap skirt;
Malay women's traditional dress"
},
{
  "name": "Baju Melayu",
  "dress_type": "fullbody dress",
  "description": "loose long-sleeved shirt
reaching mid-thigh paired with matching trousers
and a front-tied waist cloth"
},
{
  "name": "Sinhala Osariya",
  "dress_type": "fullbody dress",
  "description": "saree draped in Sri Lankan
style with pleats at front, one end over shoulder,
worn with a fitted jacket blouse"
},
{
  "name": "Hmong Pleated Dress",
  "dress_type": "fullbody dress",
  "description": "fitted long-sleeved blouse
worn with a heavily pleated skirt and a front
apron panel, knee-length"
},
{
  "name": "Ifugao Tapis",
  "dress_type": "lowerbody dress",
  "description": "rectangular wraparound skirt
secured at waist, falling to mid-thigh or knee;
Ifugao women's lower garment"
},
# =====
# MIDDLE EAST & CENTRAL ASIA
# =====

{
  "name": "Thobe",
  "dress_type": "fullbody dress",
  "description": "full-length loose robe with
long sleeves, collarless or small collar, and
button front, falling to the ankle"
},
{
  "name": "Dishdasha",
  "dress_type": "fullbody dress",
  "description": "full-length loose long-sleeved
robe with a round or stand collar and button
front, falling to ankle"
},
{
  "name": "Kandura",

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        "dress_type": "fullbody dress",
        "description": "full-length loose robe with
long sleeves and a mandarin collar, falling to
the ankle"
    },
    {
        "name": "Bisht",
        "dress_type": "fullbody flex dress",
        "description": "full-length open-front
ceremonial cloak with long sleeves, draped over
the shoulders over other garments"
    },
    {
        "name": "Abaya",
        "dress_type": "fullbody flex dress",
        "description": "full-length loose open-front
or closed over-robe with long sleeves, worn over
other clothing"
    },
    {
        "name": "Jellabiya",
        "dress_type": "fullbody dress",
        "description": "full-length loose robe with
long or short sleeves and a round or V neckline,
no collar"
    },
    {
        "name": "Chapan",
        "dress_type": "fullbody flex dress",
        "description": "full-length open-front Central
Asian robe with long wide sleeves, draped over
other garments"
    },
    {
        "name": "Kaftan",
        "dress_type": "fullbody dress",
        "description": "full-length long-sleeved loose
robe with a front opening and a round or V
neckline"
    },
    {
        "name": "Keffiyeh",
        "dress_type": "scarf",
        "description": "large square cloth folded and
draped over the head and shoulders"
    },
    {
        "name": "Shemagh",
        "dress_type": "scarf",
        "description": "large square cloth wrapped
around the head and neck"
    },
    {
        "name": "Ghutra",
        "dress_type": "scarf",
        "description": "square cloth draped over the
head and held in place with a cord ring (iqal)"
    },
    {
        "name": "Afghan Kuchi Dress",
        "dress_type": "fullbody dress",
        "description": "long-sleeved ankle-length
dress with a round neckline and very full
gathered skirt"
    },
    {
        "name": "Turkmen Don",
        "dress_type": "fullbody dress",
        "description": "full-length long-sleeved
fitted robe with a round neckline; traditional
Turkmen women's dress"
    },
    {
        "name": "Kazakh Shapan",
        "dress_type": "fullbody flex dress",
        "description": "full-length outer robe with
long wide sleeves and a straight collar, worn
over clothing"
    },

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    {
        "name": "Uyghur Doppa Outfit",
        "dress_type": "fullbody dress",
        "description": "long-sleeved fitted full-
length robe; Uyghur traditional attire worn with
a small embroidered cap"
    },

    # =====
    # AFRICA (from list)
    # =====

    {
        "name": "Agbada",
        "dress_type": "fullbody flex dress",
        "description": "very wide-sleeved full-length
open-sided outer robe with extremely voluminous
sleeves, worn over inner tunic and trousers"
    },
    {
        "name": "Dashiki",
        "dress_type": "upperbody inner dress",
        "description": "loose pullover tunic reaching
mid-thigh with a V-shaped neckline opening and
short or long sleeves"
    },
    {
        "name": "Gele",
        "dress_type": "head wrap",
        "description": "large rectangular fabric
wrapped and folded around the head into a
structured sculptural shape"
    },
    {
        "name": "Kanzu",
        "dress_type": "fullbody dress",
        "description": "full-length long-sleeved robe
with a collarless or stand collar; East African
men's formal dress"
    },
    {
        "name": "Boubou",
        "dress_type": "fullbody flex dress",
        "description": "very wide floor-length open-
sided outer robe with wide sleeves, worn over
inner garments"
    },
    {
        "name": "Djellaba",
        "dress_type": "fullbody dress",
        "description": "full-length hooded robe with
long sleeves, a pointed hood, and front opening
with toggle buttons"
    },
    {
        "name": "Shuka",
        "dress_type": "fullbody flex dress",
        "description": "rectangular cloth wrapped and
draped around the full body from shoulder to
ankle, worn over minimal undergarments"
    },
    {
        "name": "Isiagu",
        "dress_type": "upperbody inner dress",
        "description": "men's pullover top with long
sleeves and a round or V neckline; Igbo
traditional shirt"
    },
    {
        "name": "Maasai Beaded Dress",
        "dress_type": "fullbody dress",
        "description": "layered wrapped cloth draped
around the body and secured at shoulder"
    },
    {
        "name": "Zulu Isidwaba",
        "dress_type": "lowerbody dress",

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        "description": "stiff cowhide skirt flared at
hem, falling to mid-calf; Zulu women's
traditional lower garment"
    },
    {
        "name": "Xhosa Umbhaco",
        "dress_type": "fullbody dress",
        "description": "long wrapped cloth draped
around the body; Xhosa women's traditional dress"
    },
    {
        "name": "Tuareg Tagelmust",
        "dress_type": "head and face wrap",
        "description": "long cloth wrapped around head
and face, leaving only the eyes visible"
    },
    {
        "name": "Ewe Kente Cloth",
        "dress_type": "fullbody flex dress",
        "description": "wide strip-woven cloth draped
and wrapped around the full body over other
garments; Ewe ceremonial wear"
    },
    {
        "name": "Berber Amazigh Dress",
        "dress_type": "fullbody dress",
        "description": "rectangular cloth wrapped
around the body and pinned at shoulders, falling
to ankle"
    },
    {
        "name": "Herero Victorian Dress",
        "dress_type": "fullbody dress",
        "description": "full-length Victorian-style
dress with a fitted bodice, long sleeves, and a
very full gathered skirt with petticoats"
    },
    },
    # =====
    # THE AMERICAS (from list)
    # =====
    {
        "name": "Huipil",
        "dress_type": "upperbody inner dress",
        "description": "loose rectangular tunic with a
central neck opening, short or no sleeves,
reaching hip to knee"
    },
    {
        "name": "Poncho",
        "dress_type": "upperbody flex dress",
        "description": "rectangular or square cloth
with a central neck opening, draped over
shoulders falling to waist or hip"
    },
    {
        "name": "Ruana",
        "dress_type": "upperbody flex dress",
        "description": "large square or rectangular
cloth with a front slit opening, draped over
shoulders like a cloak"
    },
    {
        "name": "Pollera",
        "dress_type": "fullbody dress",
        "description": "elaborate full-length skirt
with multiple tiered ruffled layers worn with a
matching blouse"
    },
    {
        "name": "Bombín",
        "dress_type": "cap",
        "description": "small round-crowned bowler hat
with a narrow brim, worn perched atop the head"
    },
    },
    {

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        "name": "Guayabera",
        "dress_type": "upperbody inner dress",
        "description": "straight-hem shirt with two
vertical pleat rows on front and back, four
pockets, short or long sleeves"
    },
    {
        "name": "Quechua Lliclla",
        "dress_type": "shawl",
        "description": "rectangular woven cloth worn
over the shoulders and fastened at the chest with
a pin"
    },
    {
        "name": "Navajo Velvet Dress",
        "dress_type": "fullbody dress",
        "description": "long-sleeved fitted dress
reaching mid-calf with a round or square neckline"
    },
    {
        "name": "Inuit Amauti",
        "dress_type": "upperbody flex dress",
        "description": "hooded outer parka with long
sleeves and an enlarged back hood pouch for
carrying a baby, hip length"
    },
    {
        "name": "Mapuche Chamall",
        "dress_type": "upperbody flex dress",
        "description": "rectangular woven cloth
wrapped around the upper body and pinned at the
shoulder"
    },
    },
    # =====
    # MODERN (from dict list)
    # =====
    {
        "name": "T-Shirt",
        "dress_type": "upperbody inner dress",
        "description": "short-sleeved pullover top
with a round crew neckline, falling to the hip"
    },
    {
        "name": "Dress Shirt",
        "dress_type": "upperbody inner dress",
        "description": "button-front collared shirt
with long sleeves, a pointed collar, and a
straight hem"
    },
    {
        "name": "Blouse",
        "dress_type": "upperbody inner dress",
        "description": "lightweight women's top with
varied necklines and sleeve styles, falling to
the hip"
    },
    {
        "name": "Crop Top",
        "dress_type": "upperbody inner dress",
        "description": "short top ending above the
waist or at the midriff with varied necklines and
sleeve lengths"
    },
    {
        "name": "Tank Top",
        "dress_type": "upperbody inner dress",
        "description": "sleeveless top with shoulder
straps and a round or scooped neckline, falling
to the hip"
    },
    {
        "name": "Hoodie",
        "dress_type": "upperbody flex dress",
        "description": "pullover or zip-up sweatshirt"
    },
    },
    {

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with an attached drawstring hood and a front
kangaroo pocket"
},
{
  "name": "Sweatshirt",
  "dress_type": "upperbody flex dress",
  "description": "long-sleeved pullover with a
crew neckline, ribbed cuffs and hem, no hood"
},
{
  "name": "Jacket",
  "dress_type": "upperbody flex dress",
  "description": "hip-length outerwear with long
sleeves, a front zip or button closure, and a
collar"
},
{
  "name": "Blazer",
  "dress_type": "upperbody flex dress",
  "description": "single or double-breasted
tailored jacket with notched lapels, hip length,
structured shoulders"
},
{
  "name": "Suit",
  "dress_type": "fullbody dress",
  "description": "matching tailored jacket and
trousers in the same fabric with lapels and a
button front"
},
{
  "name": "Tuxedo",
  "dress_type": "fullbody dress",
  "description": "formal matching jacket and
trousers with satin lapels and a satin side
stripe on the trouser leg"
},
{
  "name": "Jeans",
  "dress_type": "lowerbody dress",
  "description": "fitted or relaxed denim
trousers with belt loops, front zip, and patch
back pockets"
},
{
  "name": "Trousers",
  "dress_type": "lowerbody dress",
  "description": "full-length tailored pants
with a waistband, belt loops, and straight or
tapered legs"
},
{
  "name": "Chinos",
  "dress_type": "lowerbody dress",
  "description": "slim or straight-cut trousers
in cotton twill with a flat front and clean
pressed finish"
},
{
  "name": "Shorts",
  "dress_type": "lowerbody dress",
  "description": "short trousers with hemline
falling above the knee, waistband, and belt loops"
},
{
  "name": "Skirt",
  "dress_type": "lowerbody dress",
  "description": "women's waist-fitted skirt of
varied length and silhouette"
},
{
  "name": "Mini Skirt",
  "dress_type": "lowerbody dress",
  "description": "very short skirt with hemline
falling well above the knee at mid-thigh"
},
{
  "name": "Midi Skirt",

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  "dress_type": "lowerbody dress",
  "description": "skirt with hemline falling
between the knee and ankle at mid-calf length"
},
{
  "name": "Maxi Skirt",
  "dress_type": "lowerbody dress",
  "description": "full-length skirt falling to
the ankle or floor"
},
{
  "name": "Bodycon Dress",
  "dress_type": "fullbody dress",
  "description": "form-fitting stretchy dress
hugging every curve, typically knee-length or
shorter"
},
{
  "name": "A-Line Dress",
  "dress_type": "fullbody dress",
  "description": "fitted at chest and waist,
flaring gradually into an A-shape from hips to
hem"
},
{
  "name": "Evening Gown",
  "dress_type": "fullbody dress",
  "description": "full-length formal dress with
a fitted bodice and flared or column skirt, often
with a low back or neckline"
},
{
  "name": "Cocktail Dress",
  "dress_type": "fullbody dress",
  "description": "semi-formal knee-length dress
in a fitted or A-line silhouette"
},
{
  "name": "Jumpsuit",
  "dress_type": "fullbody dress",
  "description": "one-piece garment with a top
and full-length trousers joined at the waist"
},
{
  "name": "Playsuit",
  "dress_type": "fullbody dress",
  "description": "one-piece garment with a top
and short shorts joined at the waist"
},
{
  "name": "Co-ord Set",
  "dress_type": "fullbody dress",
  "description": "matching top and bottom set in
the same fabric, worn together as a coordinated
outfit"
},
{
  "name": "Athleisure Set",
  "dress_type": "fullbody dress",
  "description": "matching athletic top and
leggings or shorts in stretch fabric"
},
{
  "name": "Tracksuit",
  "dress_type": "fullbody dress",
  "description": "matching zip-up jacket and
elasticated trousers in fleece or nylon"
},
{
  "name": "Kurti",
  "dress_type": "upperbody inner dress",
  "description": "straight or A-line tunic
reaching hip to knee with a round or V neckline
and long or three-quarter sleeves"
},
{
  "name": "Modern Kurta",
  "dress_type": "upperbody inner dress",
  "description": "straight-cut tunic with side

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    slits reaching knee length, long or short sleeves,
    mandarin or round collar"
  },
  {
    "name": "Palazzo Pants",
    "dress_type": "lowerbody dress",
    "description": "very wide-leg full-length
    trousers flaring dramatically from the hip in
    flowing fabric"
  },
  {
    "name": "Leggings",
    "dress_type": "lowerbody dress",
    "description": "form-fitting full-length
    stretchy trousers with no waist fastening,
    hugging legs to the ankle"
  },
  {
    "name": "Power Suit",
    "dress_type": "fullbody dress",
    "description": "matching structured blazer and
    wide or straight-leg trousers with a strong
    shoulder line"
  },
  {
    "name": "Wrap Dress",
    "dress_type": "fullbody dress",
    "description": "front wraps across the body
    tied at the waist, creating a V neckline, falling
    to knee length"
  },
  {
    "name": "Slip Dress",
    "dress_type": "fullbody dress",
    "description": "thin-strapped slinky dress
    with a bias-cut or straight silhouette, falling
    to knee or ankle"
  },
  {
    "name": "Denim Jacket",
    "dress_type": "upperbody flex dress",
    "description": "hip-length rigid denim jacket
    with button front, chest and hip pockets, and
    long sleeves"
  },
  {
    "name": "Bomber Jacket",
    "dress_type": "upperbody flex dress",
    "description": "hip-length jacket with ribbed
    collar, cuffs, and hem, a zip front, and long
    sleeves"
  },
  {
    "name": "Cargo Pants",
    "dress_type": "lowerbody dress",
    "description": "straight or relaxed-leg
    trousers with large bellowed cargo pockets on the
    thighs"
  },
  {
    "name": "Streetwear Set",
    "dress_type": "fullbody dress",
    "description": "matching oversized top and
    trousers or shorts for urban casual wear"
  },
  {
    "name": "Oversized Tee",
    "dress_type": "upperbody inner dress",
    "description": "very loose short-sleeved crew-
    neck top falling to the hip or thigh"
  },
  {
    "name": "Polo Shirt",
    "dress_type": "upperbody inner dress",
    "description": "short-sleeved collared shirt
    with a partial button placket at the front
    neckline, falling to the hip"
  },
  {

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    "name": "Peplum Top",
    "dress_type": "upperbody inner dress",
    "description": "fitted top with a short
    gathered or flared frill attached at the waist,
    extending over the hips"
  },
  {
    "name": "Maxi Dress",
    "dress_type": "fullbody dress",
    "description": "full-length women's dress
    falling to the ankle with varied sleeve and
    neckline styles"
  },
  {
    "name": "Sheath Dress",
    "dress_type": "fullbody dress",
    "description": "fitted straight knee-length
    dress following the body's silhouette closely
    with minimal flare"
  },
  {
    "name": "Cape Coat",
    "dress_type": "upperbody flex dress",
    "description": "knee-length outer coat with
    cape-style wide sleeves or an attached arm-slot
    cape draping over arms"
  },
  {
    "name": "Trench Coat",
    "dress_type": "upperbody flex dress",
    "description": "knee-length double-breasted
    outer coat with wide lapels, a storm flap, belted
    waist, and long sleeves"
  },
  {
    "name": "Sweater Dress",
    "dress_type": "fullbody dress",
    "description": "knitted dress falling to knee
    or midi length with long sleeves and a crew or
    turtleneck collar"
  },
  {
    "name": "Utility Jumpsuit",
    "dress_type": "fullbody dress",
    "description": "one-piece garment with long
    sleeves, multiple cargo pockets, and full-length
    trousers"
  },
  # =====
  # EUROPE
  # =====
  {
    "name": "Dirndl",
    "dress_type": "fullbody dress",
    "description": "tight low-cut bodice with a
    full gathered skirt below the knee, puff-sleeved
    blouse underneath, decorative apron at front"
  },
  {
    "name": "Lederhosen",
    "dress_type": "lowerbody dress",
    "description": "leather knee-length breeches
    with H-shaped suspenders crossing over the chest,
    with decorative panels on bib and thighs"
  },
  {
    "name": "Kilt",
    "dress_type": "lowerbody dress",
    "description": "knee-length pleated wrap-
    around skirt secured at front with a flat apron
    panel fastened with a large pin"
  },
  {
    "name": "Bunad (women's)",
    "dress_type": "fullbody dress",

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        "description": "long woolen skirt with a
matching fitted bodice with silver clasps, worn
with a white blouse underneath"
    },
    {
        "name": "Bunad (men's)",
        "dress_type": "fullbody dress",
        "description": "dark short jacket with silver
buttons, knee-length breeches, a white shirt, and
knee-high white stockings"
    },
    {
        "name": "Sarafan",
        "dress_type": "fullbody dress",
        "description": "sleeveless A-line pinafore
dress, floor-length, worn over a long-sleeved
blouse"
    },
    {
        "name": "Vyshyvanka (women's)",
        "dress_type": "upperbody inner dress",
        "description": "loose long-sleeved linen or
cotton blouse with a wide round collar and wide
sleeve hems"
    },
    {
        "name": "Vyshyvanka (men's)",
        "dress_type": "upperbody inner dress",
        "description": "loose linen shirt with long
sleeves, a round collar, and a front placket,
worn tucked or belted"
    },
    {
        "name": "Flamenco Dress",
        "dress_type": "fullbody dress",
        "description": "form-fitting bodice and hips
flaring dramatically from the knee into multiple
layered ruffled tiers at the hem"
    },
    {
        "name": "Gákti (women's)",
        "dress_type": "fullbody dress",
        "description": "wool dress-length tunic
falling to mid-calf with ribbon band trim at hem,
collar, and cuffs, worn with a distinctive cap"
    },
    {
        "name": "Gákti (men's)",
        "dress_type": "upperbody inner dress",
        "description": "thigh-length tunic with ribbon
borders at collar, hem, and cuffs, worn belted
at the waist"
    },
    {
        "name": "Amalia Dress",
        "dress_type": "fullbody dress",
        "description": "lace-trimmed blouse worn under
a velvet jacket, paired with a pleated ankle-
length skirt and a small tasseled cap"
    },
    {
        "name": "Kosovorotka",
        "dress_type": "upperbody inner dress",
        "description": "loose linen shirt with an
asymmetric side-buttoning collar opening to the
left, worn belted at the waist"
    },
    {
        "name": "Breton Stripe Shirt",
        "dress_type": "upperbody inner dress",
        "description": "fitted pullover with a wide
round Breton neckline and three-quarter or full-
length sleeves"
    },
    {
        "name": "Aran Sweater",
        "dress_type": "upperbody flex dress",
        "description": "thick cable-knit wool pullover
with dense cable, honeycomb, and diamond

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stitches and a round crew neckline"
    },
    {
        "name": "Fair Isle Sweater",
        "dress_type": "upperbody flex dress",
        "description": "fitted knitted pullover with
repeating geometric multi-motif band patterns in
traditional Shetland knitting style"
    },
    {
        "name": "Romanian Ie",
        "dress_type": "upperbody inner dress",
        "description": "linen blouse with a wide round
or square neckline and wide loose sleeves"
    },
    {
        "name": "Polish Krakowiak Dress",
        "dress_type": "fullbody dress",
        "description": "full ankle-length skirt with a
fitted velvet bodice and ribbon trim, worn with
an embroidered blouse underneath"
    },
    {
        "name": "Scottish Tartan Dress",
        "dress_type": "fullbody dress",
        "description": "tartan wool pinafore worn over
a blouse, or as a fitted bodice with a full
pleated midi skirt"
    },
    {
        "name": "Hungarian Matyó Embroidered Dress",
        "dress_type": "fullbody dress",
        "description": "cotton dress with a fitted
bodice and very full skirt densely covered in
raised three-dimensional surface decorations"
    },
    # =====
    # AFRICA (detailed)
    # =====
    {
        "name": "Habesha Kemis",
        "dress_type": "fullbody dress",
        "description": "ankle-length handwoven cotton
dress with a wide decorated band along the
neckline, sleeve cuffs, and hem"
    },
    {
        "name": "Netela (women's)",
        "dress_type": "shawl",
        "description": "lightweight cotton shawl
draped over the head and shoulders with a woven
border at one end"
    },
    {
        "name": "Netela (men's)",
        "dress_type": "shawl",
        "description": "lightweight cotton shawl
draped over the shoulders and wrapped around the
body with a woven border"
    },
    {
        "name": "Kanga",
        "dress_type": "fullbody dress",
        "description": "rectangular cotton cloth worn
wrapped at waist as skirt, draped over shoulder,
or tied over head"
    },
    {
        "name": "Kitenge Dress",
        "dress_type": "fullbody dress",
        "description": "tailored dress in wax-print
fabric styled as fitted peplum with pencil skirt,
wrap dress, or full-length maxi"
    },
    {

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    "name": "Gomesi",
    "dress_type": "fullbody dress",
    "description": "long dress with a square
neckline, short puffed sleeves, and a straight
skirt to the ankles, cinched with a wide sash"
},
{
    "name": "Kaba and Slit",
    "dress_type": "fullbody dress",
    "description": "two-piece: fitted blouse with
a unique tailored neckline, paired with a
matching long wrap skirt"
},
{
    "name": "Gandoura",
    "dress_type": "fullbody dress",
    "description": "loose collarless tunic with
wide or no sleeves, falling to the knee or ankle"
},
{
    "name": "Aso-Oke Wrapper (women's)",
    "dress_type": "fullbody dress",
    "description": "three-piece woven ceremonial
set: sculptural head wrap, a shoulder cloth, and
a waist wrapper"
},
{
    "name": "Moroccan Caftan",
    "dress_type": "fullbody dress",
    "description": "long formal robe with a deep V
or round collar and long sleeves, in silk or
brocade"
},
{
    "name": "Egyptian Galabiya",
    "dress_type": "fullbody dress",
    "description": "long loose robe with long
sleeves and a round or V neckline, falling to the
ankles"
},
{
    "name": "Ndebele Ceremonial Dress",
    "dress_type": "fullbody dress",
    "description": "stiff beaded rectangular front
apron worn with stacked neck and leg rings and a
beaded blanket draped over the back"
},
{
    "name": "Rwandan Mushanana",
    "dress_type": "fullbody dress",
    "description": "long fitted dress with a
fabric sash draped diagonally from one shoulder
across the body"
},
{
    "name": "Ghanaian Fugu Smock",
    "dress_type": "upperbody inner dress",
    "description": "handwoven pullover smock with
a round neckline, wide short sleeves to mid-
forearm, straight hem at hip or thigh"
},
},
# =====
# THE AMERICAS (detailed)
# =====
{
    "name": "Traje Maya (women's)",
    "dress_type": "fullbody dress",
    "description": "hand-woven huipil blouse
paired with a long hand-loomed wrap skirt secured
with a narrow woven belt"
},
{
    "name": "Ribbon Skirt",
    "dress_type": "lowerbody dress",
    "description": "wrap skirt falling waist to

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ankle with vertical ribbon strips sewn along the
full length"
},
{
    "name": "Manta (Andean shawl)",
    "dress_type": "shawl",
    "description": "rectangular woven cloth draped
over shoulders and back, fastened at the chest
with a pin"
},
{
    "name": "Tipoy",
    "dress_type": "fullbody dress",
    "description": "sleeveless one-piece dress
hanging loosely from the shoulders to the calves"
},
{
    "name": "Mexican Charro Suit",
    "dress_type": "fullbody dress",
    "description": "fitted jacket and high-waisted
flared trousers with silver button trim, worn
with a wide-brimmed hat"
},
{
    "name": "Tehuana Dress",
    "dress_type": "fullbody dress",
    "description": "wide squared-neckline blouse
with a full-length matching skirt; Isthmus of
Tehuantepec traditional dress"
},
{
    "name": "Cholita Dress",
    "dress_type": "fullbody dress",
    "description": "multiple layered petticoats
under a wide A-line skirt, fitted blouse, fringed
shoulder shawl, and a bowler hat"
},
{
    "name": "Argentine Gaucho Outfit",
    "dress_type": "fullbody dress",
    "description": "wide-leg pleated bombachas
trousers tucked into tall boots, loose shirt,
wide studded belt, and shoulder poncho"
},
# =====
# OCEANIA
# =====
{
    "name": "Lavalava (women's)",
    "dress_type": "lowerbody dress",
    "description": "rectangular cloth knotted at
waist, worn as a knee to ankle-length wrap skirt"
},
{
    "name": "Lavalava (men's)",
    "dress_type": "lowerbody dress",
    "description": "rectangular cloth knotted at
waist, worn as a knee-length wrap skirt"
},
{
    "name": "Puletasi",
    "dress_type": "fullbody dress",
    "description": "loose collarless blouse and
matching long ankle-length wrap skirt in the same
fabric"
},
{
    "name": "Hawaiian Aloha Shirt",
    "dress_type": "upperbody inner dress",
    "description": "loose short-sleeved shirt with
a notched or camp collar and a button front"
},
{
    "name": "Hawaiian Muumuu",
    "dress_type": "fullbody dress",

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        "description": "loose flowing dress from
        shoulder to ankle with a straight or slightly
        gathered silhouette"
    },
    {
        "name": "Tongan Ta'ovala",
        "dress_type": "lowerbody dress",
        "description": "pandanus-woven mat wrapped
        around the waist over clothing and tied at the
        front"
    },
    # =====
    # MODERN (detailed garments -- female)
    # =====
    {
        "name": "Pencil Skirt",
        "dress_type": "lowerbody dress",
        "description": "form-fitting straight skirt
        hugging hips and thighs, tapering to a narrow hem
        at or below the knee"
    },
    {
        "name": "Sundress",
        "dress_type": "fullbody dress",
        "description": "lightweight dress with thin
        straps or strapless bodice and a loose flowing
        skirt to the knee or mid-thigh"
    },
    {
        "name": "Shirt Dress",
        "dress_type": "fullbody dress",
        "description": "dress styled as an oversized
        button-down shirt with a collared neckline and
        front button placket, knee to midi length"
    },
    {
        "name": "Mini Dress",
        "dress_type": "fullbody dress",
        "description": "short dress with hemline at
        mid-thigh in bodycon, A-line, or wrap silhouette"
    },
    {
        "name": "Midi Dress",
        "dress_type": "fullbody dress",
        "description": "dress with hemline between
        knee and ankle at mid-calf in wrap, tiered, or
        fitted silhouette"
    },
    {
        "name": "Off-Shoulder Top",
        "dress_type": "upperbody inner dress",
        "description": "top with an elastic neckline
        sitting below both shoulders, leaving them bare,
        with a straight or ruffled edge"
    },
    {
        "name": "Wide-Leg Pants",
        "dress_type": "lowerbody dress",
        "description": "high or mid-rise trousers with
        legs flaring dramatically from the hip to a wide
        ankle opening"
    },
    {
        "name": "Cardigan",
        "dress_type": "upperbody flex dress",
        "description": "open-front knitted sweater
        with long sleeves and a button or open center-
        front fastening"
    },
    {
        "name": "Women's Overalls",
        "dress_type": "fullbody dress",
        "description": "denim or cotton dungarees with
        a chest bib, adjustable shoulder straps, and
        multiple pockets, worn over a top"
    }

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    },
    {
        "name": "Little Black Dress",
        "dress_type": "fullbody dress",
        "description": "simple fitted or semi-fitted
        short to knee-length dress with a round or V
        neckline, short or no sleeves"
    },
    {
        "name": "Yoga Pants",
        "dress_type": "lowerbody dress",
        "description": "form-fitting full-length
        athletic leggings with a wide high-rise waistband"
    },
    {
        "name": "Sports Bra",
        "dress_type": "upperbody inner dress",
        "description": "supportive cropped athletic
        top ending just below the bust with wide straps
        and a wide elastic underbust band"
    },
    {
        "name": "Denim Shorts",
        "dress_type": "lowerbody dress",
        "description": "casual shorts cut from denim
        with frayed or hemmed edges sitting at the upper
        thigh"
    },
    {
        "name": "Knit Pullover",
        "dress_type": "upperbody flex dress",
        "description": "medium-weight crew-neck
        pullover with long sleeves in ribbed or smooth
        knit"
    },
    {
        "name": "Lace Dress",
        "dress_type": "fullbody dress",
        "description": "dress made from lace over a
        lining in a fitted or A-line silhouette with
        patterning across bodice and skirt"
    },
    {
        "name": "Floral Blouse",
        "dress_type": "upperbody inner dress",
        "description": "lightweight top with button
        front or pullover design and long or short
        sleeves, relaxed or fitted cut"
    },
    {
        "name": "Wrap Skirt",
        "dress_type": "lowerbody dress",
        "description": "skirt formed by wrapping
        fabric around the waist with a V-shaped front
        opening tied at the hip, knee or midi length"
    },
    {
        "name": "Romper",
        "dress_type": "fullbody dress",
        "description": "one-piece combining a top with
        attached very short shorts, with thin straps,
        strapless, or short-sleeved bodice"
    },
    {
        "name": "Mom Jeans",
        "dress_type": "lowerbody dress",
        "description": "high-waisted relaxed-fit denim
        jeans with a wide waistband above the natural
        waist and straight or tapered leg"
    },
    {
        "name": "Women's Puffer Vest",
        "dress_type": "upperbody flex dress",
        "description": "sleeveless quilted vest filled
        with insulation divided into horizontal quilted
        channels"
    },
    {

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    "name": "Spaghetti Strap Dress",
    "dress_type": "fullbody dress",
    "description": "dress supported by two very
thin shoulder straps with a loose or fitted
bodice and flowing skirt to knee or ankle"
},
{
    "name": "Ruffle Top",
    "dress_type": "upperbody inner dress",
    "description": "blouse with layered or
cascading ruffle trim along the neckline, sleeve
edges, or front placket"
},
{
    "name": "Hoodie Dress",
    "dress_type": "fullbody dress",
    "description": "long dress styled as an
oversized hoodie with a drawstring hood and
kangaroo pocket, falling to mid-thigh or knee"
},
{
    "name": "One-Piece Swimsuit",
    "dress_type": "fullbody dress",
    "description": "single-piece swimwear covering
torso from shoulders to hips with a scooped
neckline and leg openings at the hips"
},
{
    "name": "Bikini",
    "dress_type": "fullbody dress",
    "description": "two-piece swimwear: small bra-
style top and matching brief-style bottom tied or
fastened at the sides"
},
{
    "name": "Bodysuit",
    "dress_type": "upperbody inner dress",
    "description": "form-fitting one-piece top
covering torso from shoulders to crotch,
fastening with snaps at the bottom"
},
{
    "name": "Women's Sweatpants",
    "dress_type": "lowerbody dress",
    "description": "relaxed-fit trousers with an
elastic drawstring waistband and tapered or
straight legs in fleece fabric"
},
{
    "name": "Bralette",
    "dress_type": "upperbody inner dress",
    "description": "soft unstructured top in
triangle or bandeau shape with thin adjustable
straps, no underwire"
},
# =====
# MODERN (detailed garments -- male)
# =====
{
    "name": "Oxford Shirt",
    "dress_type": "upperbody inner dress",
    "description": "button-down collar shirt in
thick textured weave with a chest patch pocket
and a straight hem"
},
{
    "name": "Flannel Shirt",
    "dress_type": "upperbody inner dress",
    "description": "casual soft-woven shirt with a
relaxed fit, two chest pockets, and a straight
hem"
},
{
    "name": "Linen Shirt",
    "dress_type": "upperbody inner dress",

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    "description": "lightweight summer shirt with
a relaxed fit, a camp or button-down collar, and
a straight untucked hem"
},
{
    "name": "Formal Waistcoat",
    "dress_type": "upperbody flex dress",
    "description": "sleeveless V-neck button-front
vest in suiting fabric with welt pockets and an
adjustable back strap"
},
{
    "name": "Men's Puffer Jacket",
    "dress_type": "upperbody flex dress",
    "description": "insulated outer jacket with
horizontal quilted channels, a zip front, and
long sleeves"
},
{
    "name": "Leather Jacket",
    "dress_type": "upperbody flex dress",
    "description": "hip-length fitted jacket with
an asymmetric front zip, a snap-tab collar, and
zippered chest pockets"
},
{
    "name": "Turtleneck Sweater",
    "dress_type": "upperbody flex dress",
    "description": "close-fitting knitted pullover
with a high cylindrical collar that folds over
to cover the neck"
},
{
    "name": "Jogger Pants",
    "dress_type": "lowerbody dress",
    "description": "tapered trousers with an
elastic drawstring waistband and ribbed
elasticated cuffs at the ankle"
},
{
    "name": "Peacoat",
    "dress_type": "upperbody flex dress",
    "description": "double-breasted hip-length
wool overcoat with wide notched lapels, two rows
of large buttons, and vertical flap pockets"
},
{
    "name": "Quarter-Zip Pullover",
    "dress_type": "upperbody flex dress",
    "description": "pullover with a short zip
running from chest to collar, long sleeves, and a
straight hem"
},
{
    "name": "Windbreaker",
    "dress_type": "upperbody flex dress",
    "description": "lightweight water-resistant
jacket with a zip front, elasticated cuffs and
hem, and a packable hood"
},
{
    "name": "V-Neck T-Shirt",
    "dress_type": "upperbody inner dress",
    "description": "short-sleeved cotton top with
a V-shaped neckline opening in a fitted or
relaxed cut"
},
{
    "name": "Swim Trunks",
    "dress_type": "lowerbody dress",
    "description": "short swimming shorts with an
elastic drawstring waistband and inner mesh
lining, falling to mid-thigh"
},
{
    "name": "Sport Coat",
    "dress_type": "upperbody flex dress",
    "description": "single-breasted tailored
jacket in tweed or herringbone with notched

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    lapels and a relaxed fit"
  },
  {
    "name": "Crew Neck Sweatshirt",
    "dress_type": "upperbody flex dress",
    "description": "pullover with a round crew
neckline, long sleeves, and ribbed hem and cuffs,
no hood"
  },
  {
    "name": "Muscle Tee",
    "dress_type": "upperbody inner dress",
    "description": "sleeveless top with very wide
deep armholes cut into the sides of the torso"
  },
  {
    "name": "Chino Shorts",
    "dress_type": "lowerbody dress",
    "description": "casual knee-length shorts in
cotton twill with a flat or pleated waistband and
a clean hemmed edge"
  },
  {
    "name": "Thermal Long-Sleeve Shirt",
    "dress_type": "upperbody inner dress",
    "description": "form-fitting base-layer top in
waffle-knit or ribbed fabric with long sleeves
and a crew or henley neckline"
  },
  {
    "name": "Men's Denim Overalls",
    "dress_type": "fullbody dress",
    "description": "denim dungarees with a square
chest bib, adjustable shoulder straps, and loose
straight-leg trousers"
  },
  {
    "name": "Basketball Shorts",
    "dress_type": "lowerbody dress",
    "description": "loose-fitting athletic shorts
in mesh fabric falling to just above or below the
knee with side vents"
  },
  {
    "name": "Dress Pants",
    "dress_type": "lowerbody dress",
    "description": "formal tailored trousers with
a flat front or single pleat and a straight or
slightly tapered leg"
  },
  {
    "name": "Men's Sweatpants",
    "dress_type": "lowerbody dress",
    "description": "relaxed-fit trousers with an
elastic drawstring waistband and straight or
tapered legs with ribbed or open cuff"
  },
  {
    "name": "Varsity Jacket",
    "dress_type": "upperbody flex dress",
    "description": "hip-length jacket with a wool
body, contrasting leather sleeves, ribbed collar,
cuffs, and hem, snap-button front"
  },
  {
    "name": "Raincoat",
    "dress_type": "upperbody flex dress",
    "description": "knee-length waterproof outer
coat with a hood, snap or zip front, sealed seams,
and large patch pockets"
  },
]

```



## I.2. Difficulty-Tier Editing Dictionaries

The complete difficulty dictionaries are provided in Listing 5. The editing space is organized into Easy, Medium, and Hard tiers, corresponding to progressively more challenging combinations of pose variation, occlusion, viewpoint changes, background complexity, and related visual conditions.

The Medium dictionary is shared across male and female source persons for its principal editing dimensions. Gender conditioning is restricted to the special-case sampling branch: the female branch additionally contains a visible-pregnancy condition, while the corresponding probabilities of the remaining special cases are normalized separately for the two branches.

Listing 5. Complete Easy, Medium, and Hard difficulty-tier editing dictionaries. The Medium tier contains gender-conditioned special-case sampling within a single dictionary definition.

```
# Unified CURVTON-205K difficulty dictionaries
#
# The difficulty configuration has exactly three top-
# level tiers: easy,
# medium, and hard. The medium tier uses one shared
# dictionary. Its only
# gender-conditioned component is the Special cases
# distribution.
#
# Source merge:
# easy <- easy_dict_new.py
# medium <- final_medium_female_dict.py +
#   final_medium_male_dict.py
#   (identical except for Special cases)
# hard <- final_hard_dict.py

DIFFICULTY_DICTIONARIES = {
  "easy": {
    "Body Shape": {
      "prob": 0.2,
      "items": {
        "very slim": 0.080000,
        "slim build": 0.170000,
        "athletic build": 0.220000,
        "average build": 0.260000,
        "curvy": 0.170000,
        "plus size": 0.100000
      }
    },
    "Background/lighting": {
      "prob": 0.7,
      "items": {
        "plain white background": 0.25,
        "solid color background": 0.1,
        "studio background": 0.15,
        "even lighting": 0.1,
        "soft natural lighting": 0.25,
        "well-lit": 0.15
      }
    },
    "Special cases": {
      "prob": 0.10,
      "items": {
        "small visible tattoo": 0.4,
        "visible tattoo": 0.3,
        "hijab": 0.15,
        "turban": 0.15
      }
    }
  },
  "medium": {
```

```
    "_sampling_config": {
      "pose_only": 0.37,
      "occlusion_only": 0.30,
      "pose_and_occlusion": 0.33,
    },
    "Body Pose": {
      "prob": 1,
      "items": [
        # --- Standing neutral / relaxed ---
        "standing straight",
        "standing casually",
        "arms at sides",
        "weight shifted to one leg",
        "arms crossed",
        "one hand on hip",
        "both hands on hips",
        "slight twist at waist",
        "looking over shoulder",
        "standing with feet shoulder-width
        apart",
        "standing with one foot forward",
        "standing with ankles crossed",
        "standing with hands clasped in front",
        "standing with hands behind back",
        "standing with thumbs in belt loops",
        "standing relaxed one knee slightly
        bent",
        "standing with arms loosely behind
        back",
        # --- Walking / moving ---
        "walking forward",
        "mid-stride",
        "stepping to the side",
        "turning to look back",
        "walking upstairs",
        "walking downstairs",
        "strolling with one hand swinging",
        "taking a step forward with arms
        relaxed",
        "simple contrapposto pose",
        "pivoting on one foot",
        # --- Arm and hand gestures ---
        "pointing forward",
        "waving hand",
        "arms raised in celebration",
        "thinking pose hand on chin",
        "listening pose hand to ear",
        "face palm",
        "one arm raised",
        "stretching arms overhead",
        "reaching forward",
        "arms spread wide",
        "both arms reaching upward slightly",
        "one hand extended palm out stop
        gesture",
        "arms loosely folded at stomach",
        "resting chin on hands",
        "snapping fingers",
        "holding out open hand palm up",
        "fingers in gun pose",
        "army salutation pose",
        "hands clasped above head",
        "one hand on chest",
        # --- Sitting poses ---
        "sitting on chair",
        "sitting cross-legged",
        "perched on stool",
        "sitting with legs stretched forward",
        "sitting with both hands on knees",
        "sitting upright arms on armrests",
        "sitting forward leaning elbows on
        knees",
        "sitting with one leg crossed over the
        other",
        "sitting on edge of chair",
        "sitting with hands in lap",
        "sitting on bench looking forward",
```

```

back",      "sitting with one arm resting on chair
            # --- Leaning poses ---
            "leaning against wall",
            "leaning on railing",
            "leaning forward with hands on knees",
            "leaning against wall arms crossed",
            "leaning on counter with both forearms
",
            "leaning on wall one shoulder touching
",
            "leaning on table with one elbow",
            # --- Floor and low poses ---
            "kneeling on one knee",
            "casual lunge",
            "squatting",
            "lying on side",
            "kneeling upright on both knees",
            "sitting on floor with knees bent feet
flat",      "sitting on floor with legs to one
side",      "crouching slightly with hands on
thighs",    # --- Active / expressive ---
            "jumping",
            "running",
            "bending slightly forward",
            "yoga tree pose",
            "yoga warrior pose",
            "namaste prayer pose",
            "dancing freestyle",
            "looking down at phone",
            "carrying object with one hand",
            "stretching one arm across chest",
            "one hand gripping opposite forearm",
            "tying shoelace kneeling",
            "reaching into bag at side",
            "holding both hands out in shrug
gesture",   "arms bent at elbow hands at waist",
            "light jog arms at sides",
            "stepping up onto curb",
            "standing with one hand raised
touching hair", "sitting on steps with elbows on knees
",
            "standing with arms gently swinging
forward",   "hands on lower back standing upright",
            "standing on tiptoes slightly reaching
up",        "crossing arms low at stomach relaxed",
            "standing profile head turned toward
camera",    "casual stance weight on back foot
arms loose",
            ],
        },
        "Occlusion": {
            "prob": 1,
            "items": [
                # --- No occlusion / minimal crop ---
                "full body visible",
                "standing behind low garden wall
blocking lower legs",
                "standing behind short reception
counter",
                "head partially behind door frame or
window edge",
                # --- Hand-held objects (small) ---
                "holding phone",
                "holding coffee cup",
                "holding water bottle",
                "holding book",
                "holding umbrella",

```

```

            "holding tablet in front",
            "holding laptop in front",
            "holding camera/DSLR in front of face",
            "holding shopping receipt in front of
torso",
            "holding keys in hand in front of
torso",
            "holding takeaway food bag in front of
torso",
            "holding reusable cup with lid in
front of torso",
            "holding folded newspaper in front of
torso",
            "holding small bouquet of flowers in
front of torso",
            "holding envelope in one hand in front
of torso",
            "holding pen or pencil in front of
torso",
            "holding sunglasses in one hand in
front of torso",
            "holding gym towel in hand in front of
torso",
            "holding hat in hand at side in front
of torso",
            "holding rolled yoga mat under arm in
front of torso",
            "holding reusable grocery bag in one
hand in front of torso",
            "holding smartphone charger cable in
front of torso",
            "holding earbuds case in front of
torso",
            "holding folded map in front of torso",
            "holding small notebook in front of
torso",
            "holding compact mirror in front of
torso",
            "holding lip balm or chapstick in
front of torso",
            "holding boarding pass in front of
torso",
            "holding business card in front of
torso",
            "holding airport gate pass in front of
torso",
            "holding small package or parcel in
front of torso",
            "holding wrapped gift (small) in front
of torso",
            "holding ice cream cone in front of
torso",
            "holding sandwich in wrapper in front
of torso",
            "holding pastry or donut in front of
torso",
            "holding single apple or fruit in
front of torso",
            "holding protein bar in front of torso
",
            "holding small juice box in front of
torso",
            "holding can of soda in front of torso
",
            "holding travel mug in front of torso",
            "holding dog leash in front of torso",
            "holding jump rope in front of torso",
            "holding resistance band in front of
torso",
            "holding tennis ball in front of torso
",
            "holding frisbee in front of torso",
            "holding small plant pot in front of
torso",
            "holding gardening gloves in front of
torso",

```

|                        |                                         |                          |                                         |
|------------------------|-----------------------------------------|--------------------------|-----------------------------------------|
|                        | "holding hand trowel in front of torso" |                          | "wearing ankle bracelet visible",       |
| "",                    | "holding spray bottle in front of       |                          | "wearing wrist sweatband",              |
| torso",                | "holding TV remote in front of torso",  | ---                      | "wearing headphones around neck",       |
|                        | "holding gaming controller in front of  |                          | # --- Hair / face partial occlusion     |
| torso",                | # --- Bags and carries ---              |                          | "hair covering part of face",           |
| torso",                | "holding small handbag in front of      |                          | "earphones with wires crossing torso",  |
|                        | "carrying shopping bags",               | jaw",                    | "hand raised near face mid-gesture",    |
|                        | "carrying backpack",                    |                          | "fingers partially covering chin",      |
| blocking lower torso", | "wearing crossbody bag",                |                          | "neckline folded up partially covering  |
| leg",                  | "stroller or baby carriage in front     |                          | "hair falling over one eye",            |
|                        | "cane or walking aid in front of lower  |                          | "bangs covering forehead",              |
|                        | "tote bag hanging at elbow",            |                          | "hair tied back with loose strands",    |
|                        | "single-strap bag over one shoulder",   |                          | "hand touching hair",                   |
|                        | "duffel bag held at side",              |                          | "hand covering mouth while speaking",   |
|                        | "laptop bag strap across chest",        |                          | "hand on cheek",                        |
|                        | "small clutch purse held in both hands  |                          | "hand adjusting glasses",               |
| "",                    | "briefcase held at side",               | obstructions ---         | "hand near ear on phone",               |
|                        | "gym bag on one shoulder",              |                          | # --- Environmental / foreground        |
| front",                | "carrying flat box in both hands in     | body",                   | "foreground plant partially blocking    |
|                        | "bicycle helmet hanging from one hand", | blocking body",          | "foreground pole or post partially      |
|                        | "messenger bag worn across body",       |                          | "standing behind railing",              |
|                        | "fanny pack worn at waist",             | reflection)",            | "standing behind glass window (minor    |
|                        | "camera bag slung over shoulder",       | blocking lower body",    | "small signboard or poster partially    |
|                        | "small rolling carry-on at side",       |                          | "hands partially hidden by an object (  |
|                        | "drawstring bag on back",               | e.g., bag, cup)",        | "neck cord card hanging over chest      |
| one hand",             | "paper shopping bag with handles in     | area",                   | "standing partially behind open door",  |
|                        | "reusable produce bag in hand",         | legs",                   | "low fence partially blocking lower     |
|                        | "small cooler bag held at side",        | blocking lower body",    | "sitting behind table with table        |
|                        | "insulated lunch bag in hand",          |                          | "standing behind half-height wall",     |
|                        | "diaper bag over shoulder",             | lower torso",            | "standing at shop counter blocking      |
|                        | "picnic basket held in one hand",       |                          | "seated at desk with desk blocking lap  |
|                        | "beach bag on shoulder",                | "",                      | "standing next to bicycle blocking one  |
|                        | "small suitcase held at side",          | leg",                    | "car door open partially blocking       |
|                        | "cosmetics bag in hand",                | lower body",             | "wheeled travel bag in front partially  |
|                        | "tool bag held at side",                | blocking legs",          | "standing in doorway frame edges        |
|                        | # --- Worn accessories ---              | clipping shoulders",     | "outdoor bench partially blocking       |
|                        | "wearing hat",                          | lower legs when seated", | "low shrub hedge partially blocking     |
|                        | "wearing sunglasses",                   | feet and ankles",        | "standing behind market stall counter", |
|                        | "wearing wristwatch",                   |                          | "window sill partially blocking lower   |
|                        | "mask covering lower face",             |                          | body from outside view",                |
| neck",                 | "loose fabric wrapped loosely around    |                          | "standing beside parked scooter         |
|                        | "wearing beanie hat",                   | blocking one leg",       | "seated at cafe table top of table      |
|                        | "wearing baseball cap",                 | blocking waist",         | "standing in shallow water obscuring    |
|                        | "wearing wide-brim hat",                | feet and ankles",        | "holding map or pamphlet partially      |
|                        | "wearing headband",                     | blocking abdomen",       | "handheld fan partially obscuring       |
|                        | "wearing earmuffs",                     | lower face",             | "carrying rolled poster tube under arm  |
|                        | "neck cord with card hanging at chest", | across torso",           | "side bag with wide strap crossing      |
|                        |                                         |                          | chest diagonally",                      |
|                        |                                         |                          | "layered chains hanging over neckline   |
|                        |                                         |                          | area",                                  |
|                        |                                         |                          | "standing behind shopping cart          |
|                        |                                         |                          | blocking lower body",                   |
|                        |                                         |                          | "standing at podium blocking lower      |

```

half of body",
  "partial shadow from overhead
structure on lower body",
  "standing behind park bench backrest",
  "standing at ATM machine blocking
lower torso",
  "standing at vending machine partially
blocking side",
  "standing behind ticket kiosk",
  "standing at gas pump partially
blocking leg",
  "standing behind baby gate blocking
lower legs",
  "standing at bar counter blocking
lower body",
  "seated on bar stool with footrest
visible",
  "standing behind luggage rack blocking
lower body",
  "standing behind display stand
partially blocking torso",
  "foreground street sign pole partially
blocking shoulder",
  "foreground hanging plant partially
blocking face",
  "foreground balloon partially blocking
upper body",
  "foreground umbrella edge clipping
frame",
  "standing behind parking meter
blocking leg",
  "standing at information desk blocking
lower half",
  "leaning against wall with shoulder
partially obscured",
  "standing behind newsstand counter",
  "standing at checkout counter blocking
waist down",
  ]
},
"Background/lighting/shadow": {
  "prob": 0.25,
  "items": {
    # sum = 1.0 (unchanged)
    "indoor setting": 0.119243,
    "room interior": 0.095632,
    "outdoor setting": 0.119243,
    "street scene": 0.095632,
    "window light": 0.143598,
    "front lighting": 0.095632,
    "side lighting": 0.095632,
    "natural setting": 0.094310,
    "overcast light": 0.047026,
    "indirect lighting": 0.023513,
    "blurred background": 0.047026,
    "bokeh background": 0.023513,
  }
},
"Camera view": {
  "prob": 0.47,
  "items": {
    # sum = 1.0 (unchanged)
    "front view": 0.284091,
    "front-facing": 0.227273,
    "facing camera": 0.170455,
    "side view": 0.113636,
    "profile view": 0.113636,
    "three-quarter view": 0.090909,
  }
},
"Age Group": {
  "prob": 0.10,
  "items": {
    # sum = 1.0 (unchanged)
    "10s": 0.12,
    "20s": 0.24,

```

```

    "30s": 0.26,
    "40s": 0.18,
    "50s": 0.11,
    "60s": 0.06,
    "70s": 0.03,
  }
},
"Special cases": {
  "prob": 0.10,
  # The medium tier is otherwise shared
  # across genders. Only the
  # special-case distribution differs,
  # because visible pregnancy is
  # included only in the female branch. Each
  # branch is normalized.
  "items_by_gender": {
    "female": {
      "visible birthmark":
        0.116279,
      "wearing hearing aid":
        0.116279,
      "visible scar on arm or hand":
        0.116279,
      "wearing neck brace":
        0.105708,
      "arm in sling":
        0.105708,
      "wearing knee brace visible":
        0.100423,
      "using service dog on leash":
        0.095137,
      "visible pregnancy":
        0.090909,
      "visible tattoos on arms":
        0.089852,
      "facial piercing (nose stud or lip
ring)": 0.063425,
    },
    "male": {
      "visible birthmark":
        0.127907,
      "wearing hearing aid":
        0.127907,
      "visible scar on arm or hand":
        0.127907,
      "wearing neck brace":
        0.116279,
      "arm in sling":
        0.116279,
      "wearing knee brace visible":
        0.110465,
      "using service dog on leash":
        0.104651,
      "visible tattoos on arms":
        0.098837,
      "facial piercing (nose stud or lip
ring)": 0.069767,
    },
  },
},
"hard": {
  "_sampling_config": {
    "pose_only": 1/3,
    "occlusion_only": 1/3,
    "pose_and_occlusion": 1/3,
  },
  "Body Pose": {
    "prob": 1,
    "items": [
      # --- Deep squat / crouch variants ---
      "deep squat heels lifted",
      "crouching low arms forward",
      "deep squat with arms wrapped around
knees",
      "low crouch one hand touching ground",

```

```

# --- Seated twisted / complex ---
"cross-legged seated torso twist",
"sitting slouched with one knee raised
",
"sitting sideways on chair backrest",
"seated twisted looking behind",
"seated hugging knees to chest",
"seated cross-leg one foot forward",
"seated with both legs dangling off
high surface",
"seated on floor spine curved over
tucked knees",
"seated reclined on hands behind body",

# --- Kneeling complex ---
"kneeling both knees leaning forward",
"kneeling with torso twist and reach",
"one-knee kneel arms overhead",
"kneeling upright arms stretched
behind back",
"kneeling low forehead near ground
arms forward",
"kneeling on both knees arms wide open
",
# --- Extreme standing / balance ---
"extreme contrapposto with shoulder
drop",
"standing on toes and arms raised",
"balancing on one leg and forward lean
",
"off-balance stumble recovery",
"standing one leg extended to the side
",
"heel-to-toe balance with stance arms
out",
"single leg stand and knee raised high
",
"standing deep forward lean from hips
arms behind",
"standing on one leg other extended
backward",
# --- Walking with complex body
dynamics ---
"walking toward camera foreshortened",
"walking away looking back over
shoulder",
"running diagonal across frame",
"sudden stop mid-run twist",
"walking fast with exaggerated arm
swing",
"walking with torso rotated opposite
to hips",
"brisk walk arms raised",
"walking with head turned sharply to
one side",
"lunging forward step with arms
swinging opposite",
# --- Jumping / airborne ---
"jumping knees tucked",
"jumping with torso twist",
"mid-air one leg extended",
"jump with arms wide both legs off
ground",
"one-legged hop arms asymmetric",
"jump with one knee tucked one leg
straight",
# --- Deep bend / reach ---
"bending deeply to pick object",
"reaching overhead extreme stretch",
"reaching down across body",
"extreme side stretch one arm over
head",
"forward fold hands reaching toward
ground",
"lateral lunge with torso over bent
knee",
"bending sideways one arm trailing
downward",
"reaching up and back arched spine",

```

```

# --- Arms / hands blocking face or
chest ---
"arms crossed overhead blocking face",
"hands covering face",
"hands behind head elbows forward",
"both arms raised with elbows bent and
fists up",
"arms wrapped tight around torso
resulting self-hug pose",
"both hands pressed flat against chest
",
"arms raised overhead with wrists
crossed",
"forearm blocking face from front",
# --- Leaning extreme ---
"leaning forward with hands on thighs",
"leaning sideways with torso bend",
"leaning backward with unsupported
torso",
"extreme lean to one side with arm
reaching floor",
"leaning far forward with body nearly
horizontal",
# --- Lying / ground complex ---
"lying supine with one knee bent",
"lying on side with upper leg raised",
"lying prone on elbows",
"lying face up with arms spread wide",
"lying on back with knees to chest and
arms hugging shins",
"lying on stomach with chin resting on
stacked hands",
"lying on side with top arm raised
toward ceiling",
"sitting on ground with leaning back
arms support",
# --- Dance / performance ---
"dancing spin",
"asymmetric freestyle dance pose",
"sharp hip-hop stance arms bent angles
",
"contemporary dance split weight arms
flowing",
"ballet-inspired arabesque arms
extended",
"jazz hands pose arms wide fingers
spread",
"pop-and-lock frozen pose angled torso
",
# --- Yoga / fitness complex ---
"yoga balance forward fold",
"yoga half-moon pose",
"yoga downward dog legs visible",
"yoga child pose arms forward",
"yoga chair pose knees bent arms
raised",
"yoga triangle pose deep lateral
stretch",
"plank position that is full body
horizontal",
"side plank with one arm raised body
angled",
# --- Stairs / environment interaction
---
"stairs climb mid-step",
"stairs descent while leaning forward
with weight on front foot",
"stepping over obstacle with wide
stance",
"ducking under low object with
compressed torso",
"sprinting start position that is
crouched forward with weight on toes",
"extreme twisting reach overhead with
curved back",
]
},

```



```

"Occlusion": {
  "prob": 1,
  "items": [
    # --- Object in front / body-level
occlusion ---
    "object occluding torso",
    "object occluding lower body",
    "object occluding upper body",
    "holding large object blocking torso",
    "bag carried in front blocking torso",
    "child held in front blocking torso",
    "carrying cardboard box blocking chest
and arms",
    "holding wide tray blocking torso and
lower face",
    "holding large framed picture covering
torso",
    "carrying surfboard vertically
blocking one side of body head to toe",
    "pushing laundry cart blocking lower
half",
    "holding oversized stuffed toy
blocking chest",
    "carrying bundled firewood across arms
blocking torso",
    "pushing full luggage trolley blocking
lower body",
    "arms laden with stacked books
blocking mid-body",
    "carrying infant car seat in front
blocking lower torso",
    "gripping large sporting equipment (
golf bag, ski poles) beside body",
    "holding long roll of fabric obscuring
one arm and torso",
    "carrying ladder diagonally blocking
upper body",
    "holding large pizza box stack
blocking torso",
    "carrying rolled carpet blocking side
of body",
    "holding wide poster board covering
chest",
    "carrying large potted plant blocking
torso",
    "holding guitar case vertically
blocking torso",
    "carrying folded table blocking lower
body",
    "holding stacked moving boxes blocking
face and chest",
    "carrying mattress on back obscuring
body shape",
    "holding large glass blocking entire
torso",
    "carrying kayak overhead blocking
upper body",
    "pushing wheelchair in front blocking
lower body",
    "carrying large cooler blocking torso
and arms",
    "holding wide whiteboard blocking
upper body",
    "carrying bundled tent poles blocking
shoulder and arm",
    "holding large flat screen TV box
blocking torso",
    "carrying yoga ball in front blocking
torso",
    "holding large drum blocking
midsection",
    "carrying stacked plastic chairs
blocking side",
    "holding large wicker basket blocking
lower torso",
    "carrying amplifier blocking one side
of body",
    "holding large art canvas blocking
chest and face",

```

```

    "carrying ski equipment bundle
blocking torso",
    "holding large umbrella blocking head
and shoulders",
    "holding large serving platter
blocking chest",
    # --- Furniture / counter ---
    "counter or desk blocking lower body",
    "furniture blocking mid body",
    "vehicle door partially blocking body",

    "sitting at kitchen island counter
visible from waist up",
    "seated behind restaurant booth table
visible from chest up",
    "standing at reception desk only upper
body visible",
    "seated at classroom desk top visible
from waist up",
    "standing behind workbench with tool
blocking forearm",
    "seated on sofa with cushion on lap
blocking lower torso",
    "leaning over car hood blocking lower
half of body",
    "seated in theater seat with seat back
visible chest up",
    "behind bakery display case visible
from shoulders up",
    "sitting inside car window framing
shoulders and head only",
    "seated at bar with bar top blocking
waist down",
    "standing behind DJ booth visible from
chest up",
    "seated at library study carrel desk
blocking lower body",
    "standing behind museum information
desk",
    "seated in cubicle with partition
blocking lower body",
    "standing behind pharmacy counter
visible from shoulders up",
    "seated at conference table visible
from chest up",
    "standing behind hotel concierge desk",

    "seated at piano bench with piano
blocking lower body",
    "standing behind lab bench with
equipment blocking torso",
    "seated at drafting table visible from
waist up",
    "standing behind checkout counter with
register blocking view",
    "seated in booth with high table
blocking chest",
    "standing behind podium with
microphone blocking chest",
    "standing behind bank teller window
visible from shoulders up",
    "seated at sewing machine with table
blocking lap",
    # --- Environmental structural ---
    "tree or foliage blocking body parts",
    "doorway frame blocking body parts",
    "window frame blocking torso",
    "partial body behind wall",
    "partial body behind pillar or pole",
    "standing behind fence or bars",
    "standing in tall grass blocking lower
legs and feet",
    "standing behind thick tree trunk
blocking torso center",
    "peeking around corner with left or
right half of body hidden by wall",
    "standing in deep doorway shadow
blocking one side",
    "behind iron gate bars creating

```

vertical stripe occlusion",  
 "standing behind densely leafed hedge  
 only head and torso visible",  
 "exiting through revolving door frame  
 bisecting body",  
 "standing at edge of crowd where  
 shoulder is behind another person",  
 "behind curtain pulled aside exposing  
 partial body",  
 "partially hidden behind open car  
 trunk lid",  
 "framed in small window cutout showing  
 torso only",  
 "standing behind small jersey barrier  
 on roadside",  
 "standing behind concrete pillar  
 blocking left or right half of body",  
 "standing in tall wheat field blocking  
 lower body",  
 "behind chain-link fence creating grid  
 pattern over body",  
 "standing behind scaffolding bars  
 blocking sections of body",  
 "partially obscured by hanging vines",  
 "standing behind brick column blocking  
 side of body",  
 "standing in tall corn stalks  
 obscuring body",  
 "behind staircase railing with  
 vertical bars",  
 "standing behind metal grating  
 blocking torso",  
 "partially hidden by large boulder",  
 "standing behind ornate room divider  
 screen",  
 "behind lattice panel creating diamond  
 pattern occlusion",  
 "standing in dense bamboo forest  
 blocking body parts",  
 "behind venetian blinds creating  
 horizontal stripe pattern",  
 "standing behind wire mesh fence",  
 "partially obscured by hanging beads  
 or curtain",  
 "standing behind pergola posts  
 blocking sections",  
 "behind tall cattails or reeds  
 blocking lower body",  
 "standing in archway with arch  
 blocking forehead",  
 "behind wrought iron decorative fence",  
 "standing behind concrete barrier wall",  
 "partially occluded by large  
 industrial pipes",  
 # --- Severe crop ---  
 "standing beside open car door with  
 door blocking one arm and hip",  
 "forehead blocked by low hanging  
 awning edge",  
 "body seen from behind through narrow  
 slit in wall",  
 "frame showing only neck to hip region",  
 "both arms extended into adjacent  
 doorways out of central view",  
 "standing inside manhole or pit with  
 upper body visible only",  
 "only legs visible under long dividing  
 table",  
 # --- Mirror / glass / reflection ---  
 "frosted glass behind which body is  
 blurred",  
 "heavy condensation on glass obscuring  
 body",  
 "car window tinting making lower body  
 very dark",  
 # --- Lighting-based ---

"strong shadow obscuring body regions",  
 "heavy cast shadow from overhead  
 structure covering torso",  
 "overexposed foreground washing out  
 body detail",  
 "dappled sunlight through leaves  
 creating irregular occlusion",  
 "neon sign glow overexposing and  
 losing body detail",  
 "lens flare obscuring portions of body",  
 "heavy fog obscuring lower body",  
 "smoke obscuring upper body",  
 "steam obscuring face and chest area",  
 "venetian blind shadows creating  
 horizontal stripes across body",  
 "tree shadow pattern breaking up body  
 outline",  
 "underexposed making dark clothing  
 merge with background",  
 "sunset backlight creating rim light  
 only on edges",  
 # --- Worn / clothing-based ---  
 "large waterproof cover obscuring body  
 shape",  
 "large tool belt and pouches wrapping  
 around hips and thighs",  
 "small mascot costume head obscuring  
 upper body",  
 # --- Foreground object blur / partial  
 ---  
 "foreground blur object blocking body",  
 "plastic wrap or cover partially  
 blocking view",  
 "advertisement board blocking body",  
 "signboard blocking torso",  
 "handheld box blocking torso and arms",  
 "blurred foreground fence post  
 partially obscuring hip",  
 "out-of-focus foreground umbrella edge  
 clipping shoulder",  
 "blurred bokeh foreground leaves  
 obscuring lower torso",  
 "foreground crowd member's arm  
 partially crossing frame",  
 "out-of-focus foreground vehicle roof-  
 line obscuring waist area",  
 "blurred foreground crowd hand  
 blocking hip and thigh",  
 "out-of-focus foreground street sign  
 partially blocking head",  
 "standing inside phone booth frame",  
 "blurred foreground railing obscuring  
 midsection",  
 "out-of-focus foreground person  
 blocking half the frame",  
 "blurred foreground hanging sign  
 blocking upper body",  
 "out-of-focus foreground branch  
 crossing face",  
 "blurred foreground architectural  
 element blocking torso",  
 "out-of-focus foreground balloon  
 cluster blocking view",  
 "blurred foreground flag partially  
 blocking body",  
 "out-of-focus foreground wire or cable  
 crossing body",  
 "blurred foreground awning edge  
 blocking upper frame",

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        "out-of-focus foreground crowd heads
blocking lower body",
        "blurred foreground shopping cart
blocking side",
        "out-of-focus foreground banner
blocking torso",
        "blurred foreground car mirror
blocking shoulder area",
        "out-of-focus foreground flower
bouquet blocking chest",
        "blurred foreground fence creating
bokeh pattern over body",
    ]
},

    "Background/lighting/shadow": {
        "prob": 0.35,
        "items": {
            # normalised from original sum of
0.982
            "dramatic lighting":          0.096741,
            "golden hour lighting":       0.091650,
            "cloth store":                0.086558,
            "neon lighting":               0.081466,
            "street light":                0.076375,
            "disco light" :                0.069246,
            "dawn lighting":               0.066191,
            "dusk lighting":               0.064155,
            "artificial stage lighting":  0.061100,
            "colored gel lighting":        0.059063,
            "incandescent lighting":       0.056008,
            "hard side light":              0.053971,
            "multiple light sources":      0.050916,
            "chiaroscuro":                  0.045825,
            "candlelight":                  0.040733,
        }
    },

    "Camera view": {
        "prob": 0.20,
        "items": {
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0.892
            "back view":                   0.013453,
            "angled view":                  0.123318,
            "turned body":                  0.112108,
            "extreme angle":                0.106502,
            "overhead view":                0.100897,
            "low angle view":                0.095291,
            "Dutch angle":                  0.089686,
            "upward looking":                0.084081,
            "turned away":                   0.078475,
            "partial back view":             0.072870,
            "diagonal view":                 0.067265,
            "worm's eye view":               0.056054,
        }
    },

    "Special cases": {
        "prob": 0.15,
        "items": {
            # normalised from original sum of
0.500
            "albinism":                     0.210000,
            "full body cast on one limb":    0.190000,
            "large-scale visible tattoo covering
arms or neck": 0.170000,
            "burn scars on visible skin":    0.160000,
            "vitiligo on arms and face":     0.150000,
            "tribal or cultural body markings": 0.120000,
        }
    },
},
}

```

```

def get_difficulty_dict(tier):
    """Return the configuration for easy, medium, or
    hard."""
    return DIFFICULTY_DICTIONARIES[tier.lower()]

def get_medium_special_cases(gender):
    """Return the gender-conditioned medium-tier
    special-case distribution."""
    return DIFFICULTY_DICTIONARIES["medium"][ "Special
cases" ][ "items_by_gender" ][gender.lower()]

```